



**INSIGHTSIAS**  
SIMPLIFYING IAS EXAM PREPARATION

# INSIGHTS

## CURRENT AFFAIRS



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# GENERAL STUDIES - 1

Topic: Indian Culture-Salient aspects of Art Forms

## ARTIFICIAL INTELLIGENCE FOR CULTURE AND LANGUAGES

- This initiative represents a strategic shift from merely **preserving** cultural resources to enabling active **cultural participation**. By leveraging AI, the Government of India aims to democratize technology, making manuscripts, monuments, and oral traditions accessible to citizens in their native languages.
- It positions AI as **Technology for Humanity**, aligning with the vision of *Welfare for All and Happiness for All*.

### Role of Artificial Intelligence in the Conservation of Culture and Language

- Digitization and Discovery of Manuscripts:** Artificial Intelligence enables **rapid scanning and cataloguing of fragile manuscripts**, protecting them from decay and improving scholarly access through automated metadata creation.  
E.g. The **Gyan Bharatam Mission** has documented over **44 lakh manuscripts**, using Artificial Intelligence tools for intelligent cataloguing.
- Multilingual and Voice-Based Access:** Artificial Intelligence removes **language and literacy barriers** through real-time speech-to-text conversion and translation services.  
E.g. At **Kashi Tamil Sangamam 2.0**, the Prime Minister's Hindi speech was translated into Tamil using **BHASHINI (Bhasha Interface for India)**.
- Preservation of Tribal Languages:** Artificial Intelligence records **oral traditions and endangered languages**, helping build digital linguistic archives.  
E.g. The **Adi Vaani platform** supports languages such as **Santali, Bhili, and Gondi**.
- Digital Access for Artisans:** Artificial Intelligence helps artisans showcase **Geographical Indication products globally** through multilingual catalogues.
- Support for Cultural Events:** Artificial Intelligence chatbots provide **multilingual guidance and navigation** during major cultural gatherings.

|                                                                                                                                                                                            |                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AI-Driven Communication &amp; Education</b><br><b>BHASHINI (National Language Translation Mission)</b><br>AI-led translation and speech-to-text services across 22 Scheduled languages. | <b>Heritage Preservation &amp; Technical Standards</b><br><b>Gyan Bharatam Mission (2024-31)</b><br>A ₹482.85 crore mission focused on digitising and disseminating India's manuscript heritage. |
| <b>Anuvadini AI Platform</b><br>Translates technical and academic textbooks into regional languages to ensure knowledge access.                                                            | <b>Technology Development for Indian Languages (TDIL)</b><br>Standardizing OCR, machine translation, and handwriting recognition for Indian scripts.                                             |
| <b>Adi Vaani for Tribal Languages</b><br>Supports subtitling for health advisories and government messages in native tribal tongues.                                                       |                                                                                                                                                                                                  |

### Key Challenges Associated

- Literacy and Digital Barrier:** Many cultural practitioners remain **excluded from advanced digital tools** due to limited formal literacy and unfamiliarity with technology.  
E.g. Despite **BHASHINI (Bhasha Interface for India)** outreach, artisans in remote clusters still struggle to manage digital storefronts without **voice-based assistance systems**.
- Scattered and Undocumented Intellectual Wealth:** Much of India's manuscript heritage remains **privately held in temples, mutts, and personal collections**, limiting centralized Artificial Intelligence processing.  
E.g. The **Gyan Bharatam Mission** faces challenges surveying manuscripts held by custodians reluctant to share collections.
- Low-Resource Language Datasets:** Endangered languages lack **sufficient digital text corpora** required to train accurate **Large Language Models**.  
E.g. Languages like **Kui and Garo** remain under development in the **Adi Vaani platform**.
- Authenticity and Provenance Issues:** Digitization raises **verification challenges for Geographical Indication products** and traditional designs.  
E.g. Mass producers often replicate traditional patterns, requiring stronger **Artificial Intelligence tagging systems**.
- Offline Access and Infrastructure Gaps:** Many heritage areas lack **stable internet and computing infrastructure** needed for Artificial Intelligence applications.  
E.g. The **NITI Aayog** highlights the need for **offline-capable Artificial Intelligence systems**.



## Way Ahead

- **Language as Digital Public Infrastructure:** Expanding the **Language Layer** so governments and startups can build inclusive multilingual applications without developing systems from scratch.
- **Verifiable Digital Credentials:** Using **Artificial Intelligence-based certification systems** to issue reliable skill credentials for artisans, improving employability and market trust.
- **Encouraging Local Innovation:** Establishing **district-level Digital Work Hubs** to promote regional language content creation and digital skill development.
- **Multi-Stakeholder Collaboration:** Linking **Indian Institutes of Technology (IITs), Indian Institutes of Information Technology (IIITs), industry, and communities** to ensure inclusive Artificial Intelligence solutions.
- **Democratization of Technology:** Promoting **open-source Artificial Intelligence models** so cultural preservation technologies remain accessible public resources.

## Conclusion

India is positioning **Artificial Intelligence** as a guardian of **civilizational heritage** through initiatives like **BHASHINI (Bhasha Interface for India)** and **Gyan Bharatam**.

## Topic: Demography and Associated issues

### INFERTILITY IN INDIA

- **Infertility** is emerging as a major public health challenge in India in **2026**, with experts noting that **mental health stress can physiologically affect reproductive capacity** in all genders.
- **Definition and Scope:** Infertility means failure to conceive after **12 months of regular unprotected intercourse**. Once viewed mainly as a women's issue, evidence shows **nearly equal male and female causes**.
- **Emerging Trend:** Attention is shifting to **male infertility** and the biological impact of **psychological stress on reproductive cells (gametes)**.

#### CURRENT TRENDS AND MARKET DATA

**15-20%**

**National Prevalence**  
Nearly 30 million Indian couples grapple with infertility, with higher rates in urban centers.

**1.9**

**TFR Dipped to 1.9**  
India's Total Fertility Rate is now well below the replacement level of 2.1.

**\$1.8 Billion**

**IVF Market by 2029**

The Indian IVF market is projected to double from \$900 million in 2024.

#### Factors Influencing Infertility

| Factor      | Primary Observation                                      |
|-------------|----------------------------------------------------------|
| Male Factor | Accounts for 40-50% of cases due to toxins and stress    |
| Urban Areas | Higher primary infertility (never conceived)             |
| Rural Areas | Higher secondary infertility due to untreated infections |

#### GOVERNMENT INITIATIVES AND REGULATION



#### ART and Surrogacy (Regulation) Acts

2025 guidelines mandate clinic registration and limit egg donation to once in a lifetime.



#### National Digital Registry

Established to track ART outcomes and ensure clinics provide transparent success rates.



#### Project Sanjivini

A collaborative pilot to disseminate reproductive health knowledge at the grassroots level.

## Causes of Rising Infertility in India

- **Delayed Parenthood:** Career priorities and financial goals push parenthood beyond the **biological reproductive prime**, reducing fertility chances.  
E.g. Urban centres like **Bengaluru and Mumbai** saw a **25% rise in women seeking fertility treatment after 35** during 2025-26.
- **Environmental Pollution: Endocrine Disrupting Chemicals (EDCs)** in air and water disturb hormonal balance affecting fertility.  
E.g. Studies in **Delhi** link severe air pollution days with **temporary decline in sperm motility** among young men.
- **Lifestyle Disorders:** Sedentary lifestyles increase **obesity and Polycystic Ovary Syndrome (PCOS)**.  
E.g. Around **one in five Indian women** suffers from PCOS causing ovulatory infertility.
- **Psychological Stress:** High cortisol disrupts the **Hypothalamic-Pituitary-Adrenal (HPA) axis**, affecting reproduction.
- **Untreated Infections: Sexually Transmitted Infections (STIs) and Pelvic Inflammatory Disease (PID)** often cause tubal infertility.

## Challenges Associated with Infertility

- **Social Stigma and Ostracization:** Women facing infertility often experience **social exclusion and derogatory labeling**, particularly in conservative communities, affecting dignity and mental wellbeing.  
E.g. In rural **Tamil Nadu**, the term "**Maladi**" is still used to stigmatize women unable to conceive.
- **Prohibitive Treatment Costs: In Vitro Fertilization (IVF) and Assisted Reproductive Technology (ART)** remain expensive for most households.  
E.g. A typical **IVF cycle costs ₹1.5-3 lakh**, while most insurance policies still exclude infertility

coverage.

- **Silence of Male Infertility:** Patriarchal norms discourage men from seeking **semen analysis**, delaying diagnosis.  
E.g. Clinical studies show men often wait **3–5 years longer** than women for fertility screening.
- **Psychological Feedback Loops:** Stress hormones reduce chances of successful conception.
- **Regulatory Gaps:** Some Tier II and Tier III city clinics lack standardized treatment protocols.

### Way Ahead

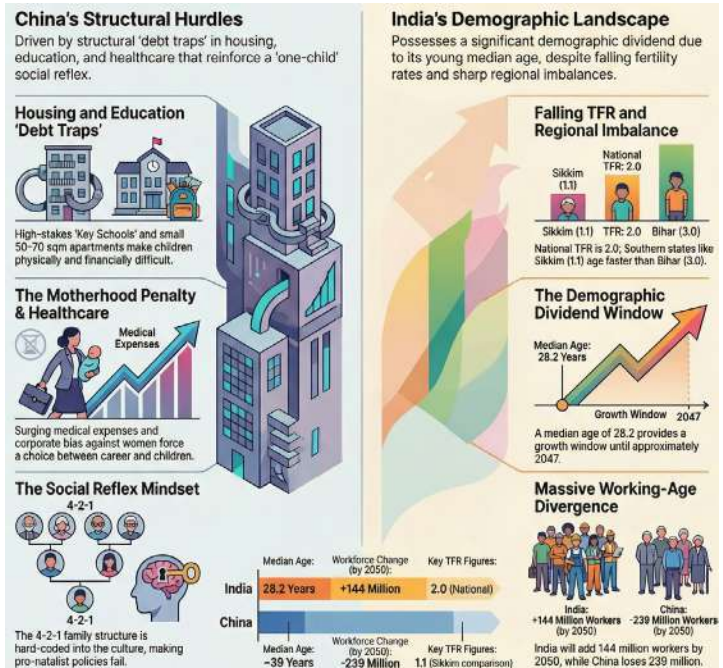
- **Integrating Mental Health:** Counselling should become a **mandatory component of In Vitro Fertilization (IVF)** cycles to help couples manage emotional stress during treatment.
- **Insurance Inclusion:** Insurance Regulatory and Development Authority of India (IRDAI) should mandate partial infertility coverage in health insurance.
- **Workplace Sensitivity:** Companies should introduce **fertility leave and egg-freezing insurance support** for employees delaying parenthood.
- **Male-Centric Campaigns:** Public health campaigns must **de-stigmatize male infertility**, encouraging early semen analysis.
- **Community Education:** Accredited Social Health Activist (ASHA) workers should spread awareness that infertility is **a medical condition, not a curse**.

### Conclusion

Infertility is a **biological, social, and psychological challenge** requiring gender-neutral care and empathetic support.

## CHINA'S POPULATION WOES

- China is currently facing a **demographic death spiral** where the population is not only shrinking but aging rapidly. Decades of the [One-Child Policy](#) created a "4-2-1" family structure (four grandparents, two parents, one child), placing an immense economic burden on the youth.
- The crisis has transitioned from a lack of permission to have children to a lack of ability or desire, driven by the high costs of housing, education, and healthcare.



### Challenges Associated

- **Job Creation Gap:** A large youth population becomes a dividend only when **sufficient employment opportunities** exist; otherwise it risks turning into a demographic liability.  
E.g. High **youth unemployment exceeding 20% in some age groups** contributes to social unrest and skilled migration abroad.
- **Skilling Mismatch:** The education system often fails to create **industry-ready graduates**, widening the employability gap.  
E.g. Studies show **only about 50% of Indian graduates are employable**, highlighting the importance of the **Skill India Mission**.
- **Ageing Before Affluence:** Some regions are **ageing before achieving high income levels**.  
E.g. **Tamil Nadu** faces rising demand for geriatric healthcare and pensions.
- **Resource Strain:** Rapid urbanization pressures **water and air quality**.  
E.g. Crises in **Bengaluru and Delhi–National Capital Region**.
- **Dependency Ratio Shifts:** Rising elderly population will **increase the support burden on workers**.

### Lessons for India from China's Woes

- **Avoid Coercion:** Population management should rely on **education, healthcare, and awareness**, rather than strict legislative controls.
- **Subsidize Social Pillars:** Governments must ensure **affordable education, healthcare, and housing**, so family decisions are not constrained by rising living costs.
- **Support Working Women:** Reduce the



**motherhood penalty** through paid maternity leave and stronger childcare systems like the **National Creche Scheme**.

- **Plan for Regional Divergence:** Policies should address **aging southern states** and **youthful northern regions** differently.
- **Focus on Quality Education:** Expand **equitable, high-quality schooling** nationwide rather than creating scarcity-driven elite systems.

### Conclusion

China's experience shows **low fertility is difficult to reverse once entrenched**. India must utilize its **demographic window** by investing in human capital, inclusive growth, and policies supporting families sustainably.

# GENERAL STUDIES - 2

Topic: Federal Structure

## INDIA'S FEDERALISM IS IN NEED OF A STRUCTURAL RESET

The debate on India's federalism has been reignited following the submission of **Part I of the Justice Kurian Joseph Committee Report**.

Commissioned by the Tamil Nadu government, the report advocates for a structural reset to restore state autonomy and right-size the Union's role in governance.

About India's federalism is in need of a structural reset:

- A structural reset refers to a fundamental recalibration of the power balance between the Union and the States. It moves away from the **centralizing bias** of the 1940s—born out of post-Partition anxieties—toward a model where States have autonomous legislative and fiscal space.



### Need for a Structural Reset

- Erosion of Fiscal Autonomy:** The Union's rising dependence on **cesses and surcharges**, excluded from devolution under **Article 270**, steadily reduces States' constitutionally mandated share in the divisible tax pool.  
**E.g.** By 2025, non-divisible levies exceeded **18% of gross tax revenue**, constraining State finances.
- Centrally Sponsored Schemes (CSS) Micromanagement:** Centrally Sponsored Schemes (CSS) impose uniform templates, disregarding diverse regional needs and capacities.  
**E.g.** Replacement of the **Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)** with the **Viksit Bharat (Gramin) Act** limited flexible rural planning.
- Encroachment on Concurrent List:** Increasing use of subordinate legislation overrides State-specific

laws in shared subjects.

**E.g.** The **National Education Policy (NEP)** and central health norms often conflict with regional priorities.

- Overburdened Union Executive:** Excessive central supervision weakens administrative effectiveness.  
**E.g.** The **National Institution for Transforming India (NITI Aayog)** benchmarks local issues like water management.
- Political Maturity of States:** Modern States are politically consolidated and policy innovative.  
**E.g.** Tamil Nadu's Noon Meal Scheme evolved into a national model.

### Challenges Associated with the Reset

- Delimitation Concerns (2026):** Reallocation of **Lok Sabha seats** based on post-2026 population data may disadvantage States that effectively controlled demographic growth, raising concerns over federal equity and representation balance.  
**E.g.** Southern States fear reduced parliamentary strength compared to northern States after delimitation.
- The Governor's Interventions:** Frequent use of discretionary powers, especially delays in granting assent to Bills, has strained Centre-State relations and invited judicial review.  
**E.g.** Kerala and Tamil Nadu saw prolonged gubernatorial delays (2024–25), leading to Supreme Court scrutiny.
- Fiscal Deficit Caps:** Union-imposed borrowing ceilings can restrict State welfare spending and infrastructure investment during economic transitions.  
**E.g.** Kerala's 2025 challenge questioned limits on State borrowing autonomy.
- One-Size-Fits-All Digital Mandates:** Centralized digital governance models may weaken State administrative flexibility and create exclusion risks.  
**E.g.** Aadhaar-based payment systems caused benefit gaps in remote rural regions.
- National Security vs. State Policing:** Expansion of Central agency jurisdiction can overlap with State law-and-order powers, intensifying federal tensions.  
**E.g.** Punjab and West Bengal opposed enhanced roles of the Border Security Force and Central investigative agencies.

### Way Ahead

- Reform the Finance Commission Terms of Reference:** Include cesses and surcharges in the divisible pool to ensure States receive genuine, untied fiscal resources and strengthen cooperative federalism.
- Revitalize the Inter-State Council:** Convert it into a mandatory consultative forum for legislation con-

cerning the Concurrent List, enhancing structured Centre–State dialogue.

- **Institutionalize Subsidiarity:** Adopt the principle that governance functions should be performed by the lowest competent authority, allowing the Union government to focus on strategic national priorities.
- **Governor Reform:** Implement Punchhi Commission recommendations to curb discretionary powers and prescribe a time frame for granting assent to Bills.
- **Protect Political Parity:** Decouple delimitation from pure population metrics to safeguard representation of performing States.

### Conclusion

- A structural reset seeks a **right-sized Union**, aligning authority with accountability.
- Balanced federalism strengthens partnership governance and sustains India's diversity amid global competition.

Topic: Constitutional, Statutory, Quasi-Judicial, Executive Bodies

## INDEPENDENCE OF THE ELECTION COMMISSION

The independence of the **Election Commission of India (ECI)** has come under debate following controversies related to voter roll revisions and opposition moves seeking removal of the [Chief Election Commissioner](#) (CEC).

About Independence of the Election Commission:

- Independence of the [Election Commission](#) means its ability to conduct elections free from executive, political, or external influence, ensuring impartial supervision, direction, and control of elections in India.

### Safeguarding the Vote: Constitutional and Judicial Oversight of the Election Commission

The Election Commission (EC) is protected by specific constitutional articles and Supreme Court rulings designed to prevent executive interference. These safeguards range from tenure security to the establishment of elections as a "Basic Structure" of the constitution.

#### Constitutional Safeguards

##### Article 324: Vested Authority

Vests the superintendence, direction, and control of elections entirely in the Election Commission.

##### Security of Tenure and Service Conditions

CEC removal requires Supreme Court judge standards; service conditions cannot be negatively varied after appointment.

##### Multi-Member Body (Since 1993)

A three-member body ensures collective decision-making and prevents individual bias in election oversight.

#### Landmark Supreme Court Judgments

##### Free Elections as "Basic Structure"

Indira Gandhi v. Raj Narain (1975) and Mohinder Singh Gill (1978) established elections as foundational; ruled Article 324 allows the EC to act even where law is silent.

##### Institutional Status and Equality

T.N. Seshan v. Union of India upheld the multi-member status and equated ECs with the CEC.

##### Appointment Neutrality and the CJI

Anoop Baranwal (2023) and Jaya Thakur (2024-2026) examine the CJI's role; the court held that the CJI should be part of the selection committee.

### Need for Independence of the Election Commission of India

- **Ensuring Level Playing Field:** Prevents misuse of State machinery and enforces the Model Code of Conduct during elections.  
**E.g.** In the 2024 General Elections, strict action was demanded against high-profile speeches.
- **Maintaining Public Trust:** Protects ballot credibility and democratic stability through transparency.  
**E.g.** The 2025 vote theft allegations required detailed clarification to prevent unrest.
- **Accuracy of Electoral Rolls:** Independent oversight avoids targeted voter exclusion.  
**E.g.** The 2025 Bihar Special Intensive Revision controversy raised deletion concerns.
- **Countering Executive Overreach:** Resists manipulation of election timing for political gain.  
**E.g.** One Nation, One Election debates required neutral feasibility assessment.
- **Effective Adjudication:** Functions as a quasi-judicial authority in party disputes.  
**E.g.** Decisions on Shiv Sena and Nationalist Congress Party splits (2023–24) underscored impartiality.

### Challenges Associated with Election Commission of India Independence

- **Appointment Process:** The 2:1 selection panel—Prime Minister and Union Minister versus Leader of Opposition—enables executive dominance in appointments.  
**E.g.** Rapid appointment of two Election Commissioners in March 2024, before Supreme Court review, triggered criticism.
- **Lack of Financial Autonomy:** Expenditure is voted by Parliament instead of being charged on the Consolidated Fund of India.  
**E.g.** In 2025, Bihar Special Intensive Revision funding required Ministry of Finance approval.
- **Vulnerability of Election Commissioners:** Unlike the Chief Election Commissioner, others lack equal removal safeguards, affecting independence.  
**E.g.** This may encourage alignment for eventual promotion.
- **Electoral Roll Discrepancies:** Technology-based roll purification risks wrongful deletions.  
**E.g.** Haryana's 2025 automated de-duplication reportedly removed many legitimate voters.
- **Information Transparency:** Restricted access to polling data raises accountability concerns.  
**E.g.** A June 2025 directive mandated destruction of closed-circuit television footage within 45 days.

### Way Ahead



- **Restoring Judicial Oversight:** Reintroduce the **Chief Justice of India** in the selection committee to ensure impartial appointments.
- **Equal Protection for all Commissioners:** Amend **Article 324(5)** to grant Election Commissioners safeguards equal to the **Chief Election Commissioner**.
- **Independent Secretariat:** Create a dedicated administrative cadre, reducing reliance on government-deputed officials.
- **Financial Independence:** Charge expenditure on the **Consolidated Fund of India** to prevent Ministry-level control.
- **Standardized Deletion Protocols:** Codify Model Electoral Roll Rules mandating physical hearings before voter deletion.

### Conclusion

- Independence of the **Election Commission of India** safeguards electoral legitimacy.
- Judicial and legislative reforms are vital to curb executive dominance and protect democratic mandates.

Topic: Role of Civil Services in a Democracy

## BRIDGING A DIVIDE WITH AN INDIAN SCIENTIFIC SERVICE

The proposal for an Indian Scientific Service (ISS) is in the spotlight following the Economic Survey 2025-26 and recent high-level meetings of the Empowered Technology Group, which highlighted the need for a specialized cadre to manage [India's Deep-Tech](#) and AI-first governance transition.

### About Bridging a divide with an Indian Scientific Service:

The ISS is envisioned as a **permanent, all-India specialized cadre** of scientists and technocrats. Unlike the generalist [Indian Administrative Service](#) (IAS), the ISS would:

- Integrate scientific expertise directly into the **decision-making hierarchy** of ministries.
- Operate under **distinct service rules** that prioritize scientific integrity and peer review over traditional administrative neutrality.
- Provide a structured career path for researchers to contribute to policy without being stifled by colonial-era conduct rules.



### The Need for a Dedicated Indian Science Service

- **Complexity of Modern Governance:** Emerging fields like biotechnology and Artificial Intelligence demand technical expertise beyond generalist administration.  
**E.g.** Drafting the Digital India Act 2025 required understanding algorithmic bias and advanced data governance.
- **Bridging the Valley of Death:** Translating Technology Readiness Levels 1–3 research into Levels 7–9 market products needs sustained technical leadership.  
**E.g.** Green Hydrogen innovations face scaling delays due to fragmented oversight.
- **Scientific Integrity and Independence:** Central Civil Services Conduct Rules, 1964 may restrict candid scientific advice.  
**E.g.** Himalayan ecological warnings encountered bureaucratic resistance.
- **Global Competitiveness:** Scientist-diplomats strengthen Government-to-Government technological negotiations.  
**E.g.** Semiconductor talks with the United States and European Union require lithography expertise.
- **Long-term Foresight:** Scientific missions demand continuity beyond short administrative tenures.  
**E.g.** The National Supercomputing Mission needs decade-long planning stability.

### Challenges Associated with Indian Science Service

- **Generalist vs. Specialist Friction:** Integration may trigger turf conflicts between **Indian Administrative Service** and proposed **Indian Science Service** officers over authority and seniority.  
**E.g.** Friction in the Ministry of Health between medical experts and administrators has slowed policy execution.
- **Lateral Entry Resistance:** Mid-career induction of

scientists faces institutional pushback.

**E.g.** Early 2020s lateral entry reforms saw internal resistance.

- **Salary Parity:** Government pay scales struggle to attract global talent.  
**E.g.** Indian Space Research Organisation and Defence Research and Development Organisation lose experts to higher-paying tech firms.
- **Rigid Promotion Structures:** Seniority often outweighs scientific merit in career advancement.  
**E.g.** Council of Scientific and Industrial Research scientists cite research sidelining.
- **Defining the Boundary:** Balancing scientific advice with political priorities remains complex.  
**E.g.** Climate negotiations require reconciling emission goals with economic growth.

### Way Ahead

- **Pilot Cadres:** Launch in priority sectors like **Indian Environmental and Ecological Service** and **Indian Public Health Service**.
- **Structural Protection:** Legally record scientific assessments, even if final policy diverges.
- **Dynamic Pay Scales:** Introduce performance-linked, market-competitive salaries to prevent brain drain.
- **Collaborative Training:** Joint programs at **Lal Bahadur Shastri National Academy of Administration, Mussoorie** for **Indian Administrative Service** and **Indian Science Service** officers.
- **National Initiative for Developing and Harnessing Innovations-style Grants:** Provide financial autonomy for high-risk research.

### Conclusion

- Institutionalizing expertise strengthens technology-driven governance.
- Science must anchor policymaking for a future-ready India.

### Topic: Governance

## DIGITAL GOVERNANCE

India's draft amendments to the [IT Rules, 2021](#) have reignited debate in February 2026 over how to effectively curb deepfakes and synthetically generated content.

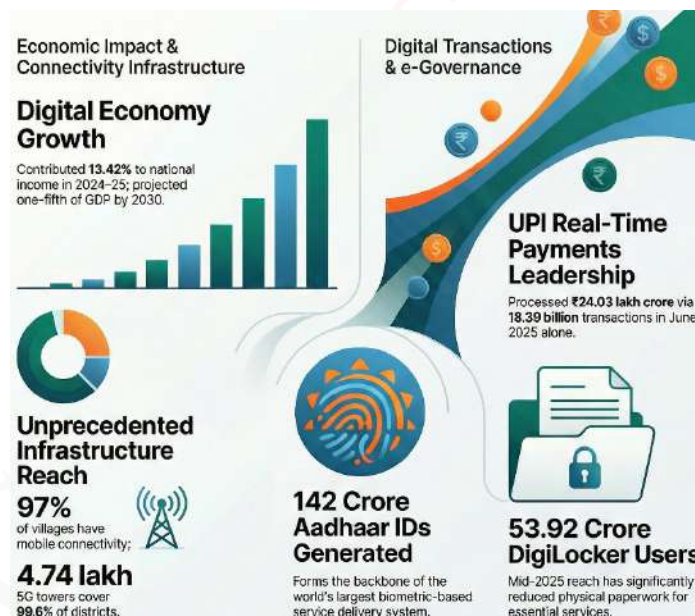
- The discussion has now shifted from platform takedowns to the urgent need for content provenance, mandating traceable digital tags or watermarks on all AI-generated media.

### About Digital Governance:

- Digital governance is the application of digital technologies (like AI, blockchain, and cloud computing) and constitutional principles (like

transparency, [accountability](#), and the rule of law) to transform public administration.

- It is not just about digitizing paperwork; it is a fundamental shift in how the state, market, and citizens interact to ensure services are fast, inclusive, and future-ready.



### Need for Digital Governance in India:

- **Ensuring Democratic Integrity:** Artificial Intelligence-generated deepfakes distort public discourse and may incite unrest during elections.  
**E.g.** In 2025, the **Indian Cyber Crime Coordination Centre (I4C)** intervened when fabricated political videos triggered communal tension.
- **Combating Gendered Cyber-Abuse:** Synthetic media disproportionately targets women through non-consensual explicit content.  
**E.g.** The **National Cyber Crime Reporting Portal** recorded a surge in deepfake complaints; over **90% of global deepfakes are pornographic**, reflecting similar trends in India.
- **Financial Security & Fraud Prevention:** Deepfakes enable biometric fraud and identity theft.  
**E.g.** In 2026, one in five biometric fraud attempts used animated selfies to bypass **Know Your Customer (KYC)** verification.
- **Inclusive Service Delivery:** Bridging language barriers ensures equitable access.  
**E.g.** **BHASHINI** supported 35+ languages at Maha Kumbh 2025.
- **Administrative Efficiency & Accountability:** Digital platforms curb leakages.



**E.g. Direct Benefit Transfer (DBT)** removed ghost beneficiaries in **Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)**.

### Challenges Associated with Digital Governance

- **Algorithmic Opacity:** Automated welfare and policing systems often lack transparency, limiting citizens' ability to challenge decisions.  
**E.g.** In 2026, concerns emerged over black-box **Artificial Intelligence (AI)** tools in predictive policing reinforcing social bias.
- **Surveillance & Privacy Erosion:** Extensive biometric and facial recognition monitoring may chill free expression.  
**E.g.** The 2026 Digital Constitutionalism debate questioned data collection under security applications and possible state overreach.
- **Digital & Literacy Divide:** Limited internet access and low digital skills exclude vulnerable groups.  
**E.g.** In 2025, rural elderly citizens struggled with biometric-only ration access despite 5G rollout.
- **Concentration of Power:** Big technology firms influence speech and markets without oversight.  
**E.g.** The Digital Competition Bill, 2024–25 targeted platform dominance.
- **High Infrastructure Costs:** Cybersecurity upgrades demand sustained funding.  
**E.g.** The National Cyber Security Strategy seeks major forensic investment.

### Ethical Principles Associated

- **Accountability (Dharma):** Every **Artificial Intelligence (AI)**-assisted decision must trace back to a clearly identifiable human authority to ensure responsibility.
- **Justice (Nyaya):** Systems should remain **fair, explainable, and culturally inclusive**, respecting India's linguistic and social diversity.
- **Transparency:** Citizens must know when interacting with **synthetic media or algorithms**, safeguarding informed participation.
- **Decisional Autonomy:** Technology should empower rational choice, not manipulate perceptions through deepfakes or behavioural targeting.

### Way Ahead

- **Designate a Lead Regulator:** Establish an autonomous AI regulator to unify oversight across ministries.
- **Implement 'CrediMark':** Introduce mandatory, tamper-proof provenance tags for synthetic content.
- **Risk-Tiered Obligations:** Impose stricter compliance in elections, governance, and finance.

- **Regulatory Sandboxes:** Enable supervised innovation in watermarking tools.
- **Public Awareness & Infrastructure:** Expand media-forensics labs and digital literacy drives.

### Conclusion

India's **Viksit Bharat 2047** vision depends on balancing innovation with constitutional safeguards, ensuring trustworthy and responsible digital governance.

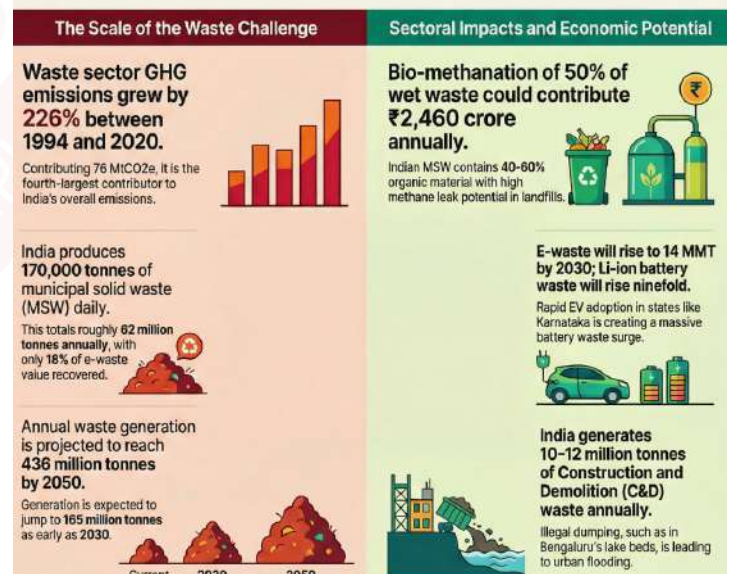
## NITI AAYOG STUDY: SCENARIOS TOWARDS VIKSIT BHARAT AND NET ZERO (WASTE SECTOR)

**NITI Aayog** released a landmark series of 11 reports detailing India's first government-led integrated study to align the Viksit Bharat 2047 vision with the Net Zero 2070 climate commitment.

- Volume 8 of this series specifically provides a strategic roadmap for transforming the waste sector into a resource recovery powerhouse.

### Viksit Bharat: Pathways to Net Zero Waste

Based on a NITI Aayog study, India faces a massive surge in waste generation that contributes significantly to GHG emissions. The pathway to "Viksit Bharat" (Developed India) requires decoupling economic growth from emissions through universal collection, 100% door-to-door segregation, and bio-methanation.



### Challenges in the Waste Sector

- **Limited Processing Capacity:** Formal recycling systems cannot match rapid waste generation, causing **massive economic losses**.  
**E.g.** Only **10% of electronic waste (e-waste)** is processed through authorized channels; the rest is handled informally in places like Moradabad.
- **Inefficient Source Segregation:** Poor door-to-door segregation lowers recyclable quality and raises costs.  
**E.g.** In tier-2 cities, mixed dry and wet waste makes bio-methanation unviable.



- **Lack of Quality Standards:** Absence of national certification reduces recycled product value.  
E.g. The waste tyre sector loses about ₹7,500 crore annually due to uncertified recycled rubber.
- **Urban–Rural Disparity:** Rural areas lack organized collection systems.  
E.g. Villages in Uttar Pradesh and Bihar resort to plastic burning.
- **Workforce & Skill Gaps:** Informal workers lack technology and safety training.  
E.g. Scrap dealers in Jaipur struggle to extract rare earth metals safely.

### NITI Aayog Recommendations

- **Universal Collection:** Achieve **100% door-to-door solid waste collection** and ensure treatment of all collected wastewater by 2047.
- **Waste-to-Resource:** Shift from disposal to **high-value recovery**, converting waste into **Bio-Compressed Natural Gas (Bio-CNG)**, compost, and treated water.
- **Policy Reform:** Ban pyrolysis of imported **end-of-life tyres** to promote domestic recycling and curb pollution.
- **Circular Economy:** Formalize informal workers and strengthen **Extended Producer Responsibility (EPR)** for batteries and electronic waste (e-waste).
- **Behavioural Change:** Use **Mission Lifestyle for Environment (LiFE)** to promote source segregation as a **Jan Andolan**.

### Conclusion

Reaching **Net Zero by 2070** is an economic imperative. Waste as embodied energy can secure supply chains, raise **Gross Domestic Product (GDP)**, and demands governance reforms with active household participation.

## IT RULES AMENDMENT 2026

The Union Government notified the Information Technology (Intermediary Guidelines and [Digital Media Ethics Code](#)) Amendment Rules, 2026 on February 10, 2026.

- The amendment specifically targets the rising threat of deepfakes and misinformation by mandating a swift 3-hour takedown window and compulsory labelling for AI-generated content.

### About IT Rules Amendment 2026:

- The amendment is a revision of the existing **IT Rules, 2021**, aimed at bringing Synthetically Generated Information (SGI) under strict legal oversight. It defines SGI as any audio, visual, or audio-visual content created or altered algorithmically to appear real or

indistinguishable from a natural person or real-world event.

## India's Digital Integrity: The 2026 AI and IT Amendments

Excerpts from India's Digital Integrity: The 2026 AI and IT Amendments detail new legal definitions for synthetic media and establish strict transparency, filtering, and takedown requirements for digital platforms and users.

### TRANSPARENCY & USER ACCOUNTABILITY

- **Legal Recognition of "Synthetically Generated Information"**  
Covers media created/modified by algorithms to look authentic; targets deepfakes while exempting basic filters, accessibility features, or research.
- **Mandatory Labelling & Digital Fingerprints**  
AI-generated videos require visible watermarks; audio requires spoken disclaimers; platforms must embed provenance markers to trace original AI tools.
- **User Declaration & Platform Verification**  
Users must disclose AI-made content; platforms must use technical tools to verify if "real" video is actually a deepfake.

### ENFORCEMENT & PLATFORM LIABILITY

- **Accelerated Takedown Timelines**  
Content must be removed within 3 hours for illegal court/govt orders and 2 hours for sensitive violations like non-consensual deepfake nudity.
- **Automated Blocking of Illegal AI Content**  
Intermediaries must use automated filters to block CSAM, non-consensual revenge porn, explosive instructions, and impersonation of high-ranking officials.
- **Safe Harbour & Section 79 Clarification**  
Platforms lose legal protection and can be sued for user content if they fail to label AI or miss the 3-hour takedown window.

### NEW EXPEDITED RESPONSE WINDOWS

| Previous Standard (Context) |          | NEW REQUIREMENT       |         |
|-----------------------------|----------|-----------------------|---------|
| Legal/Govt Takedown         | 36 Hours | Legal/Govt Takedown   | 3 HOURS |
| Non-Consensual Nudity       | 36 Hours | Non-Consensual Nudity | 2 HOURS |

Grievance Acknowledgment: 7 Days (Compared to previous 15 Days)

### Significance of the Amendment

- **Combating Viral Misinformation:** Prevents rapid spread of fake content before real-world harm occurs, strengthening the **Cyber Crime Coordination Centre (C4)** response.  
E.g. During the 2025 state elections, fabricated candidate videos were swiftly removed through C4 intervention.
- **Protecting Individual Dignity:** Enables fast removal of non-consensual **Artificial Intelligence (AI)**-generated imagery to prevent irreversible reputational loss.  
E.g. The 2024 Rashmika Mandanna deepfake incident exposed urgent regulatory gaps.
- **Ensuring Electoral Integrity:** Safeguards elections during the **Model Code of Conduct (MCC)** period from AI-driven manipulation.  
E.g. AI-cloned voices of deceased leaders were used in local campaigns.
- **Strengthening Business Accountability:** Compels global platforms like **Meta** and **X** to enhance India-specific grievance redressal.  
E.g. Companies expanded Indian Grievance Officer teams in 2025.
- **Alignment with New Criminal Laws:** Replaces **Indian Penal Code (IPC)** with **Bharatiya Nyaya Sanhita (BNS), 2023**, ensuring legal coherence.  
E.g. Delhi Police invoked BNS with Information Technology rules in AI-based financial scam cases.

## Challenges Associated with Implementation

- **Technical Accuracy of Detection:** Automated **Artificial Intelligence (AI)** tools struggle to distinguish sophisticated deepfakes from authentic footage, risking wrongful removals.  
E.g. In 2025, genuine satire videos were mistakenly flagged on major platforms.
- **Resource Constraints for 3-Hour Takedowns:** Smaller intermediaries lack 24/7 legal capacity to meet the **180-minute compliance** mandate.  
E.g. Regional apps in South India reported high real-time moderation costs in 2024.
- **Censorship Concerns:** Short takedown windows may enable suppression of legitimate dissent.  
E.g. **Karnataka High Court** case *X Corp v. Union of India* questions broad powers.
- **Traceability vs Privacy:** Metadata rules may weaken **end-to-end encryption** safeguards.  
E.g. Privacy advocates in 2025 warned of risks to anonymous whistleblowers.
- **Blended Content Complexity:** Mostly real videos with minor AI alterations are difficult to regulate.  
E.g. A 2024 protest video had AI-generated hate subtitles.

## Way Ahead

- **Standardized Watermarking:** Develop global standards for **invisible digital watermarks** that survive compression and re-uploads.
- **Capacity Building:** Train local police beyond **Deputy Inspector General (DIG)** rank to detect synthetic harms.
- **Public Awareness:** Educate citizens to identify **deepfakes**, reducing sole dependence on takedowns.
- **Independent Review:** Create an autonomous expert body to assess takedown orders and prevent misuse.
- **Incentivizing Research:** Fund Indian startups building **Artificial Intelligence (AI)** detection tools for regional languages.

## Conclusion

The **2026 Information Technology (IT) Rules** amendment shifts accountability to platforms. The **3-hour takedown window** prioritizes safety over safe harbour, requiring balance between enforcement, free speech, and privacy.

## 50TH YEAR OF ENDING BONDED LABOUR

February 2026 marks the **50th anniversary** of the [Bonded Labour System](#) (Abolition) Act, 1976.

- Despite five decades of the law, recent reports

from Odisha highlight a disturbing trend where rescued labourers are slipping back into bondage due to delayed rehabilitation

### LEGAL FEATURES OF THE 1976 ACT

#### Abolition of Liability and Freedom

All debt repayment obligations are extinguished and labourers are discharged from forced labour.

#### District Responsibility for Rehabilitation

District Magistrates and Vigilance Committees are mandated to identify, release, and rehabilitate labourers.

#### Cognizable Offense and Caste Scope

Forcing bondage is a punishable crime covering economic debt and hereditary, caste-based servitude.

### DATA AND IMPLEMENTATION REALITIES

#### National and Regional Release Figures

**1.65 lakh** labourers released nationally; over **8,304 rescued in Odisha** (mostly tribal).

#### Financial Corpus and Rehabilitation Scale

**50%** of Odisha districts lack the **₹10 lakh** corpus; assistance ranges from **₹1–3 lakh**.

#### Work Hours and Restricted Movement

Labourers often work **14–15 hours** a day in makeshift shelters with restricted movement.

## Challenges Associated with the Elimination of Bonded Labour

- **Relapse into Bondage:** Rescue without immediate financial rehabilitation pushes victims back to exploiters due to economic vulnerability.  
E.g. Panchanan Muduli from Odisha returned to a brick kiln within five months of rescue as promised assistance was not delivered.
- **Bureaucratic Delays & Accountability Gaps:** Poor coordination between source and destination states delays Release Certificates and financial aid.  
E.g. Workers rescued in Andhra Pradesh or Telangana often wait years for Odisha authorities to process documents.
- **Lack of Awareness & Monitoring:** Mandatory district surveys are rarely conducted, resulting in outdated data.  
E.g. The Socio-Economic and Caste Census 2011 remains the last major data source.
- **Caste-Based Institutionalization:** Traditional servitude linked to caste hierarchies is often denied by officials.  
E.g. In Puri district, 1,283 Release Certificates were revoked, ignoring systemic caste-based bondage.
- **Debt Trap & Lack of Livelihood Alternatives:** Absence of land and skill-based employment forces re-migration.  
E.g. Jayaraj Jagat received ₹19,000 but migrated again due to unsustainable village income opportunities.

## Way Ahead

- **Immediate Relief:** Activate the ₹10 lakh district corpus fund to ensure spot-payment within 48 hours of rescue, preventing re-exploitation.



- **Convergence of Schemes:** Link survivors to **Mahatma Gandhi National Rural Employment Guarantee Scheme, Pradhan Mantri Awas Yojana (Housing)**, and ration cards immediately to curb distress migration.
- **Digital Tracking:** Establish a real-time inter-state portal to monitor release certificates and fund transfers transparently.
- **Skill Development:** Provide vocational training (tailoring, masonry) with sustained handholding to promote local entrepreneurship.
- **Strengthening Vigilance Committees:** Reconstitute District and Sub-divisional committees with active Civil Society Organization participation.

## Conclusion

The Bonded Labour Abolition Act's 50th year demands rehabilitation over mere rescue. Sustainable freedom requires economic dignity, social justice, and permanent reintegration.

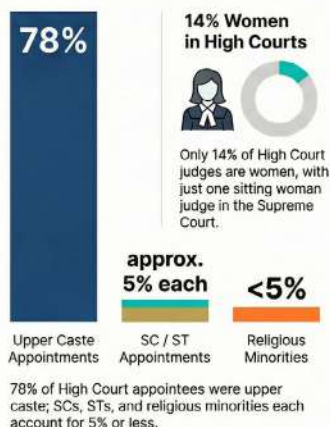
## Topic: Judiciary

# THE NEED FOR DIVERSITY IN THE JUDICIARY

- Judicial diversity refers to the representation of various social, gender, and regional groups within the court system. It ensures that the bench reflects the **diverse fabric** of Indian society, thereby enhancing the legitimacy and inclusivity of judicial interpretations.

## Judicial Demographics & Representation

Significant Caste & Minority Imbalance (High Court Appointees 2018-2024)



## Systemic Delays & Constitutional Mandates

Pendency Crisis & Vacancy Rates (as of 2026)



## Constitutional Appointment Framework



**Article 130: Regional Benches**  
Allows the CJI and President to appoint other places as seats for the Supreme Court.

## Need for Diversity in Judiciary

- **Enhanced Public Trust:** A diverse judiciary strengthens institutional legitimacy by reflecting India's social diversity and constitutional commitment to representation.  
E.g. The elevation of Justice B.R. Gavai strength-

ened confidence among marginalized communities in judicial fairness and inclusiveness.

- **Inclusivity in Interpretation:** Judges' lived experiences shape constitutional interpretation, enabling better understanding of social realities behind legal disputes.  
E.g. Women judges have contributed to greater sensitivity in gender-related rulings, particularly in marital and domestic violence cases before the Supreme Court.
- **Correcting Historical Under-representation:** Diversity reforms address systemic exclusion and ensure equitable opportunities in higher judicial appointments.  
E.g. India's long absence of a woman Chief Justice highlights structural barriers within the higher judiciary.
- **Democratization of the Bar:** Representation at senior levels motivates aspiring lawyers from diverse backgrounds to pursue litigation and judicial careers.  
E.g. The Supreme Court Bar Association advocates one-third representation for women to broaden participation.

## Challenges Associated with Diversity

- **Opaque Collegium System:**  
The lack of transparency in the collegium system of judicial appointments often perpetuates elite networks, limiting opportunities for marginalized candidates.  
E.g. Delays or rejections of women candidates since 2020, without clear reasons, reveal how opacity undermines diversity.
- **"Old Boys' Club" Mentality:** Informal networks and patriarchal norms favour established male circles, restricting advancement for others.  
E.g. Women lawyers face greater scrutiny and career breaks, creating a funnel effect toward senior positions.
- **Lack of Formal Reservation:** Absence of constitutional quotas in the higher judiciary makes diversity dependent on discretion rather than structured inclusion.  
E.g. Caste reservations in lower courts do not extend to High Courts.
- **Geographical Barriers:** Centralisation around Delhi reduces regional diversity in elevations to the Supreme Court.  
E.g. High litigation costs limit lawyers from Tamil Nadu and the Northeast.
- **Structural Barriers for Women:** Inadequate workplace infrastructure discourages sustained participation of women.



E.g. Lack of separate washrooms in many district courts affects retention.

### Way Ahead

- **Reviving the National Judicial Appointments Commission (NJAC):**  
Reintroduce a transparent appointments body including the executive and civil society, while safeguarding judicial independence.
- **Establishing Regional Benches:**  
Create permanent **Supreme Court (SC)** benches in Chennai, Mumbai, and Kolkata to decentralize access and reduce costs.
- **Institutionalizing Diversity Metrics:**  
Embed demographic criteria within the **Memorandum of Procedure (MoP)** for judicial appointments.
- **Time-bound Appointments:**  
Require government clearance of collegium recommendations within fixed timelines.
- **Mentorship Programs:**  
Develop structured pipelines for first-generation and marginalized lawyers.

### Conclusion

A diverse judiciary strengthens merit, deepens constitutional guardianship, and promotes inclusive justice.

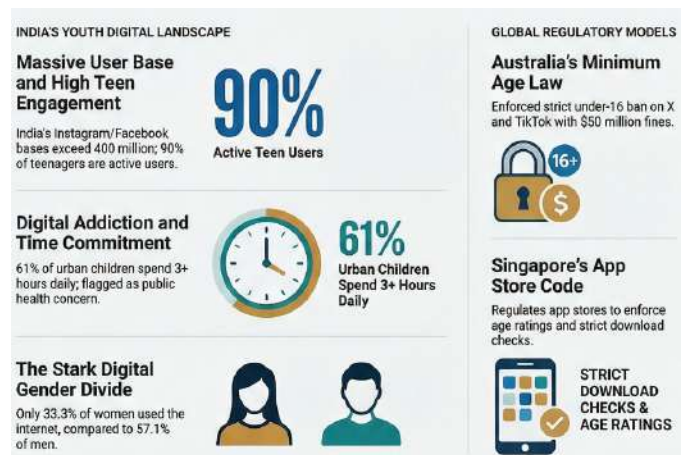
### Topic: Vulnerable Sections

## SOCIAL MEDIA BAN FOR CHILDREN

The tragic suicide of three sisters in Ghaziabad, sparked by screen addiction and parental conflict over mobile phone use, has reignited an intense national debate on banning [social media for minors](#) in India.

### About Social Media Ban for Children:

- A social media ban for children refers to legal or regulatory prohibitions that prevent individuals under a certain age (typically **16**) from creating or maintaining accounts on major digital platforms.
- It involves shifting the burden of age verification onto tech companies, often requiring government-issued IDs or [biometric verification](#) to ensure minors do not access these digital Wild West.



### Need for Social Media Ban for Children

- **Combating Extreme Addiction:** Algorithm-driven platforms trigger **compulsive usage patterns**, causing severe behavioural changes and emotional dependency.  
E.g. The 2026 Ghaziabad triple suicide was linked to a Korean task-based love game the sisters felt unable to exit.
- **Mental Health Protection:** Excessive screen exposure correlates with **anxiety, depression, and body image distress**, especially among youth.  
E.g. The Economic Survey 2025–26 connects high screen time with deteriorating mental health in ages 15–24.
- **Prevention of Cyber-Grooming:** Restrictions lower risks of **online exploitation** in poorly moderated digital spaces.  
E.g. Reports in 2026 exposed Artificial Intelligence chatbot failures enabling inappropriate interactions with minors.
- **Reducing Exposure to Self-Harm Content:** Limiting access curbs the **contagion effect** of viral self-harm challenges.  
E.g. Uttar Pradesh investigators linked the Ghaziabad case to obsessive online task engagement.
- **Restoring Academic Focus:** Constant notifications impair **sleep, concentration, and cognitive development**.  
E.g. In January 2026, the Chief Economic Adviser warned that cheap data and endless scrolling weaken students' attention spans.

### Challenges to Banning Social Media

- **Technical Porosity:** Digitally adept children bypass restrictions using **Virtual Private Networks (VPNs)** and alternative accounts, weakening enforcement.  
E.g. Despite age-gating, many Indian minors use VPNs to access restricted platforms such as TikTok.
- **Privacy and Surveillance Risks:** Age verification may require linking accounts to government identification, threatening **data privacy and civil liberties**.

E.g. Critics warn that enforcing the Digital Personal Data Protection (DPDP) Act through Aadhaar-linked logins could enable mass surveillance.

- **Loss of Digital Lifelines:** Marginalized youth depend on platforms for **community, identity, and emotional support**.  
E.g. Queer and differently-abled rural youth rely on online peer networks unavailable offline.
- **Exacerbating Gender Inequality:** Device policing may disproportionately restrict **girls' digital access**, widening gender gaps.  
E.g. Age mandates can justify stricter control over female internet use in patriarchal households.
- **Push to Darker Corners:** Prohibitions may shift users to **unregulated and encrypted platforms**, increasing risks.  
E.g. Previous bans triggered migration from Instagram to Telegram groups hosting unchecked extremist content.

### Way Ahead

- **Legally Enforceable Duty of Care:** Impose binding obligations on Big Tech to ensure **safety-by-design algorithms** instead of symbolic bans.
- **Independent Expert Regulator:** Create a specialized digital safety authority beyond the Ministry of Electronics and Information Technology (MeitY).
- **Localized Research:** Support longitudinal studies on impacts across **castes, regions, and socio-economic groups**.
- **Democratic Inclusion:** Involve children as **active stakeholders** in shaping digital policies.
- **Healthy Media Ecology:** Promote digital literacy in schools to strengthen **critical and responsible usage**.

### Conclusion

A blanket prohibition creates an **illusion of control** while ignoring structural inequalities. India must enforce accountability, ensure systemic reform, and build a balanced digital ecosystem.

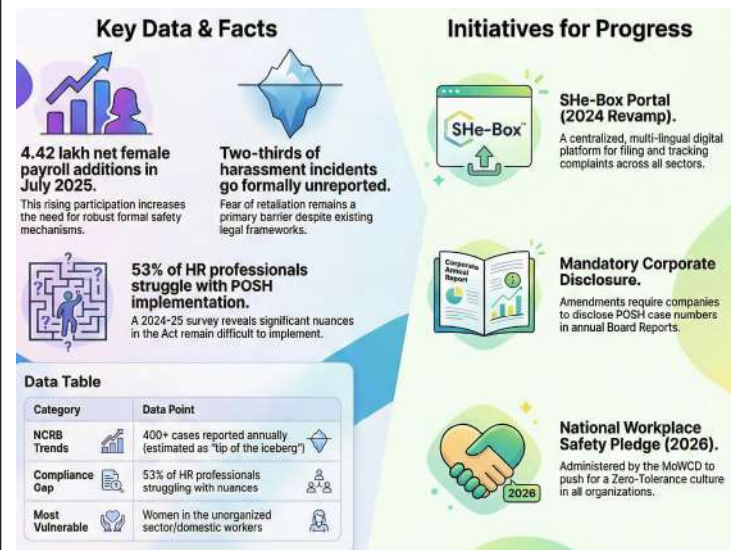
## SAFEGUARDING WOMEN AT WORK

The Ministry of Women and Child Development (MoWCD) recently organized a National Conference on Safety of Women at Workplace to strengthen the implementation of the SH Act 2013 and promote the revamped [SHe-Box portal](#) as a single-window digital solution for complaint redressal.

### About Safeguarding Women at Work:

- Safeguarding women at work refers to the legal and institutional framework designed to prevent, prohibit, and redress sexual harassment. It is anchored by the [Sexual](#)

### [Harassment of Women at Workplace \(Prevention, Prohibition and Redressal\) Act, 2013](#) (SH Act).



### Need for Safeguarding Women at Workplace

- **Constitutional Mandate:** Protecting women's rights under **Articles 14, 15 and 21** ensures equality, non-discrimination and dignity. The **Sexual Harassment (Prevention, Prohibition and Redressal) Act, 2013** must be strictly enforced.  
E.g. In 2023–24, the Supreme Court in *Aureliano Fernandes* held that weak enforcement of the Sexual Harassment Act violates constitutional dignity.
- **Economic Empowerment:** A safe workplace supports **Viksit Bharat** and higher **Female Labour Force Participation Rate (FLFPR)**.  
E.g. Studies link safety concerns to declining female workforce participation in parts of North India.
- **Retention of Talent:** Bias in **Prevention of Sexual Harassment (POSH)** systems increases attrition of skilled women.  
E.g. Tech-sector reports (2025) show mid-senior exits when redressal appears unfair.
- **Inclusive Growth:** Protection in the **unorganized sector** is essential for domestic and agricultural workers.  
E.g. Surveys (2025) found limited awareness about the **District Officer's** grievance role.
- **Mental and Physical Well-being:** Harassment causes severe trauma, requiring a **trauma-informed response**.  
E.g. *Project Stree Manoraksha* (2025) addresses survivors' mental health.

### Challenges Associated

- **Procedural Awareness Gap:** Many employees lack clarity about the **three-month limitation period** and the **Internal Committee (IC)** under the Sexual Harassment of Women at Workplace Act.

E.g. A 2024 report found only 8% fully understood their **Prevention of Sexual Harassment (POSH)** policy.

- **Fear of Retaliation:** Victims fear blacklisting and livelihood loss.  
E.g. Sports cases (2024–25) showed athletes avoided reporting senior coaches.
- **Ineffective Local Committees (LCs):** District **Local Committees** for the unorganized sector remain underfunded.  
E.g. 2025 probes revealed outdated member lists in Eastern India.
- **Digital Divide:** Limited rural access hampers use of the **Sexual Harassment electronic-Box (SHe-Box)** portal.  
E.g. Agricultural workers rely on undocumented oral complaints.
- **Gender-Specific Limitations:** Law excludes male and some transgender victims.  
E.g. 2025 debates urged gender-neutral reforms.

### Way Ahead

- **Universal Training:** Ensure **mandatory, periodic sensitization workshops** for all employees beyond one-time onboarding sessions.
- **Strengthening Local Committees:** State governments must adequately fund **Local Committees (LCs)** and publicize contacts in every panchayat and district office.
- **Incentivizing Compliance:** Link contracts to verified ratings on the **Sexual Harassment electronic-Box (SHe-Box)** portal.
- **Support for Unorganized Sector:** Deploy mobile safety vans and awareness camps for domestic and construction workers.
- **Strict Penalty Enforcement:** Enforce license cancellation for repeat violations to prioritize workplace safety.

### Conclusion

The **Sexual Harassment (Prevention, Prohibition and Redressal) Act, 2013** and **SHe-Box** form a strong legal-digital framework. Effective implementation and judicial vigilance are essential for **inclusive and sustainable growth**.

Topic: Social Sector Services relating to Health, Education

## IOM GLOBAL APPEAL 2026 REPORT

The International Organization for Migration (IOM) has requested USD 4.7 billion to assist 41 million people worldwide in 2026, marking a shift toward sharper prioritization amid a

[Humanitarian Reset](#) and a constrained global funding landscape.

### Reasons for Migration:

- **Economic Opportunity:** 168 million people migrate primarily to fill labour shortages and stimulate innovation.
- **Conflict and Violence:** Ongoing crises, such as those in Sudan and DR Congo, continue to push millions into displacement.
- **Environmental Degradation:** Extreme weather and climate shocks are forcing families to choose between staying and searching for safety.
- **Protracted Crises:** Displacement often lasts for years, driving onward migration as local resources and protection fail.
- **Inequality and Hardship:** Persistent economic disparities drive people toward dangerous, irregular journeys in search of dignity.

### Challenges Associated with Migration

- **Lack of Social Security:** Many migrants work in insecure jobs without access to **welfare schemes or health benefits**.  
E.g. Despite the **eShram (Electronic Shram)** portal, unorganized workers in **Mumbai** struggle to access state welfare.
- **Hazardous Living Conditions:** Overcrowding pushes migrants into **informal settlements** with poor sanitation.  
E.g. In **Dharavi**, migrants face acute **water and waste management** issues.
- **Exploitation and Low Wages:** Weak legal safeguards lead to **unsafe workplaces and wage theft**.  
E.g. Seasonal workers from **Bihar and Uttar Pradesh** report delayed payments.
- **Restrictive Global Policies:** Tighter immigration laws limit opportunities.  
E.g. **H-1B (specialty occupation work visa)** reforms in the **United States** raised costs and eligibility barriers.
- **Harmful Narratives and Misinformation:** Disinformation fuels **anti-immigrant sentiment and isolation**.  
E.g. Policy shifts in the **United Kingdom and Canada** capped student intake and reduced migration.

### Way Ahead

- **Route-Based Approach:** Adopt **cohesive, data-informed strategies** across migration corridors instead of fragmented responses.
- **Enhanced Localization:** Empower **local and national institutions** to lead humanitarian and development efforts.



- **Regular Pathway Expansion:** Create **flexible legal migration channels** aligned with labour market needs.
- **Data-Driven Solutions:** Expand the **Displacement Tracking Matrix (DTM)** for actionable insights.
- **Resilience Funding:** Ensure **flexible, unearmarked funding** for agile field operations.

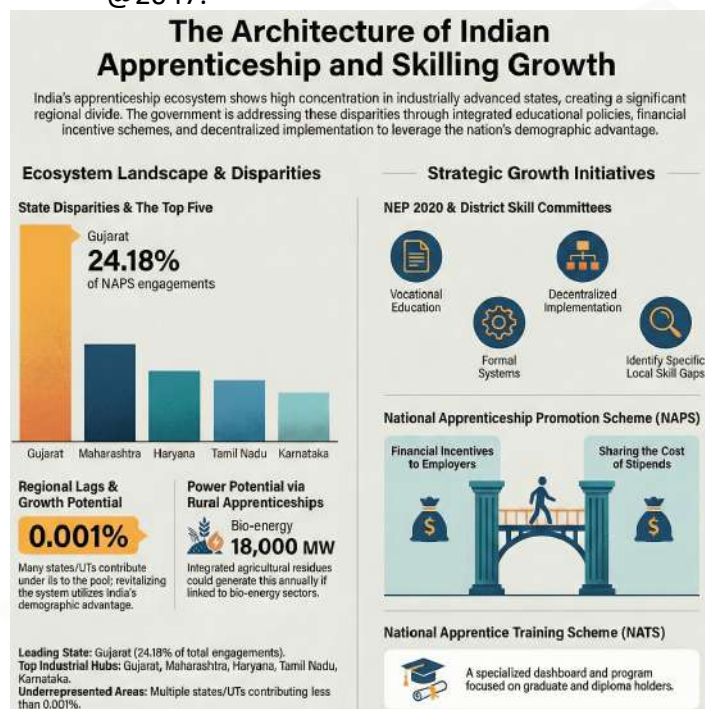
## Conclusion

The **International Organization for Migration (IOM) Global Appeal 2026** views migration as key to addressing **poverty and climate change**, requiring collective action and funding.

## NITI AAYOG RELEASES REPORT ON REVITALIZING APPRENTICESHIP ECOSYSTEM

NITI Aayog launched a seminal policy report, titled “Revitalizing Apprenticeship Ecosystem: Insights, Challenges, Recommendations and Best Practices.”

- The report aims to position apprenticeships as a strategic pillar for [India’s human capital](#) development and the vision of Viksit Bharat @2047.



## Importance of Apprenticeship Ecosystem

- **Bridging the Skill Gap:** Enables smooth transition from academic knowledge to practical industrial skills, aligning education with market needs. E.g. The **National Education Policy 2020** integrates vocational education with formal degrees, ensuring structured hands-on training.
- **Enhancing Employability:** Follows a learning-by-doing approach, making youth job-ready and industry-aligned.

E.g. The **National Apprenticeship Promotion Scheme** supports structured training leading to manufacturing employment in Gujarat.

- **Driving Enterprise Productivity:** Provides businesses access to pre-trained talent familiar with operational systems. E.g. The **Micro, Small and Medium Enterprises** cluster consortia model reduces costs and fosters innovation.
- **Promoting Social Mobility:** Creates alternative career pathways for rural and marginalized youth. E.g. Empowered District Skill Committees address localized skill demands.
- **International Competitiveness:** Aligns labour standards with global benchmarks, enhancing work-force mobility. E.g. Certification portability enables participation in cross-cultural global markets.

## Challenges in the Apprenticeship Ecosystem

- **Aspirational Bias:** Strong cultural preference for conventional academic degrees over vocational pathways reduces apprenticeship acceptance. E.g. Data shows higher unemployment among graduates compared to technically skilled youth.
- **Inter-State Disparities:** Industrial concentration in select states limits balanced national skilling participation. E.g. Gujarat records high engagement, while Assam and Madhya Pradesh report limited involvement.
- **Complex Regulatory Framework:** Multiple laws and portals discourage Micro, Small and Medium Enterprises participation. E.g. Confusion between National Apprenticeship Promotion Scheme and National Apprenticeship Training Scheme documentation lowers uptake.
- **Inadequate Infrastructure:** Remote districts lack sufficient training facilities and institutional capacity. E.g. District Skill Committees often face funding and technical constraints.
- **Industry-Aspiration Misalignment:** Student preferences sometimes diverge from evolving technological industry needs. E.g. Artificial Intelligence growth outpaces curriculum updates in Industrial Training Institutes.

## Initiatives Taken So Far:

- **NEP 2020 Integration:** Aligning vocational education and apprenticeships within the formal school and higher education system.
- **National Apprenticeship Promotion Scheme (NAPS):** Providing financial incentives to employers to engage apprentices and sharing the cost of

stipends.

- **National Apprentice Training Scheme (NATS):** A specialized dashboard and program focused on graduate and diploma holders.
- **District Skill Committees (DSCs):** Decentralizing implementation to allow districts to identify and address their specific skill gaps.

### Way Ahead

- **Common Digital Platform:** Create a unified interface for apprenticeship registration and tracking to reduce Micro, Small and Medium Enterprises compliance burden.
- **Apprenticeship Engagement Index:** Use a performance index to foster competitive federalism and strengthen state skilling systems.
- **Deepening MSME Participation:** Promote cluster-based consortia enabling small industries to jointly train apprentices.
- **Global Portability:** Align certifications with international standards to enhance workforce mobility.
- **Awareness Campaigns:** Elevate vocational education as a mainstream career pathway nationwide.

### Conclusion

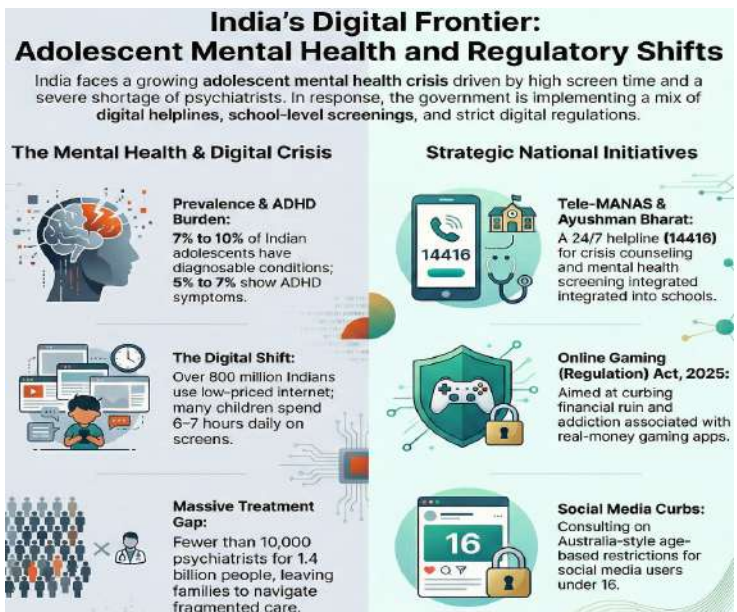
The National Institution for Transforming India Aayog blueprint converts demographic potential into dividend through inclusive, future-ready skilling.

## THE QUIET CRISIS OF ADOLESCENT MENTAL HEALTH

Recent incidents involving adolescent deaths in Ghaziabad have reignited national concern over the growing [mental health crisis](#) among children and adolescents in India.

### What it is?

- The quiet crisis refers to the pervasive but often invisible mental health struggles—such as anxiety, depression, and [digital addiction](#)—that begin as early as childhood (age 4–5) but are frequently dismissed by families and schools as mere behavioral phases or adult issues.



### Reasons for the Quiet Crisis

- **Unregulated Digital Environments:** Excessive screen time causes brain fatigue, sleep disruption, and emotional instability.  
E.g. The **Economic Survey 2025–26** identified digital addiction as a serious public health threat affecting India's long-term productivity.
- **Academic Pressure over Wellbeing:** Schools emphasise competitive examinations over emotional regulation and stress management.  
E.g. The **Annual Status of Education Report 2024** noted high social media use alongside rising academic anxiety.
- **The Displacement Effect:** Screen exposure replaces sensory play and direct human interaction vital for healthy brain development.  
E.g. Psychologists warned excessive gaming delays cognitive processing in toddlers.
- **Social Media Comparison:** Platforms intensify Fear of Missing Out and body image concerns among adolescents.  
E.g. A 2025 study reported distress among adolescent girls due to online comparisons.
- **Lack of Early Recognition:** Early emotional disorders remain untreated due to stigma and delayed intervention.  
E.g. The February 2026 Ghaziabad tragedy highlighted absence of timely mental health screening.

### Challenges Associated with It

- **Severe Manpower Shortage:** Limited child psychiatrists and psychiatric social workers restrict access to care nationwide.  
E.g. At **Annual National Conference of Indian Psychiatric Society 2026**, experts reported an 85 percent treatment gap.
- **Technological Workarounds:** Minors bypass

age-gating using Virtual Private Networks and fake profiles.

E.g. Officials noted easy evasion of the **Digital Personal Data Protection Act** safeguards.

- **Pervasive Stigma:** Mental illness is misjudged as weakness, discouraging timely help.  
E.g. Families conceal self-harm symptoms to avoid exclusion.
- **Institutional Resistance:** Technology corporations oppose stricter compliance measures.  
E.g. **Meta** and **Google** questioned Aadhaar-based verification proposals.
- **Fragmented Referral Pathways:** Schools lack structured specialist linkages.  
E.g. Manodarpan initiatives show weak follow-up for high-risk students.

### Way Ahead

- **Digital Wellness Curricula:** Move beyond Information Technology literacy toward structured lessons on screen-time regulation and cyber-safety awareness.
- **Mandatory Physical Activity:** Enforce daily play to counter sedentary habits and strengthen neuro-plasticity development.
- **Parental Support Groups:** Build community forums promoting trauma-informed parenting and shared coping strategies.
- **Routine School Screening:** Introduce universal mental health assessments alongside pediatric growth monitoring.
- **Age-Based Access Policies:** Establish balanced social media limits while safeguarding vulnerable youth access.

### Conclusion

Adolescent mental health underpins India's demographic dividend. Preventive, collective action ensures resilient, connected childhoods.

### Topic: Poverty & Inequality

## THE CYCLE OF DEPRIVATION AND AFFLUENCE

A recent longitudinal analysis of [income mobility](#) in India between 2014 and 2025 reveals a troubling trend where downward mobility is outpacing upward climbs.

- The study highlights that the share of households slipping into lower income brackets nearly doubled, reaching **26.8% by 2025**.

### About the Cycle of Deprivation and Affluence

- It denotes volatile household movement across income groups, reflecting structural economic insta-

bility.

- It explains income mobility—upward financial progress and downward vulnerability to economic shocks.

### Key Data & Facts

- **Doubling of Downward Mobility:** Households shifting to lower income groups increased sharply between 2015 and 2025.
- **Rural Distress:** Nearly one-third rural families became poorer compared to 2014 levels.
- **Stagnant Middle:** Stable income households declined drastically, signalling intense social churn.

### Key Reasons for the Rise of Inequality

- **Informal Sector Neglect:** Weak revival of agriculture and small industries leaves most workers insecure.  
E.g. Persistent stress in the **Micro, Small and Medium Enterprises** sector reduced living-wage prospects.
- **Impact of Coronavirus Disease 2019:** Poor post-pandemic management deepened structural divides.  
E.g. A K-shaped recovery enriched technology and finance, hurting tourism and retail workers.
- **Educational Barriers:** Unequal higher education access blocks entry into productive sectors.  
E.g. Contractual teaching weakens quality for non-elite students.
- **Social Discrimination:** Bias against marginalized communities limits mobility.  
E.g. Underrepresentation in corporate leadership signals barriers.
- **Urban-Centric Growth:** Gains cluster in metropolitan hubs.  
E.g. Bengaluru's technology boom contrasts stagnant agrarian incomes.

### Challenges Associated with Reduced Mobility

- **Social Instability:** Downward shifts fuel frustration and unrest among youth.  
E.g. Protests over recruitment under **Agnipath** and Railway examinations reflect anxiety.
- **Human Development Setbacks:** Income decline raises infant mortality and morbidity risks.  
E.g. Malnutrition rises where volatile incomes reduce protein intake.
- **Weak Aggregate Demand:** Survival-focused households weaken consumption-driven growth.  
E.g. Low entry-level vehicle sales contrast luxury demand.
- **Entrenched Poverty Traps:** Inequality restricts merit-based mobility.



E.g. Expensive private coaching favours affluent families.

- **Policy Paralysis:** Headline growth hides deprivation. E.g. Dependence on free food grains under **Pradhan Mantri Garib Kalyan Anna Yojana** reveals vulnerability.

### Way Ahead

- **Strengthen Public Infrastructure:** Invest in quality public health and education to reduce out-of-pocket expenditure triggering downward mobility.
- **Revive the Informal Sector:** Expand credit and technology access for **Micro, Small and Medium Enterprises** to generate stable employment.
- **Social Protection Reform:** Transform honorarium-based roles into salaried positions with social security coverage.
- **Address Spatial Inequality:** Promote tier-two and tier-three cities for balanced regional growth.
- **Anti-Discrimination Frameworks:** Enforce Equal Pay for Equal Work.

### Conclusion

Broad-based inclusion beyond **Gross Domestic Product** growth ensures durable social mobility and stability.

Topic: International Relations

## FROM STRATEGIC AUTONOMY TO VIKSIT BHARAT: REFRAMING INDIA'S FOREIGN POLICY

Prime Minister Narendra Modi formally acknowledged a new world order in Parliament, signaling a transition from the traditional [policy of strategic autonomy](#) toward a more proactive, interest-based framework centered on the vision of Viksit Bharat 2047.

### The Decay of Global Multilateralism and the Rise of Transactional Diplomacy

Traditional global cooperation is being replaced by transactional diplomacy, where "America First" and China-centric strategies prioritize national interests over shared values. This shift is characterized by the paralysis of global bodies, the weaponization of economic policy, and a retreat from global public goods.

#### The Erosion of Global Institutions

##### Dysfunctional Global Institutions & WTO Paralysis

The U.S. rejected the WTO dispute mechanism in 2025, forcing India into bilateral trade deals.

##### China's Institutional Capture

Beijing heads principal UN agencies, using influence to skew global standards in digital governance.

#### Transactional Diplomacy & Weaponized Trade

##### Weaponization of Trade via 50% Tariffs

In 2025, the U.S. imposed 50% tariffs on Indian steel, linked to Russian oil purchases.

##### Retreat from Global Public Goods

U.S. exits from the International Solar Alliance and WHO (2025–26) underscore transactionalism.

### The Limits of Strategic Autonomy

- **Cold War Obsolescence:** Strategic autonomy originated in the **Cold War era**; in today's **technology-driven geopolitical competition**, remaining unaligned can risk exclusion from critical defence and innovation networks. E.g. Pressure to choose between the **United States Patriot missile system** and the **Russian S-400 air defence system** shows the growing difficulty of balancing rival powers.
- **Economic Vulnerability:** Without strong domestic industrial capacity, autonomy becomes symbolic during **global supply chain disruptions**. E.g. Heavy dependence on **semiconductor imports from East Asia** constrains independent positions on **Indo-Pacific technology standards**.
- **The Swing State Label:** The **United States and its allies** increasingly view India as a strategic variable requiring incentives rather than a consistently neutral actor. E.g. Recent military strategies reflect a shift toward managed competition.
- **Fragmented Global South:** Developing nations now pursue **divergent priorities**, complicating unified leadership claims in multilateral forums. E.g. Climate negotiations reveal differences on coal use and climate finance among African and Pacific island countries.

### The New Strategic Reality for India

- **Asymmetric Power Politics:** Global relations increasingly resemble a "**might is right**" order, compelling smaller nations into unequal trade arrangements. E.g. The **India–United States Interim Trade Agreement (February 2026)** required higher Indian industrial imports to obtain tariff reductions.
- **Technological Dominance as Sovereignty:** Future power rests on **Artificial Intelligence (AI), cyber capabilities, and space infrastructure**, redefining sovereignty beyond territory. E.g. The **India–Russia (2025) pact** to deploy **Navigation with Indian Constellation (NavIC)** and **Global Navigation Satellite System (GLONASS)** ground stations strengthens a non-Western navigation grid.
- **Competitive Manufacturing:** India must expand manufacturing as earlier **multilateral trade pathways narrow**. E.g. Under the **China Plus One strategy**, electronics exports surged, yet competition from Vietnam and Mexico remains intense.
- **Neighborhood Volatility:** Rising extremism and expanding Chinese influence complicate regional security. E.g. The proposed **Bangladesh–Pakistan mutual**

**defense understanding (2025)** signals a potential “2.5-front” diplomatic challenge.

### Reframing Indian Foreign Policy

- **Bide Time and Build Internally:** Maintain a **measured global profile** while strengthening **Production Linked Incentive (PLI) schemes** and infrastructure under **Viksit Bharat 2047**.  
E.g. The **Union Budget 2026–27** prioritized **Rare Earth Corridors** to reduce dependence on Chinese processing and secure domestic supply chains.
- **Aggressive Trade Diversification:** Reduce reliance on the United States by concluding **Free Trade Agreements (FTAs)** with major partners.  
E.g. The **India–European Union FTA (2026)** expanded market access, covering most export sectors.
- **New Tech-Centric Alliances:** Deepen cooperation in **space, quantum computing, and cyber technologies** with Russia and emerging powers.  
E.g. The **Kazan (2024)** and **Tianjin (2025)** summits explored linking official digital currencies for **BRICS (Brazil, Russia, India, China, South Africa)** trade.
- **Passive Regional Posture:** Frame neighbourhood tensions as **diplomatic challenges**, safeguarding economic momentum.  
E.g. **India–Kuwait dialogues under Viksit Bharat 2047** emphasized energy and investment diplomacy.

### Conclusion

India’s foreign policy marks its sharpest shift since 1991, evolving from cautious strategic autonomy to an assertive **Viksit Bharat 2047** vision—seeking technological strength, diversified trade, and status as an independent global pole.

## INDIA–FRANCE SPECIAL GLOBAL STRATEGIC PARTNERSHIP

French President Emmanuel Macron visited India, to participate in the AI Impact Summit and inaugurate the 2026 [India-France Year of Innovation](#).

- During this visit, both nations elevated their bilateral ties to a “**Special Global Strategic Partnership**” to guide cooperation until 2047.

### About India–France Special Global Strategic Partnership:

The “Special Global Strategic Partnership” is a high-level diplomatic upgrade that shifts the relationship from sectoral co-operation to a comprehensive, long-term alliance aimed at global stability. It focuses on:

- **Strategic Autonomy:** Strengthening the sovereignty and independent decision-making of both nations.
- **Global Governance:** Acting as a force for global good to address macroeconomic imbalances and climate crises.

- **Security & Innovation:** Deepening the co-development of advanced technologies (AI, Space, Nuclear) while securing resilient, trusted supply chains.



### Challenges Associated with the Partnership

- **Divergent Views on Global Conflicts:** Strategic autonomy and regional priorities create diplomatic nuances.  
E.g. India’s stance on the Ukraine conflict differs from Western positions, complicating joint messaging.
- **Trade and Regulatory Barriers:** Negotiations face hurdles over labour, environment, and data standards.  
E.g. The **India–European Union Free Trade Agreement** talks remain prolonged.
- **Implementation Delays in Nuclear Energy:** Liability and technical issues slow projects.  
E.g. The Jaitapur Nuclear Power Project awaits progress amid civil liability debates.
- **Technological Protectionism:** Transfer of sensitive defence technologies remains restricted.  
E.g. Jet engine Transfer of Technology demands high-level intervention.
- **Rising Regional Instability:** West Asian tensions threaten connectivity corridors.  
E.g. Red Sea risks endanger the **India–Middle East–Europe Economic Corridor**.

### Way Ahead

- **Operationalizing India–Middle East–Europe Economic Corridor:** Convene the 2026 ministerial meeting to accelerate corridor implementation.
- **Democratizing Artificial Intelligence:** Bridge global divides through secure, inclusive technological



access.

- **United Nations Security Council Reforms:** Strengthen advocacy for India's permanent membership.
- **Green Energy Transition:** Expand the **International Solar Alliance** for climate resilience training.
- **Deepening People-to-People Ties:** Broaden French education access for diverse Indian students.

### Conclusion

The Special Global Strategic Partnership advances cooperative security, innovation, and democratic multipolar stability.

## INDIA-ISRAEL BILATERAL RELATIONS

Prime Minister of India conducted a historic state visit to Israel, where both nations elevated their ties to a **Special Strategic Partnership for Peace, Innovation, and Prosperity**.

### About India-Israel Bilateral Relations:

- India and Israel share a multi-dimensional relationship characterized by deep security cooperation, a knowledge-hub partnership in technology, and a shared vision for regional stability. The relationship has evolved from hesitant [diplomatic beginnings](#) to a robust, open alliance based on mutual strategic interests and civilizational ties.

### India & Israel: A Special Strategic Partnership for Peace, Innovation, and Prosperity

The relationship between India and Israel has evolved from limited engagement in 1950 to a "Special Strategic Partnership" by 2025. This summary highlights historical milestones, current economic statistics, and the outcomes of recent high-level bilateral meetings focused on AI, cybersecurity, and labor mobility.



### Challenges to Bilateral Relations

- **Regional Stability and Conflict:** West Asian tensions threaten energy routes and diaspora security. E.g. Suspension of Druzhba oil flows and Gaza volatility complicate strategic balancing.

- **The Balancing Act (Palestine):** Commitment to a two-state solution requires diplomatic caution. E.g. Support for peace initiatives while maintaining autonomy at the United Nations reflects nuance.
- **Cross-Border Terrorism:** Divergent global definitions hinder coordinated counter-terror policy. E.g. Recent attacks highlight the need for deeper intelligence-sharing.
- **Cyber Vulnerabilities:** Linked digital payments raise exposure to state-sponsored attacks. E.g. A Cyber Policy Dialogue addressed infrastructure threats.
- **Trade Barriers:** Complex tariffs delay the Free Trade Agreement. E.g. The 2026 Terms of Reference safeguarded domestic manufacturing interests.

### Way Ahead

- **India–Middle East–Europe Economic Corridor Implementation:** Fast-track corridor projects, positioning Israel as a Mediterranean gateway for Indian trade and logistics integration.
- **Semiconductor Diplomacy:** Combine Israeli chip-design expertise with India's self-reliant manufacturing push in Dholera and emerging industrial clusters.
- **Water Security Cooperation:** Expand river rejuvenation and desalination initiatives using advanced waste-to-water technologies nationwide.
- **Academic Integration:** Activate the India–Israel–Innovation Academic Cooperation Forum to translate research into market-ready solutions.
- **Deep-Tech Joint Ventures:** Shift from buyer-seller defence ties toward co-development and co-production under Make in India.

### Conclusion

A Special Strategic Partnership now drives collaboration across artificial intelligence, space, agriculture, and innovation ecosystems.



# GENERAL STUDIES - 3

Topic: Economy

## THE STATES AND THE 16TH FINANCE COMMISSION

- The 16th FC is in the news because it has submitted its recommendations for the 2026-31 period, introducing a landmark “Contribution to GDP” criterion and enforcing strict [fiscal discipline](#) on States by capping deficits and ending off-budget borrowings.

The share for States

Till the 13th FC (2010-2015), the devolution involved specific transfers for Centrally Sponsored Schemes (CSS) with extensive conditionalities

Table 1: Criteria for horizontal devolution amongst States

| Criteria                    | 13th FC<br>2010-15 | 14th FC<br>2015-20 | 15th FC<br>2020-26 | 16th FC<br>2026-31 |
|-----------------------------|--------------------|--------------------|--------------------|--------------------|
| Income distance             | 47.5               | 50                 | 45                 | 42.5               |
| Population (1971)           | 25                 | 17.5               | -                  | -                  |
| Population (2011)           | -                  | 10                 | 15                 | 17.5               |
| Area                        | 10                 | 15                 | 15                 | 10                 |
| Forests                     | -                  | 7.5                | 10                 | 10                 |
| Fiscal discipline           | 17.5               | -                  | -                  | -                  |
| Demographic performance     | -                  | -                  | 12.5               | 10                 |
| Tax effort                  | -                  | -                  | 2.5                | -                  |
| State's contribution to GDP | -                  | -                  | -                  | 10                 |
| <b>Total</b>                | <b>100</b>         | <b>100</b>         | <b>100</b>         | <b>100</b>         |



- The Finance Commission is a constitutional body (Article 280) that recommends how the Divisible Pool of central taxes should be shared between the Union and the States (Vertical Devolution) and among the States themselves (Horizontal Devolution).

### Taxes Shared:

- These include [Corporation Tax](#), Personal Income Tax, Central GST (CGST), and the Centre's share of Integrated GST (IGST).

### Major Recommendations of the 16th FC:

- Vertical Devolution:** Retained at **41%** for the period 2026-31.
- New Horizontal Criterion:** Introduced “Contribution to GDP” with a **10% weight**, rewarding states with higher economic output.
- Fiscal Deficit Caps:** States' fiscal deficit is capped strictly at **3% of GSDP**.
- End of Off-Budget Borrowings:** All liabilities of state-owned entities must now be shown in the main State budget.
- Discom Reforms:** Recommended the **privatisation of DISCOMs** (Power Distribution Companies) to reduce state debt.

### Positives of the Recommendations

- Reward for Growth:** Gross Domestic Product contribution criterion motivates states to

strengthen industry and ease of doing business. E.g. Tamil Nadu and Karnataka may receive higher shares due to strong output.

- Transparency in Debt:** Ending off-budget borrowings ensures clearer fiscal accountability. E.g. Telangana and Andhra Pradesh will reflect actual liabilities transparently.
- Urban Focus:** Urbanisation Premium grants support structured city expansion. E.g. Ahmedabad and Pune can manage peri-urban integration.
- Forest Protection:** Incentives for increasing forest cover promote conservation. E.g. Chhattisgarh and Madhya Pradesh can expand green cover.

### Negatives of the Recommendations

- Vertical Stagnation:** Retaining **41 percent** central tax share limits states' fiscal autonomy and expansion capacity. E.g. Bihar struggles to finance essential infrastructure without greater devolution.
- Penalising Welfare:** Caution against unconditional cash transfers intrudes into state policy flexibility. E.g. Gruha Lakshmi and Majhi Ladki Bahin may face funding scrutiny.
- One-Size-Fits-All Power Reforms:** Mandatory Distribution Company privatisation overlooks regional sensitivities. E.g. Punjab may witness unrest over free power concerns.
- Equity Concerns:** Lowering Income Distance weight weakens support for poorer states. E.g. Uttar Pradesh could see marginal share reduction.

### Conclusion

- Compliance-Driven Federalism:** Sixteenth Finance Commission prioritises **economic efficiency** and **fiscal discipline** over equity.
- Constrained Autonomy:** Retaining **41 percent devolution** and excluding **cesses** limits states' fiscal flexibility despite growth recognition.

### Creative Industries as Growth Engines

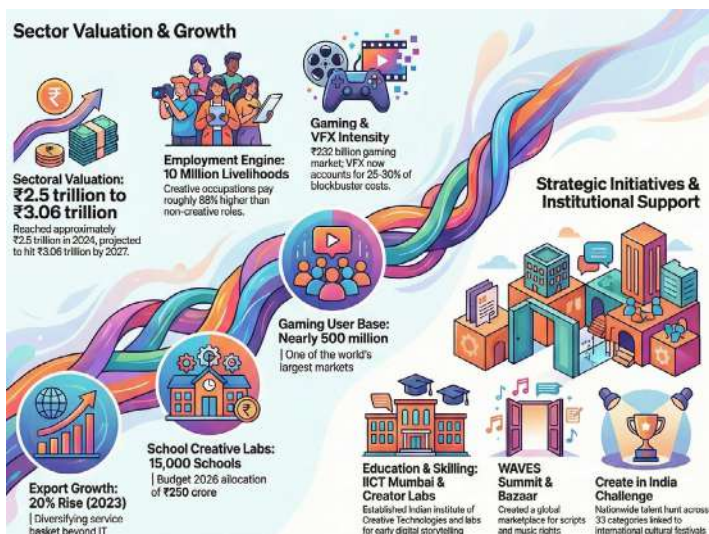
- The creative industries are in the spotlight following the [Union Budget 2026-27](#) and the Economic Survey 2025-26, which projected a requirement of 2 million professionals in the AVGC sector by 2030 and announced the establishment of 15,000 Content Creator Labs in schools.

### What is the Orange Economy?

- Creative Value Ecosystem:** The Orange Economy, coined by Iván Duque and Felipe Buitrago, generates value from **creativity, culture, and**

## intellectual property.

- **Heritage-Digital Bridge:** It connects **arts, crafts, and festivals** with gaming, Visual Effects, and Over-The-Top platforms, making imagination globally tradeable.



## Creative Economy as a Growth Engine

- **Massive Job Creation:** Animation, Visual Effects, Gaming, Comics and Extended Reality sector generates employment for youth beyond metropolitan cities. E.g. Government projects two million jobs by 2030, boosting studios in Pune and Indore.
- **Soft Power Diplomacy:** Cultural exports enhance global influence and tourism appeal. E.g. Project K and RRR popularised cinematic locations internationally.
- **Urban Multiplier Effect:** Festivals stimulate hospitality, transport and retail sectors. E.g. Ahmedabad and Navi Mumbai concerts increased hotel demand.
- **Technological Spillover:** Gaming innovations aid healthcare, education and defence applications. E.g. Unreal Engine supports immersive medical simulations.
- **Democratisation:** Digital platforms empower rural creators economically. E.g. DD National's Creator's Corner mainstreamed micro-influencers nationally.

## Challenges Associated with the Creative Economy

- **Platform Trap:** Dependence on global algorithms risks sudden **demonetisation** and revenue instability. E.g. Policy shifts on short-video platforms sharply reduced earnings of Indian micro-influencers.
- **Intellectual Property Financing Gaps:** Micro, Small and Medium Enterprises lack collateral for formal

credit access.

E.g. Bengaluru studios rely on costly private loans as digital assets lack recognition.

- **Skill-Industry Mismatch:** Training emphasises software skills over storytelling foundations. E.g. India Game Developer Conference highlighted shortage of original designers.
- **Infrastructure Bottlenecks:** High rendering and cloud costs burden small studios. E.g. Firms outsource Computer-Generated Imagery due to limited High-Performance Computing clusters.
- **Regulatory Complexity:** Multiple clearances delay live events. E.g. International festivals scaled back amid permission hurdles.

## Way Ahead

- **Intellectual Property Lending:** Collaborate with Reserve Bank of India to enable copyrights and trademarks as institutional collateral.
- **National Animation, Visual Effects, Gaming, Comics and Extended Reality Policy:** Implement uniform state incentives for creative clusters nationwide.
- **Artificial Intelligence Tools:** Develop domestic animation and dubbing solutions for cost-efficient global localisation.
- **Single-Window Clearance:** Operationalise Live Entertainment Development Cell for streamlined event permissions.
- **Original Intellectual Property Focus:** Transition from outsourcing hub to global content creator.

## Conclusion

- **Scalable Creative Engine:** India's Orange Economy converts imagination into structured growth.
- **Demographic to Creative Dividend:** Linking skilling with markets can globalise the Created in India brand.

## INDIA'S SURPLUS LABOUR TRAP

- The recent nationwide strikes by [gig workers](#), contractual teachers, and ASHA/Anganwadi workers have highlighted the surplus labour trap, where millions remain in low-productivity, precarious jobs.

## What is India's Surplus Labour Trap?

- **Structural Imbalance:** A large workforce remains in **low-wage, low-productivity jobs** like gig work and subsistence farming due to inadequate formal employment generation.



## Data and Facts on India's Labour Market

- **Informality:** Nearly **90 percent** of 650 million workers operate in the unorganised sector without contracts or safety nets.
- **Low Participation:** Around **350 million working-age individuals** remain outside active labour markets.
- **Low Compensation:** Accredited Social Health Activists and Anganwadi workers earn modest honoraria below minimum wages.
- **Public Sector Rush:** Graduates compete intensely for limited Group D government posts.

## Potential of the Indian Labour Market

- **Demographic Dividend:** India's **young population** offers vast productivity gains with proper skilling. E.g. Global Capability Centres in Bengaluru employ Indian talent in advanced global services.
- **Digital Economy Leadership:** Expanding digital platforms enable **flexible employment opportunities** nationwide. E.g. Open Network for Digital Commerce empowers small retailers and delivery workers.
- **Manufacturing Hub Opportunity:** China Plus One shifts can absorb surplus agricultural labour into factories. E.g. Apple suppliers Foxconn and Tata create formal jobs in Tamil Nadu and Karnataka.
- **Green Energy Transition:** Renewable expansion generates a **green workforce demand**. E.g. Prime Minister Surya Ghar Muft Bijli Yojana boosts solar technicians.
- **Social Infrastructure Expansion:** Health and education growth formalises care economy employment. E.g. All India Institute of Medical Sciences-like institutions expand roles in tier-two cities.

## Challenges Associated

- **Job Precarity:** Gig-ification and algorithmic targets prioritise speed over **worker safety**. E.g. Zomato and Blinkit strikes in Gurgaon exposed accident risks and weak coverage.
- **Stagnant Real Wages:** Inflation erodes earnings of vulnerable workers. E.g. Mahatma Gandhi National Rural Employment Guarantee Act wages trail market agricultural rates.
- **Skill Mismatch:** Graduates lack industry-ready technical capabilities. E.g. India Skills Report found limited employability.
- **Gender Discrimination:** Women face informal, underpaid roles. E.g. Anganwadi protests demand recognition as

formal workers.

- **Weak Enforcement:** Sub-contracting dilutes labour protections. E.g. Silkyara tunnel collapse revealed safety lapses.

## Way Ahead

- **Formalise Informal Work:** Implement **Social Security Code** ensuring gig workers receive health and accident insurance.
- **Invest in Human Capital:** Prioritise teachers and healthcare staff over physical infrastructure spending.
- **Living Wage Transition:** Recognise Accredited Social Health Activists and Anganwadi workers as formal employees.
- **Collective Bargaining:** Safeguard unionisation rights without administrative repression.
- **Decentralised Industrialisation:** Promote Micro, Small and Medium Enterprises in rural regions.

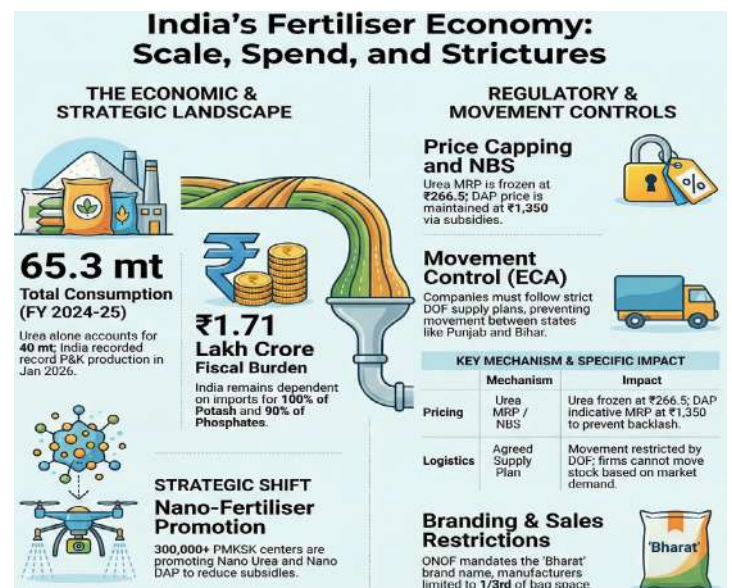
## Conclusion

- **Dignified Employment Shift:** Empowering labour through fiscal commitment can transform demographic potential into sustainable growth.

## Topic: Agriculture, its Marketing & Livestock

## THE FERTILISER INDUSTRY IN INDIA

- The fertiliser industry is the backbone of India's food security, acting as a bridge between [industrial manufacturing](#) and agricultural productivity. While technically a commercial sector, it functions as a **policy-supported backbone** where the government dictates every major variable—from the price at which a product is sold to the specific district where a railway rake must be unloaded.





## Implications of Excessive Control

- **Soil Health Deterioration:** Underpriced **Urea** encourages overuse, distorting Nitrogen:Phosphorus:Potassium ratio to 11:4:1 against 4:2:1.  
E.g. Punjab and Haryana report salinity and declining fertiliser efficiency.
- **Stifled Innovation:** Bans on premium inputs weaken research and development incentives.  
E.g. Uttar Pradesh restrictions deter high-efficiency specialty nutrients.
- **Negative Investor Sentiment:** Policy uncertainty discourages private and foreign manufacturing investment.  
E.g. Limited cost recovery stalls new urea plants.
- **Fiscal Fragility:** Subsidy burdens rise with global price volatility.  
E.g. 2025–26 overshoot reflects import disruptions.
- **Operational Inefficiency:** Logistic micromanagement triggers shortages and black-marketing.  
E.g. Chitrakoot farmers faced queues despite adequate stocks.

## Way Ahead

- **Urea under Nutrient Based Subsidy:** Correct price distortions and promote balanced nutrient application.
- **Direct Benefit Transfer:** Transfer subsidies to farmers' bank accounts for market-linked retail pricing.
- **Decriminalise Sector:** Remove non-subsidised fertilisers from Essential Commodities Act to reduce Inspector Raj.
- **Green Incentives:** Provide Production Linked Incentive for nano-fertilisers and green ammonia.
- **Rationalise Taxation:** Fix inverted Goods and Services Tax structure.

## Conclusion

- **Market Transition Needed:** Shifting from Permit Raj controls to direct farmer support ensures innovation, fiscal sustainability and Aatmanirbhar fertiliser security.

## CIRCULAR ECONOMY IN AGRICULTURE

- The Government of India recently highlighted the progress of the **"Waste-to-Wealth"** mission, the [GOBARdhan scheme](#) has covered over 50% of India's districts.

## About Circular Economy in Agriculture:

- A circular economy in agriculture is a regenerative system where waste is minimized and resources are kept in use for as long as possible. Instead of the traditional take-make-dispose model, it focuses on the **6 Rs (Reduce, Reuse, Recycle, Refurbish, Recover, and Repair)** to transform crop residues, animal manure, and food waste into valuable inputs like [bio-CNG](#), compost, and biochar.



## Waste-to-Wealth: India's Agricultural Circular Economy

India generates **350 million tonnes** of agricultural waste annually, presenting a significant opportunity for energy production and economic growth. Through targeted schemes like **GOBARdhan** and massive infrastructure funds, the nation is transitioning toward a **\$2 trillion** circular economy.

### Key Data & Market Potential



### Strategic Initiatives & Infrastructure

- **GOBARdhan & CRM Schemes:** Converts dung into CBG/manure and provides subsidies for machines like Happy Seeders.
- **Specialized Infrastructure Funds:** AHIDF (₹15,000 cr) supports meat/dairy processing and scientific animal by-product management.
- **Rural Waste Management (ODF Plus):** Jal Jeevan and Swachh Bharat 2.0 focus on solid and liquid waste in villages.

## Importance of Circular Economy in Agriculture

- **Soil Health Restoration:** Reusing organic waste lowers dependence on chemical fertilisers and rebuilds soil carbon.  
E.g. Indian Council of Agricultural Research promotes biogas slurry to restore depleted soils.
- **Climate Mitigation:** Methane capture cuts greenhouse gas emissions from decomposition.  
E.g. Unified GOBARdhan Portal monitors

Compressed Biogas supporting Net Zero goals.

- **Enhanced Farm Income:** Waste-to-wealth models create additional farmer revenue.  
E.g. Punjab farmers sell crop residue for ex-situ management.
- **Water Security:** Treated wastewater reduces groundwater exploitation.  
E.g. Jal Shakti Mission promotes greywater reuse.
- **Resource Efficiency:** Supports Sustainable Development Goal 2 and 12.  
E.g. Biochar improves carbon sequestration and water retention.

#### Challenges Associated with Circular Agriculture

- **High Initial Investment:** Biogas plants and residue machinery demand substantial capital from small farmers.  
E.g. Agriculture Infrastructure Fund support still requires collateral for bio-Compressed Natural Gas units.
- **Logistical Bottlenecks:** Transporting bulky biomass from fragmented farms is costly.  
E.g. Short sowing windows hinder ex-situ stubble collection.
- **Technological Gaps:** Advanced biochar systems and modular reactors remain nascent.  
E.g. Small Modular Reactors face regulatory and expertise barriers.
- **Behavioral Inertia:** Traditional burning practices persist despite incentives.  
E.g. Indo-Gangetic farmers prefer fire for quick clearing.
- **Weak Market Linkages:** Organic manure struggles against subsidised urea competition.  
E.g. Waste-to-Wealth plants face profitability challenges.

#### Way Ahead

- **Carbon Credit Incentives:** Integrate farmers into carbon markets for biochar and biogas adoption.
- **Strengthen Farmer Producer Organizations:** Enable village-level Custom Hiring Centres and bio-Compressed Natural Gas clusters.
- **Research and Development Push:** Invest in engineered biochar and microbial solutions for decomposition.
- **Policy Parity:** Extend Fertiliser Control Order support to organic inputs.
- **Awareness Campaigns:** Expand Jan Andolan on waste segregation benefits.

#### Conclusion

- **Circular Imperative:** Scaling GOBARDhan and Agriculture Infrastructure Fund can convert waste into sustainable prosperity.

#### Topic: Energy

### WOMEN LED DECENTRALISED RENEWABLE ENERGY INDIA

- The India Distributed Renewable Energy Summit (IDRES) 2026 recently highlighted women-led decentralised [renewable energy](#) (DRE) as a strategic pillar for India's net-zero transition.
- Concurrently, Chhattisgarh launched "**Anjor Vision 2047**," aiming to establish 5,000 women-led DRE solutions and create 50,000 green jobs by 2030.

#### About Women led decentralised renewable energy India:

- Women-led DRE is a transformative model that moves rural women from being passive last-mile consumers to active **designers, owners, and operators** of small-scale energy systems (like solar pumps, mini-grids, and solar dryers).
- It integrates energy access with [gender equity](#), ensuring that clean energy infrastructure is managed by local women collectives, such as Self-Help Groups (SHGs), to power both domestic needs and rural livelihoods.

| THE VITAL NEED FOR WOMEN-LED DRE                                                                                                                                                                                                                                     |                |                    | SPECIFIC REGIONAL EXAMPLES                                                  |                                                                                                                                                                                                                                          | STRATEGIC INITIATIVES FOR A GREEN FUTURE                                                                                                                                                                          |                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                      | REGION         | APPLICATION        | IMPACT                                                                      |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                   |                                                                                                                                                                                                                       |
| <b>RELIABILITY &amp; CLIMATE RESILIENCE</b><br><br>DRE ensures power for health centers and emergency services during outages or extreme weather.                                 | Chhattisgarh   | Health Centers     | Refrigeration for life-saving vaccines during grid outages.                 | <b>PM SURYA GHAR &amp; LAKHPATI DIDI SCHEME</b><br><br>Targeting 10,000 solar villages by 2030 and 3 crore women entrepreneurs via DRE integration. | <b>ANJOR VISION 2047 (CHHATTISGARH)</b><br><br>A state roadmap aiming for 66% renewable energy through women-led green jobs. | <b>SOLAR URJA LAMP (SoUL) PROJECT</b><br><br>An IIT Bombay initiative training rural women to assemble and maintain solar lamps. |
| <b>MITIGATING TIME POVERTY &amp; ENHANCING PRODUCTIVITY</b><br><br>Solar-powered automation frees hours from drudgery, enabling women to run competitive, mechanized enterprises. | Uttar Pradesh  | Micro-cold Storage | Preserves produce for farmers, reducing post-harvest losses.                |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                   |                                                                                                                                                                                                                       |
| <b>ENHANCING RURAL SAFETY &amp; MOBILITY</b><br><br>Reliable solar streetlighting increases women's mobility and participation in community meetings after dark.                  | Andhra Pradesh | Solar Water Pumps  | Ensures reliable irrigation for agriculture, increasing crop yields.        |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                   |                                                                                                                                                                                                                       |
|                                                                                                                                                                                                                                                                      | Bihar          | E-Rickshaws        | Provides clean transport, increasing mobility and income for women drivers. |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                   |                                                                                                                                                                                                                       |

#### Key Challenges Associated

- **High Upfront Costs:** Decentralized Renewable Energy appliances demand significant capital beyond rural women's reach.  
E.g. Solar bulk milk chillers costing ₹25 lakh exceed Self-Help Group affordability without green credit.
- **Technical Skill Gap:** Shortage of trained women technicians weakens maintenance networks.  
E.g. Jharkhand installations remain defunct due to distant service providers.
- **Deep-Seated Patriarchy:** Limited land ownership restricts women's financial autonomy.  
E.g. Agriculture Census shows 13.9 percent female ownership, constraining loan access.
- **Limited Market Awareness:** Entrepreneurs lack



information on clean energy opportunities.

E.g. Himachal fruit processors rely on diesel over solar dryers.

- **Service Infrastructure Deficit:** Village-level spare parts remain scarce.  
E.g. Western Odisha irrigation systems face peak-season downtime.

### Way Ahead

- **Asset Ownership:** Mandate women as primary or joint owners of clean energy assets.
- **Green Credit Access:** Provide dedicated loans and **First Loss Default Guarantees** for women-led enterprises.
- **Solar Didis:** Expand Science, Technology, Engineering and Mathematics training for local maintenance cadres.
- **Panchayat Integration:** Enable Gram Panchayats to deploy Own Source Revenue for Energy-as-a-Service partnerships.
- **Mainstreaming Decentralized Renewable Energy:** Embed technologies within Deendayal Antyodaya Yojana–National Rural Livelihoods Mission.

### Conclusion

- **Women-Led Transition:** Empowering last-mile women ensures inclusive energy security, climate resilience and equitable Viksit Bharat progress.

## INDIA'S ENERGY SHIFT THROUGH THE GREEN AMMONIA ROUTE

- India recently conducted a landmark auction through the Solar Energy Corporation of India (SECI), discovering record-low prices for [green ammonia](#) that are 40%–50% lower than European benchmarks.
- Green ammonia is produced by combining nitrogen from the air with green hydrogen (generated by splitting water using renewable energy).
- Unlike traditional grey ammonia, which relies on natural gas and emits significant CO<sub>2</sub>, green ammonia has a [near-zero carbon footprint](#).

### STRATEGIC POTENTIAL OF GREEN AMMONIA

#### Decarbonizing Agriculture and Maritime Shipping



Replaces fossil-fuel feedstock in fertilizers (e.g., Paradeep Phosphates) and serves as zero-carbon marine fuel for green shipping corridors.

#### Hydrogen Carrier and Energy Storage Medium



Acts as a stable medium for global exports to Japan/South Korea and provides long-duration grid storage for renewable energy.

#### Key Hubs: Kandla, Tuticorin, and Paradeep



Coastal ports are being transformed into Hydrogen Hubs to operationalize maritime decarbonization and global green ammonia supply.

### INITIATIVES AND NATIONAL FRAMEWORK

#### National Green Hydrogen Mission & SIGHT Programme



A **₹17,490 crore** SIGHT outlay supports a **5 MMTPA** production target and **₹8 lakh crore** investment by 2030.

#### Strategic Port Recognition and Infrastructure



Deendayal, Paradip, and V.O. Chidambaranar ports are designated hubs for hydrogen derivatives and maritime decarbonization hubs.

#### SECI Auction Model for Market Certainty



Implementing 10-year fixed-price offtake agreements and aggregated procurement to provide long-term certainty for developers.

### Challenges Associated

- **Cost Gap:** Green ammonia remains costlier than grey alternatives, limiting competitiveness.  
E.g. Price-sensitive industries hesitate without blending mandates.
- **Infrastructure Deficit:** Ports lack specialised bunkering and storage ecosystems.  
E.g. VOC Tuticorin demand insufficient for dedicated facilities.
- **Regulatory Fragmentation:** Divergent state policies on grid banking and subsidies create uncertainty.  
E.g. States struggle funding green hydrogen incentives.
- **Technology Readiness:** Electrolysers and ammonia cracking units remain capital intensive.  
E.g. Import dependence persists despite Production Linked Incentive schemes.
- **Safety Concerns:** Toxicity demands strict handling standards.  
E.g. Marine fuel transition requires engine redesign and training.

### Way Ahead

- **Regulatory Harmonisation:** Ensure uniform state rules for grid access, banking and wheeling charges.
- **Blended Finance:** Mobilise low-interest multilateral capital to enhance project viability.
- **Global Certification:** Align standards with European Union norms for export readiness.
- **Mandatory Blending:** Require green hydrogen and ammonia use in refineries and fertiliser plants.
- **Safety Research:** Develop indigenous monitoring and sensor systems for ammonia handling.



## Conclusion

- **Energy Independence:** Competitive markets and sustained investment can anchor green ammonia in India's Net Zero transition.

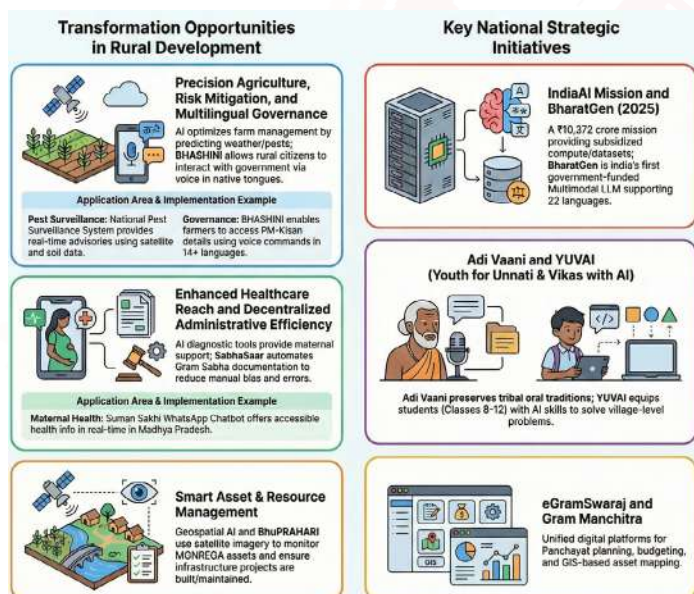
## Topic: Science and Technology- Developments & Applications

### ARTIFICIAL INTELLIGENCE (AI) TRANSFORMING RURAL INDIA

- The India-AI Impact Summit 2026 recently highlighted AI as a structural pillar for Viksit Bharat@2047, transitioning from pilot projects to system-wide implementation in rural governance and livelihoods.
- Additionally, the launch of BharatGen and the [India AI Governance](#) Guidelines have formalized the roadmap for inclusive, multilingual AI on Indian soil.

#### About Artificial Intelligence (AI) Transforming Rural India:

- In the rural context, AI is a social-purpose technology designed to perform cognitive tasks like crop disease reasoning and multilingual translation. It functions as a **public good**, integrated into [Digital Public Infrastructure](#) (DPI) to bridge service gaps in areas where physical infrastructure is limited.



## Challenges Associated

- **Digital Infrastructure Deficit:** Limited electricity and broadband hinder **real-time Artificial Intelligence deployment**.  
E.g. Remote hilly Shadow Areas face patchy

connectivity, restricting cloud tools.

- **Low Digital Literacy:** Users lack skills to identify misinformation or deepfakes.  
E.g. Rural communities struggle verifying Artificial Intelligence advice.
- **Data Bias:** Western-trained models misread Indian dialects and soil contexts.  
E.g. Large Language Models overlook tribal speech variations.
- **High Costs:** Graphics Processing Units and data annotation burden local budgets.  
E.g. Rural bodies prioritise sanitation over technology.
- **Ethical Risks:** Facial recognition raises privacy concerns.  
E.g. Data Protection and Digital Personal Data Protection Act debates highlight surveillance fears.

## Way Ahead

- **Hybrid Connectivity:** Deploy Low Earth Orbit satellites for reliable Artificial Intelligence access in off-grid villages.
- **Artificial Intelligence Data Labs:** Establish 570+ labs in Tier-2 and Tier-3 cities for India-centric datasets.
- **Explainable Artificial Intelligence:** Ensure transparent welfare decisions, including Mahatma Gandhi National Rural Employment Guarantee Act payments.
- **Grassroots Literacy:** Integrate Artificial Intelligence modules within Pradhan Mantri Gramin Digital Saksharta Abhiyan.
- **Sovereign Artificial Intelligence Stack:** Promote BharatGen for culturally relevant data governance.

## Conclusion

- **Inclusive Digital Public Good:** Ethical governance and local-language innovation can transform rural quality of life sustainably.

## Topic: Environment & Conservation

### WETLANDS AS A NATIONAL PUBLIC GOOD

- India celebrated [World Wetlands Day](#) with the theme **Wetlands and traditional knowledge**, notably adding two new Ramsar sites—Patna Bird Sanctuary (UP) and Chhari-Dhand (Gujarat)—bringing the national total to 98.

#### About Wetlands as a National Public Good:

- Wetlands are land areas saturated or flooded with

water, either permanently or seasonally. They act as national public goods because they provide essential services—like flood control and [water purification](#)—that benefit the entire community and cannot be easily restricted to private owners.

### Key Characteristics of Wetlands:

- **Hydrology:** Presence of water at or near the soil surface for at least part of the year.
- **Hydric Soils:** Soils developed under anaerobic (low oxygen) conditions due to prolonged saturation.
- **Hydrophytes:** Specialized vegetation, such as mangroves or reeds, adapted to grow in water-logged soil.
- **Ecotone Status:** They serve as transition zones (buffer areas) between dry land and deep-water aquatic systems.



### Threats and Challenges

- **Urban Encroachment:** Rapid real estate expansion leads to wetland filling for construction, shrinking natural flood buffers and biodiversity habitats.
- E.g. Pallikaranai Marsh (Chennai) has declined due to high-rise townships and infrastructure projects.
- **Pollution and Dumping:** Wetlands are misused for solid waste and sewage disposal, degrading water quality and aquatic life.
- E.g. Ashtamudi Lake (Kerala) witnessed fish deaths in 2025 from industrial effluents and organic waste dumping.
- **Hydrological Disruption:** Dams and sand mining obstruct natural water inflows, altering ecological balance.
- E.g. Harike Wetland (Punjab) shrank as paddy cultivation expanded.
- **Invasive Species:** Non-native plants reduce biodiversity and water storage capacity.
- E.g. Water Hyacinth dominates Loktak Lake (Manipur).
- **Climate Change:** Sea-level rise and increasing

salinity endanger coastal and high-altitude wetlands.

E.g. Sundarbans (West Bengal) faces accelerating mangrove loss.

### Way Forward

- **Notification and Demarcation:** Legally notify wetland boundaries and publish maps to curb illegal conversion.
- **Watershed-Scale Governance:** Manage wetlands within entire river basins, not isolated projects.
- **Capacity Building:** Launch a national mission training managers in Geographic Information Systems, hydrology, and community restoration.
- **Mainstreaming Traditional Knowledge:** Integrate indigenous systems like Kenis of Wayanad into planning.
- **Nature-Based Infrastructure:** Embed wetlands in Smart City designs as flood buffers.

### Conclusion

- Wetlands ensure water security and climate resilience through ecological functionality and shared stewardship.

## ILLEGAL MINING CRISIS

- A massive explosion at an illegal rat-hole coal mine in Meghalaya's [East Jaintia Hills](#) killed at least **18 labourers**, reigniting the national debate over the persistence of banned mining practices and lack of enforcement.

### About Illegal Mining Crisis

- **Illegal mining** involves extraction without valid licenses, environmental clearances, or despite bans. In Meghalaya, it mainly occurs as **rat-hole mining**, where narrow horizontal tunnels or vertical pits are dug unsafely.
- Despite the 2014 ban by the **National Green Tribunal** and Supreme Court support, high-value low-ash coal sustains this hazardous practice.





### Implications of Illegal Mining in India

- **Loss of Human Life:** Absence of ventilation and structural safeguards causes collapses and toxic explosions.  
 E.g. The Thangsku blast killed 18 workers due to unsafe dynamite use in rat-hole mines.
- **Severe Environmental Degradation:** Acid mine drainage contaminates rivers and destroys aquatic ecosystems.  
 E.g. The Kopili River in Meghalaya and Assam turned highly acidic, eliminating aquatic life.
- **Revenue Leakage:** Evasion of royalties and taxes weakens public finances.  
 E.g. A 2025 Comptroller and Auditor General report in Uttar Pradesh flagged massive losses.
- **Funding Organized Crime:** Coal mafias exploit profits to sustain criminal networks.  
 E.g. Activist Agnes Kharshiing faced attacks for exposing illegal coal transport.
- **Ecological Destabilization:** Unscientific extraction triggers subsidence and deforestation.  
 E.g. Land sinking in Jharia, Jharkhand, collapsed houses.

### Challenges in Controlling Illegal Mining

- **Politico-Criminal Nexus:** Influential mine owners backed by politicians weaken enforcement and ignore oversight mechanisms.  
 E.g. The High Court-appointed Katakay Committee reported repeated executive inaction in Meghalaya.
- **Difficult Terrain and Remote Locations:** Forested hills obstruct drone and satellite monitoring, delaying emergency response.  
 E.g. The Thangsku blast site's remoteness slowed National Disaster Response Force deployment.
- **Socio-Economic Dependence:** Limited livelihoods

compel communities to rely on coal income despite bans.

E.g. Families in East Jaintia Hills continue illegal mining for higher wages.

- **Ineffective Technological Adoption:** Mining Surveillance System alerts lack timely ground verification and prosecution.  
 E.g. Right to Information records showed zero convictions despite multiple triggers.
- **Legal Ambiguity and Grandfathering:** Disputes over pre-ban coal enable fresh extraction to be misrepresented as old stock.  
 E.g. Meghalaya High Court hearings highlighted misuse claims.

### Way Ahead

- **Satellite-to-Action Mandate:** Legally mandate action on **Mining Surveillance System** alerts within forty-eight hours, with departmental accountability for negligence.
- **Scientific Mining Policy:** Transition toward regulated, safety-compliant mining aligned with Supreme Court directives and environmental norms.
- **Alternative Livelihoods:** Strengthen Meghalaya's **Bioeconomy (2024–2026)** and eco-tourism to reduce dependence on illegal coal income.
- **Internet of Things and Smart Sensors:** Install Smart Weighbridges and **Global Positioning System** tracking to ensure coal moves only with digital transit permits.
- **Special Environmental Courts:** Establish fast-track courts for swift prosecution of mining violations.

### Conclusion

- Prevent tragedies through technology-driven enforcement, institutional accountability, and sustainable, scientific mining reforms.

## GLOBAL CLIMATE GOVERNANCE

- Climate governance refers to the web of international treaties, domestic laws, and institutional frameworks (like the [UNFCCC](#)) designed to coordinate global action to mitigate greenhouse gas emissions and adapt to climate impacts.





## Challenges Associated

- **Implementation Disease:** Ambitious climate pledges lack binding timelines and assured financing, slowing execution.  
E.g. India's 500 gigawatt non-fossil target faces delays from unsigned Power Sale Agreements and transmission gaps.
- **Growth vs. Sustainability:** Development priorities frequently override ecological safeguards in emerging economies.  
E.g. Great Nicobar Island clearance highlighted tensions between infrastructure expansion and biodiversity protection.
- **Coal Dependency:** Base-load coal reliance complicates rapid decarbonization pathways.  
E.g. Approval of additional coal capacity until 2032 affects transition goals.
- **Urban Climate Vulnerability:** Unplanned urbanization intensifies Urban Heat Islands.  
E.g. Delhi and Mumbai record higher temperatures due to concrete expansion.
- **Short-termism in Politics:** Immediate socio-economic concerns overshadow structural climate reforms.  
E.g. Wayanad landslides were treated mainly as disaster response issues.

## Way Ahead

- **Legally Binding Roadmaps:** Establish time-bound fossil-fuel phase-down schedules for major emitters instead of voluntary pledges.
- **Reform of International Finance:** Redesign global finance to deliver low-interest, long-term climate loans for emerging economies.
- **Subnational Governance:** Empower states and cities to lead adaptation, reflecting localized

climate risks.

- **Nature-Based Solutions:** Integrate Blue-Green infrastructure, including mangroves and urban forests, into planning to curb heatwaves and floods.
- **Universal Insurance:** Expand Climate Risk Insurance for smallholder farmers and coastal communities.

## Conclusion

- Climate governance must shift from diplomatic symbolism to rapid financing, binding commitments, and ecosystem protection.

## NATURE-BASED SOLUTIONS (NBS)

- Nature-based Solutions are defined by the IUCN as actions to protect, sustainably manage, and restore natural or [modified ecosystems to](#) address societal challenges (like climate change, food security, or water safety) effectively and adaptively.
- They are unique because they provide simultaneous benefits for both human well-being and biodiversity.



## Challenges to Nature-based Solutions

- **Lack of Standardized Frameworks:** Absence of uniform standards enables greenwashing and flawed ecological design.  
E.g. Afforestation in Central India faced criticism for monoculture Eucalyptus plantations harming groundwater.
- **Financing and Budgetary Constraints:** Nature-based Solutions receive limited, inconsistent funding compared to grey infrastructure.  
E.g. Only a fraction of climate policies allocate explicit budgets for ecological components.
- **Complex Governance and Land Tenure:**

Overlapping forest, water, and urban authorities delay restoration initiatives.

E.g. Aravalli Green Wall efforts face multi-state land-right disputes.

- **Disconnect from Urban Planning:** Projects remain peripheral instead of central to city design.  
E.g. Bengaluru developers still prioritize concrete drainage.
- **Knowledge and Expertise Gaps:** Inadequate site-specific data undermines project success.  
E.g. Mangrove restoration in Tamil Nadu failed due to ignored hydrological changes.

### Way Ahead

- **Mainstreaming Nature-based Solutions:** Embed Blue-Green infrastructure within **Prime Minister Gati Shakti** planning.  
E.g. Integrate rain gardens and bioswales under the Smart Cities Mission.
- **Unlocking Green Finance:** Launch Sovereign Forest Bonds and transparent carbon markets.  
E.g. Use **Digital Personal Data Protection Act, 2023** systems for carbon verification.
- **Community-Led Governance:** Empower Gram Sabhas and Indigenous groups.  
E.g. Expand women-led committees under Jal Jeevan Mission.
- **Science-Based Monitoring:** Apply satellite and geospatial tracking.  
E.g. Meri LiFE portal monitors sapling survival.
- **Scaling Agroforestry:** Promote tree-based farming transitions.  
E.g. Follow South Asian Agroforestry Congress roadmap.

### Conclusion

- **Nature-based Solutions** encourage partnership with ecosystems to address climate and societal challenges effectively.
- Funding shortages and weak standardization restrict large-scale implementation.
- Strengthening finance and empowering communities can convert natural capital into durable climate resilience nationwide sustainably.

Topic: Defence, ISRO & Navy

## INCREASED CAPITAL SPENDING FOR THE DEFENCE SECTOR

- The government announced a historic 15.2% hike in the total defence budget for FY2026-27, primarily to address tactical gaps exposed by the recent

Operation Sindoor conflict.

- The allocation aims to strengthen India's "security-development-self-reliance" balance amid rising geopolitical tensions.
- Capital spending (or capital outlay) is the portion of the budget dedicated to **modernization and the acquisition of new assets** such as advanced weapons, fighter jets, ships, and submarines.
- For FY2026-27, capital outlay has surged by **21.8%** to reach **₹2,19,306 crore**, marking a strategic pivot from manpower costs to technology-intensive capability building.

### Need for Increasing the Defence Budget

- **Replenishment of War Reserves:** Replace depleted ammunition and precision-guided munitions after active operations.  
E.g. High capital outlay followed Operation Sindoor in May 2025.
- **Countering Two-Front Threats:** Address simultaneous challenges from adversarial neighbours along sensitive borders.  
E.g. Modernization sustains deterrence at the Line of Actual Control against China.
- **Maritime Security in Indian Ocean Region:** Enhance undersea and surface surveillance capabilities.  
E.g. Project 75(I) strengthens stealth submarine capacity.
- **Technological Superiority:** Invest in Artificial Intelligence, cyber, and unmanned systems.  
E.g. Increased Agnipath allocation supports modernization.
- **Border Connectivity:** Build all-weather infrastructure for rapid mobilization.  
E.g. Border Roads Organisation funding accelerates strategic tunnels.

### Issues Associated with the Defence Sector

- **Structural Imbalance:** High manpower expenses, including salaries and pensions, constrain modernization funding.  
E.g. Defence pensions significantly rival capital acquisition allocations in recent budgets.
- **Absorption Capacity:** Domestic industries face challenges executing large contracts within strict timelines.  
E.g. Fragmented planning and delayed trials reduce capital utilization efficiency.
- **Import Dependency:** Critical technologies like aero-engines remain externally sourced.  
E.g. The Indian Air Force 114 Multi-Role Fighter Aircraft project depends on foreign Original Equipment Manufacturers.



- **Execution and Delivery Delays:** Indigenous platforms experience prolonged development cycles.  
E.g. Project 75(I) submarine acquisition delays affect naval strength.
- **Budget as Percentage of Gross Domestic Product:** Spending remains below strategic recommendations.  
E.g. Allocation hovers near two percent of Gross Domestic Product.

### Way Ahead

- **Institutionalizing Emergency Procurement:** Convert crisis-driven fast-track acquisitions into permanent systems for critical technologies and rapid capability gaps.
- **Theaterisation and Jointness:** Increase funding for Joint Staff structures to strengthen integrated planning and inter-service resource optimization.
- **Intellectual Property-led Design:** Prioritize domestic ownership of Intellectual Property in at least fifty percent of defence contracts.
- **Developing Export Hubs:** Expand modular manufacturing to achieve ₹35,000 crore annual defence exports.
- **Nurturing Micro, Small and Medium Enterprises:** Integrate domestic firms into global supply chains for advanced components.

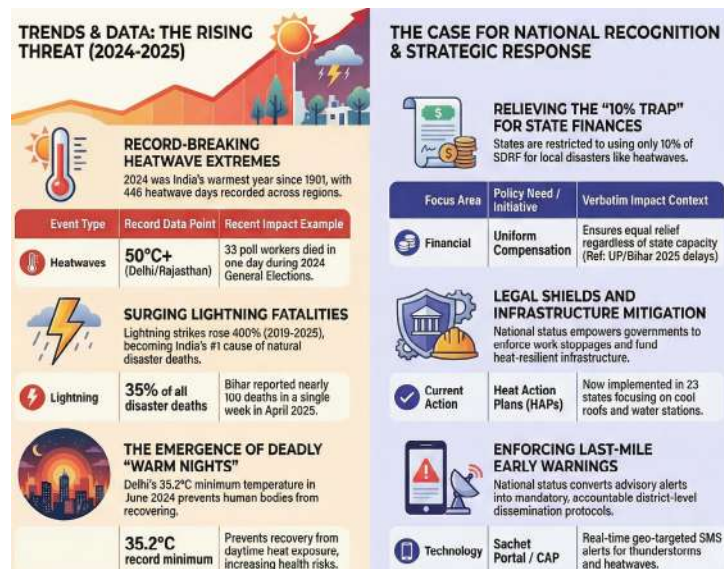
### Conclusion

- The defence budget strengthens immediate readiness while advancing strategic autonomy.
- Long-term success depends on rapid technology absorption, efficient execution, and globally competitive domestic defence manufacturing capabilities.

Topic: Disaster Management

## 16TH FINANCE COMMISSION RECOMMENDS HEATWAVES AND LIGHTNING AS NATIONAL DISASTERS

- The [16th Finance Commission](#) has formally recommended the inclusion of heatwaves and lightning in India's list of nationally notified disasters, a move triggered by record-breaking temperatures and rising fatalities in 2024-25.



### Challenges Associated with Inclusion

- **Attribution of Death:** Proving heat as the primary cause remains medically complex.  
E.g. During 2024 heatwaves, cases were recorded as fever, not heatstroke.
- **Financial Strain on Centre:** Annual compensation payouts could escalate significantly.  
E.g. Thousands of lightning deaths would raise fiscal burdens sharply.
- **Widespread Occurrence:** Heatwaves span vast regions, complicating disaster-zone identification.  
E.g. Northwest India experiences prolonged temperatures above forty-five degrees Celsius.
- **Data Accuracy:** Rural under-reporting distorts fund allocation.  
E.g. Media figures understate actual rural fatalities.
- **Incentive for Misreporting:** Expanded categories may inflate compensation claims.  
E.g. Disaster claims historically surge after new inclusions.

### Way Ahead

- **Dynamic Thresholds:** Establish state-specific temperature and Wet Bulb limits for declaring heatwaves.
- **Parametric Insurance:** Design insurance models protecting farmers and outdoor workers from wage losses.
- **Last-Mile Connectivity:** Deploy Aapda Mitra volunteers for real-time lightning alerts in fields.
- **Urban Cooling:** Integrate Cool Roof policies and Green Corridors under the Smart Cities Mission.
- **Scientific Death Audits:** Form medical boards to certify heat-related fatalities transparently.

### Conclusion

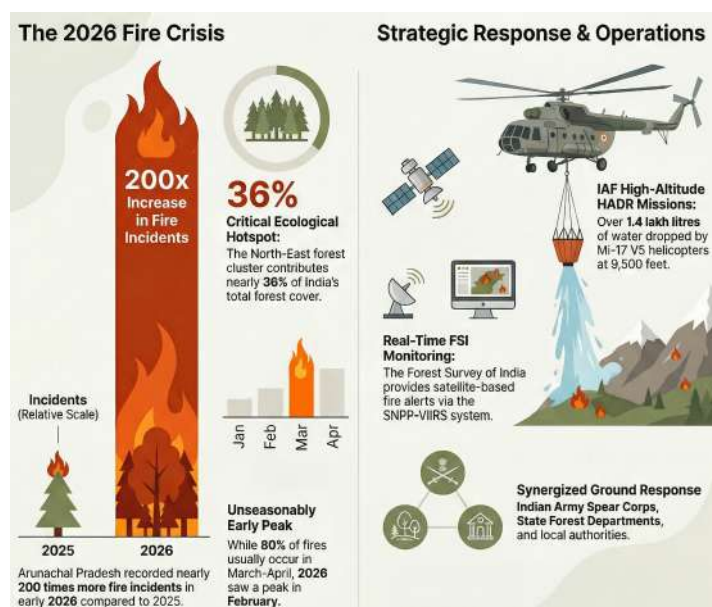
- Recognition of heatwaves and lightning strengthens climate-responsive disaster governance.



- Structured fiscal backing and transparent certification can safeguard vulnerable communities and shift policy from reactive relief toward proactive resilience.

### Forest Fires in North East India

- Forest fires in the North East are primarily **surface fires** that feed on dry leaf litter, grass, and forest debris. Unlike the crown fires seen in the West, these spread rapidly across steep slopes during the dry winter months (December–March).
- The region is a [global biodiversity hotspot](#), and such fires threaten rare species, indigenous livelihoods, and the stability of the mountain soil.



### Reasons for Forest Fires

- **Dry Weather and Climate Variability:** Prolonged dry spells and weak Western Disturbances dry forest floors, increasing ignition risk. E.g. Severe moisture deficit in Lohit Valley intensified wildfire spread.
- **Shifting Cultivation (Jhum):** Slash-and-burn practices can escape control and engulf nearby forests. E.g. Unchecked Jhum fires reportedly triggered the Walong blaze.
- **Accumulation of Fuel Load:** Dry bamboo, pine needles, and grasses act as inflammable material. E.g. Dzukou Valley vegetation fueled flames toward Japfu Peak.
- **Accidental Human Ignition:** Negligence by visitors or locals sparks fires. E.g. Trekkers' carelessness worsened Dzukou Valley fire.
- **Topographic Wind Patterns:** Steep slopes and valley winds accelerate uphill spread.

E.g. Strong winds near Japfu Peak intensified flames.

### Challenges Associated with Elimination

- **Inaccessible Terrain:** Rugged Himalayan ridges prevent fire engines from reaching hotspots quickly. E.g. In Anjaw, teams required days to access remote fire locations.
- **Rarefied Air and Poor Visibility:** Thin air and dense smoke restrict aerial firefighting efficiency. E.g. Indian Air Force helicopters faced lift constraints near Padumpokhiri Lake.
- **Shortage of Water Sources:** Limited nearby water bodies delay repeated water drops. E.g. Dzukou operations required long trips to Dimapur lakes.
- **Communication Gaps:** Delays between satellite alerts and field response escalate fire intensity. E.g. Surface fires in Nagaland evolved into slope fires before intervention.
- **Socio-Cultural Factors:** Regulating Jhum cultivation involves sensitive livelihood considerations. E.g. Traditional practices persist in Arunachal Pradesh.

### Way Ahead

- **Community-Led Fire Brigades:** Empower village councils as first responders under Joint Forest Management frameworks.
- **Technology Integration:** Deploy drones and Artificial Intelligence-based fire spread modelling for prediction and night surveillance.
- **Infrastructure Development:** Build high-altitude static water tanks to support helicopter bucket operations.
- **Fuel Management:** Encourage commercial collection of dry biomass like pine needles for bio-fuel use.
- **Inter-State Coordination:** Create a North-East Fire Task Force for coordinated response.

### Conclusion

- Strengthening community participation and modern technology is essential for sustainable wildfire management.
- Protecting fragile North-Eastern ecosystems safeguards climate stability, biodiversity conservation, and long-term regional environmental security.

Topic: Cyber Crimes & Digital Security**CYBERCRIME IN INDIA**

- Cybercrime refers to criminal activities carried out using computers, networks, or digital devices. In the Indian context, it has evolved from simple phishing to sophisticated digital arrests, sextortion, and AI-driven fraud.
- It targets individuals, businesses, and [critical infrastructure](#), threatening both economic stability and personal privacy.

**Reasons for the Rise in Cybercrime**

- **Increased Internet Penetration:** Expanding connectivity enlarges the digital attack surface, especially among new rural users.  
E.g. Fraudsters target first-time users in Tier-2 and Tier-3 cities lacking cyber hygiene awareness.
- **Desire for Easy Money:** Economic pressures and quick-profit temptations fuel fraudulent investment schemes.  
E.g. Fake stock trading applications surged amid speculative enthusiasm.
- **Adoption of Artificial Intelligence and Deepfakes:** Generative tools enable convincing impersonation scams.  
E.g. Digital Arrest frauds used synthetic police visuals during calls.
- **Cross-Border Scam Centers:** Organized foreign networks coordinate large-scale fraud operations.  
E.g. Several scams trace to Southeast Asian compounds.
- **Loopholes in Digital Banking:** Unified Payments Interface misuse exploits limited security awareness.  
E.g. Victims shared Personal Identification Numbers during collect requests.

**Challenges Associated with Cybercrime**

- **Low First Information Report Conversion:** Portal complaints rarely translate into formal investigations due to jurisdictional complexities.  
E.g. Lakhs of reports in 2025 resulted in limited First Information Reports nationwide.
- **Anonymity and Technical Sophistication:** Virtual Private Networks and encrypted layers conceal criminal identities.  
E.g. Ransomware-as-a-Service models stalled hospital attack probes.
- **Jurisdictional Hurdles:** Multi-state bank accounts and mobile numbers complicate arrests.  
E.g. Delhi fraud linked to mule accounts elsewhere.

- **Forensic Talent Gap:** Shortage of trained cyber forensic experts delays digital evidence analysis.  
E.g. Dedicated cyber stations face device backlogs.
- **Social Stigma:** Victims hesitate reporting sextortion or digital arrest scams.  
E.g. Delayed helpline contact increased financial losses.

**Way Ahead**

- **Mandatory Electronic First Information Reports:** Expand nationwide electronic First Information Report registration for all digital complaints.
- **Artificial Intelligence-Powered Defense:** Use Machine Learning to pre-emptively flag mule bank accounts.
- **International Cooperation:** Strengthen Indian Cyber Crime Coordination Centre collaboration with Interpol and Southeast Asian authorities.
- **Cyber Hygiene Education:** Introduce compulsory digital literacy for students and senior citizens.
- **Strengthening National Critical Information Infrastructure Protection Centre:** Secure power grids, banks, and vital networks.

**Conclusion**

- Cybercrime now poses a serious socio-economic threat beyond technical boundaries.
- Sustainable prevention demands coordinated enforcement, international partnerships, resilient infrastructure, and widespread digital awareness among citizens.

Topic: Internal Security & its Challenges**DECLINING NAXALISM IN INDIA**

The surrender of **Thippiri Tirupathi (alias Devuji)**, a top-ranking Maoist leader with a ₹1 crore bounty, marks a historic unravelling of the CPI (Maoist) leadership.

- Naxalism, or Left-Wing Extremism ([LWE](#)), is an armed Maoist insurgency that seeks to overthrow the state through guerrilla warfare. For decades, it was described as India's greatest internal security threat.
- However, it is now in a state of terminal decline, transitioning from a Red Corridor that once spanned ten states to isolated pockets in the [Bastar region](#) and the Jharkhand-Bihar border.

**Reasons for the Decline**

- **Leadership Crisis:** Senior Telugu leaders' surrenders and deaths weakened ideological and tactical direction.  
E.g. The surrender of Devuji and strategist Sonu

left cadres disoriented.

- **Aggressive Security Posture:** Greyhounds and Commando Battalion for Resolute Action units adopted proactive offensives.  
E.g. Forces entered former no-go zones in South Bastar.
- **Vacuum in Ideology:** Tribal youth prefer connectivity and employment over insurgent doctrine.  
E.g. Recruitment shortages forced reliance on lower-rung militia.
- **Infrastructural Penetration:** Roads and mobile towers eroded geographical advantages.  
E.g. Road Requirement Plan improved rapid troop access.
- **Enhanced Intelligence:** Drone surveillance and technical tracking restricted platoon mobility.  
E.g. Precision operations targeted senior commanders.

#### Key Initiatives Taken:

- **Operation SAMADHAN:** A comprehensive strategy focusing on Smart Leadership, Aggressive Strategy, Motivation, Actionable Intelligence, and Dashboard-based KPIs.
- **Aspirational Districts Programme:** Focused development in 35 LWE-hit districts to improve healthcare, education, and financial inclusion.
- **ROSHNI Scheme:** A special placement-linked skill development initiative specifically targeting tribal youth in LWE-affected areas.
- **Surrender and Rehabilitation Policy:** Offering high cash incentives (e.g., ₹1 crore for top leaders) and land to encourage insurgents to join the mainstream.

#### Remaining Challenges

- **Hardened Pockets:** Small guerrilla units persist in dense Abujmad forests where terrain favors ambush tactics.  
E.g. Sporadic attacks in Bastar indicate residual sniper activity.
- **Inter-State Coordination Gaps:** Insurgents exploit porous borders between neighboring states.  
E.g. Leaders shift between Jharkhand and Bihar to evade capture.
- **Governance Deficit:** Remote hamlets still lack clean water, healthcare, and fair forest produce pricing.  
E.g. Grievances in Sukma remain recruitment triggers.
- **Overground Fronts:** Urban networks provide ideological and logistical cover.  
E.g. Radicalization efforts target university spaces.

- **Weaponry and Improvised Explosive Devices:** Low-cost explosives disrupt security operations.  
E.g. Dantewada convoy blasts highlight persistent threats.

#### Way Ahead

- **Sustaining Development:** Establish permanent schools, hospitals, and welfare delivery after insurgent retreat.
- **Tribal Language Outreach:** Communicate in Gondi and Halbi to counter extremist narratives effectively.
- **Strengthening Local Police:** Gradually transition from Central Reserve Police Force deployment to culturally rooted local policing.
- **Mine Protected Vehicles:** Upgrade Mine Protected Vehicles to reduce Improvised Explosive Device casualties.
- **Border Sealing:** Employ satellite mapping across Odisha–Chhattisgarh–Maharashtra tri-junctions.

#### Conclusion

- The decline of Communist Party of India (Maoist) leadership signals strategic erosion; sustained governance and inclusive development remain essential to prevent resurgence.



# GENERAL STUDIES - 4

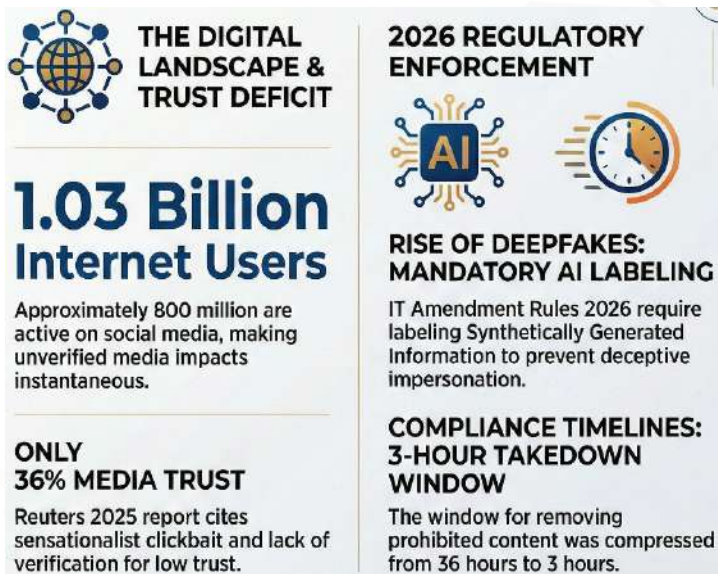
Topic: Code of ethics and code of conduct

## DIGITAL MEDIA AND CODE OF ETHICS

The Uttarakhand High Court issued a stern warning to journalists and digital creators, emphasizing that media operations on [social and digital platforms](#) must strictly adhere to a prescribed code of ethics to avoid legal consequences, including criminal charges for defamation or extortion.

### About Digital Media and Code of Ethics:

- A **Code of Ethics** for digital media is a set of guidelines and regulatory standards designed to ensure accuracy, fairness, and accountability in online journalism.
- In India, this is primarily governed by the [Information Technology \(Intermediary Guidelines and Digital Media Ethics Code\) Rules, 2021](#), which mandate that digital news publishers follow the Norms of Journalistic Conduct.



### Need for Digital Media to Follow Code of Ethics

- **Prevention of Character Assassination:** Unverified digital content can **damage reputations before facts emerge**, causing irreversible professional and social harm.  
 E.g. In the **Uttarakhand High Court** case, an unverified post led to action against a **State Bank of India (SBI)** branch manager despite the complaint being withdrawn.

- **Curbing Viral Misinformation:** The **speed of digital platforms amplifies false information**, spreading panic rapidly.  
 E.g. False death rumours about actor **Dharmendra** circulated widely before verification.
- **Maintaining Public Order:** Misleading reports on sensitive issues can **trigger unrest and communal tensions**.  
 E.g. A **Myanmar video mislabelled as Manipur violence** spread online in 2025.
- **Protecting Vulnerable Groups:** Ethical rules prevent **secondary victimization of children and crime victims**.  
 E.g. The **Arunachal Press Club** apologized after a minor's interview was published.
- **Preserving Democratic Integrity:** Ethical journalism ensures **fact-based voter information**.

### Challenges in Implementation by Digital Media

- **Clicks Economy:** Advertisement-driven revenue models reward **sensational headlines and viral content** instead of careful reporting.  
 E.g. Several digital paparazzi portals in **2025** leaked **hospital footage of celebrities** to increase engagement and advertising income.
- **Anonymity and Ghost Portals:** Many online outlets operate **without registered editors or addresses**, complicating accountability.  
 E.g. Authorities in **2026** struggled to serve notices to **WhatsApp-based news groups** sharing defamatory content.
- **Speed vs Accuracy:** The race to break news encourages **publishing before verification**.  
 E.g. The **Uttarakhand High Court** noted allegations were circulated without verifying with concerned parties.
- **Technological Barriers: Artificial Intelligence (AI) and deepfakes** blur authenticity.  
 E.g. Regional creators shared **AI-cloned political audio** assuming it genuine.
- **Weak Self-Regulation:** Smaller publishers often bypass the **three-tier regulatory framework**.

### Way Ahead

- **Mandatory Registration:** All digital news publishers should register with the **Ministry of Information and Broadcasting (MIB)** to ensure accountability and access to a formal **grievance redressal mechanism**.
- **Verification-First Protocol:** Digital newsrooms must adopt a **pause-before-publish policy**, especially for stories involving criminal allegations or private individuals.

- **Artificial Intelligence Labelling Compliance:** Platforms must follow **Information Technology (IT) Rules 2026**, clearly watermarking **synthetic or Artificial Intelligence-generated media**.
- **Legal Literacy for Journalists:** Governments should train journalists on **Norms of Journalistic Conduct** and **Bharatiya Nyaya Sanhita (BNS)** defamation provisions.
- **Tiered Penalty Structure:** Stronger **financial and legal penalties** should deter repeat offenders.

#### Conclusion

- Digital media must uphold **accuracy, accountability, and ethical responsibility**. Protecting freedom of speech while preventing misinformation and reputational harm is essential to sustain **public trust and democratic integrity** in the information ecosystem.

# FACTS FOR PRELIMS

## GS-1

Topic: Indian Culture-Salient aspects of Art Forms

### UNION BUDGET 2026-27 ANNOUNCES SCHEME TO DEVELOP BUDDHIST CIRCUIT IN NORTHEAST

- The Buddhist Circuit (Northeast) is a heritage-tourism initiative aimed at conservation, connectivity, and promotion of Buddhist monasteries and pilgrimage sites across northeastern States, integrating culture with livelihoods and regional development.

#### Key budget announcements:

- Buddhist Circuit (Northeast):** New scheme to preserve and develop temples and monasteries.
- Tourism infrastructure:** Creation of five tourism destinations in [Purvodaya States](#) and deployment of 4,000 e-buses for tourist mobility.
- Regional growth:** ₹5,000 crore allocated over five years for CITY Economic Regions (CERs) to boost urban-industrial ecosystems.
- Connectivity thrust:** Complemented by the [East Industrial Corridor](#) to link tourism with logistics and markets.

Here is a list of the most important monasteries in the region, categorized by state:

#### Arunachal Pradesh

Arunachal houses the largest concentration of Mahayana Buddhist sites, many located in the Tawang and West Kameng districts.

- Tawang Monastery (Galden Namgyal Lhatse):** The largest monastery in India and the second-largest in the world. Perched at 10,000 feet, it was founded in the 17th century and houses a massive 18-foot gilded Buddha statue.
- Bomdila Monastery:** Established in 1965, it is a prominent center for the Mahayana sect and offers panoramic views of the Himalayas.
- Urgelling Monastery:** Famous as the birthplace of the [6th Dalai Lama](#). It is one of the oldest in the region, dating back to the 15th century.
- Golden Pagoda (Namsai):** A beautiful Burmese-style pagoda known for its gold-plated structure and serene gardens, reflecting the Theravada tradition.
- Taktsang Gompa:** Often called the Tiger's Nest of India, it is located near the Bhutan border and is where Guru Padmasambhava is said to have

meditated.

#### Sikkim

Sikkim's monasteries are predominantly of the Nyingma and Kagyu schools of [Tibetan](#) Buddhism.

- Rumtek Monastery:** The largest in Sikkim and the seat-in-exile of the **Gyalwang Karmapa** (Kagyu sect). It is a major center for Buddhist studies.
- Pemayangtse Monastery:** One of the oldest and most prestigious monasteries in Sikkim. It historically controlled all other Nyingma monasteries in the state.
- Enchey Monastery:** Located in Gangtok, this 200-year-old monastery is built on a site blessed by the tantric master Lama Druptob Karpo.
- Tashiding Monastery:** Considered the holiest monastery in Sikkim; it is believed that even a glimpse of the monastery can cleanse one of their sins.
- Dubdi Monastery:** Known as the Hermit's Cell, it is the [oldest monastery in Sikkim](#), established in 1701 during the consecration of the first Chogyal (King).

#### Other Notable Monasteries

- Ghoom Monastery (West Bengal):** While technically in North Bengal (Darjeeling), it is a vital part of the North East circuit. It is famous for its 15-foot high statue of the Maitreya Buddha.
- Namphake Monastery (Assam):** Located near Dibrugarh, it belongs to the Tai-Phake community and is one of the oldest and most beautiful Theravada monasteries in the state.
- Venuvan Vihara (Tripura):** Located in Agartala, this is a significant site for the local Buddhist community and features a metal Buddha statue brought from Myanmar.

Topic: Modern Indian History

### SERENGZIA BATTLE

- The Serengzia battle was a fierce armed resistance by Ho Adivasis against the East India Company in 1837, fought in the Serengzia valley of present-day Jharkhand (West Singhbhum).
- It represents one of the earliest organised tribal military challenges to British expansion in eastern India.

#### Historical background:

- The Kolhan region (East & West Singhbhum, Seraikela-Kharsawan) was traditionally governed by the Ho community.
- In **1820-21**, the British brought Kolhan under the [Bengal Presidency](#) to secure trade routes between



Bengal and Madras.

- British policies imposed taxes, allowed non-tribal settlement, and enforced alien languages and authority systems.
- Exploitation by zamindars and officials led to repeated unrest, including the **Kol uprising of 1831**.
- By **1836**, [British forces](#) established the **Kolhan Estate Government**, capturing Ho villages and pirs, intensifying resistance.

#### Causes of the battle:

- **Loss of autonomy:** Imposition of British administration over sacred Ho land believed to be granted by Sing-Bonga (supreme deity).
- **Economic exploitation:** Forced taxation and land alienation.
- **Cultural suppression:** Linguistic imposition and social domination.
- **Military repression:** Arrests, village occupations, and coercive policing by British forces.

#### Leaders involved:

- The Ho resistance was led by: Poto Ho (principal leader, from Rajabasa), Berai Ho, Pundua (Pandua) Ho, Badai Ho, Nara Ho, Devi Ho, and Sugni Ho.

#### The battle (1837):

- **Ho strategy:** Guerrilla-style warfare using terrain advantage in the narrow Serengsia valley.
- **Weapons & tactics:** Bows and arrows, obstacles on valley paths, burning cow dung mixed with ash and chilli powder, and [coordinated attacks](#) from hill slopes.
- **Outcome:** Over **100 British soldiers** killed; about **26 Ho fighters** lost their lives. British forces were forced to retreat initially.

#### Outcomes and aftermath:

- British retaliation followed with village burnings, mass arrests, and collective punishment.
- By **December 8, 1837**, all major Ho leaders were captured.
- **Executions:**
  - **January 1, 1838:** Poto Ho, Berai Ho, and Nara Ho hanged publicly at Jagannathpur.
  - **January 2, 1838:** Bora Ho and Pandua Ho hanged near Mundasai, Serengsia.
- Around **79 Ho fighters** were imprisoned.
- Despite repression, the resistance influenced later recognition of Kolhan's distinct administrative status and [tribal self-governance](#) traditions.

## DENOTIFIED TRIBES (DNTS)

- Denotified Tribes, also known as Vimukta Jatis, are communities that were collectively branded as criminal tribes during British rule. These groups are largely nomadic or semi-nomadic, historically dependent on occupations [such as pastoralism](#), salt trading, acrobatics, snake charming, transport services, and artisanal work.

#### Historical background:

- In **1871**, the colonial government enacted the **Criminal Tribes Act**, declaring entire communities as addicted to crime by birth.
- Members were subjected to **constant surveillance**, **restricted movement**, **forced registration**, and social ostracisation.
- The law was **repealed in 1952**, after Independence, and the communities were officially **denotified**, giving rise to the term [Denotified Tribes](#).
- However, the **Habitual Offenders framework** replaced the colonial law, allowing police to continue targeting these groups, perpetuating stigma.

#### Key characteristics:

#### [Extreme social exclusion: Persistent stigma of being born criminals continues in policing and society.](#)

- **Administrative invisibility:** Many DNTs lack caste certificates, domicile records, or land titles.
- **Educational deprivation:** In several regions, entire communities have negligible secondary-level education.
- **Economic precarity:** Seasonal migration, informal labour, and absence of stable livelihoods dominate.
- **Political misclassification:**
  - Most DNTs have been absorbed into SC, ST, or [OBC lists](#), where they are unable to compete with relatively advanced groups.
  - Around 260 communities are not classified anywhere, leaving them outside reservation and welfare frameworks.

## MAHARSHI DAYANAND SARASWATI



- Maharshi Dayanand Saraswati (1824–1883), born Mool Shankar, was a Hindu ascetic, Vedic scholar and social reformer who founded the [Arya Samaj](#) (1875) to reform Hindu society and restore what he saw as the original Vedic spirit.

### Early life:

- Born on **12 February 1824** at Tankara (Kathiawar region, present-day Gujarat) in a Brahmin family.
- A childhood temple incident (questioning idol worship) and the deaths of close family members deepened his search for truth.
- Left home to avoid an arranged marriage and spent years as a wandering ascetic; later became a disciple of **Swami Virajananda**, who guided him toward a return to the Vedas mission.

### His philosophies:

- **Back to the Vedas:** Vedas as the primary and most authentic source of dharma/knowledge.
- **Rational reform:** opposed blind ritualism, superstition, and practices he considered irrational.
- **Social equality:** challenged caste-by-birth and supported dignity of all individuals; emphasized merit-based social roles.
- **Women's upliftment:** supported **women's education**, opposed social practices like child marriage, and encouraged reforms in family and society.
- **Ethical life & discipline:** stressed truth, self-control ([brahmacharya](#)), yoga, and moral conduct.

### Contribution to socio-religious movement:

- Founded **Arya Samaj (1875)**—a reform movement promoting Vedic learning, social reform, education, and national-cultural awakening.
- Encouraged debate, public reasoning, and scriptural study accessible beyond narrow priestly control.

- Inspired later reform and nationalist currents through emphasis on **self-respect, reform, and indigenous awakening**.

### Literary works:

- Dayanand wrote **60+ works**, including major texts such as:
  - **Satyarth Prakash** (his most influential work).
  - **Sanskarvidhi** (on sacraments/rites in a reformed framework).
  - **Rigvedadi Bhashya Bhumika** and partial Vedic commentaries (Rigveda/Yajurveda).
- He also established/linked efforts for publishing and spreading Vedic literature through institutions like **Paropakarini Sabha (Ajmer)**.

## 80TH ANNIVERSARY OF THE 1946 ROYAL INDIAN NAVY REVOLT

- The Royal Indian Navy (RIN) Revolt was a **five-day armed uprising (February 18–23, 1946)** by Indian naval ratings against British colonial authority.
- It began as a protest over poor conditions but soon evolved into a wider anti-colonial rebellion involving sailors, workers, and civilians.

### Historical Background:

- Indian ratings faced [racial discrimination](#), **poor food, low wages, and harsh treatment**.
- Inspired by the **Quit India Movement (1942)** and the trials of the **Indian National Army (INA)**.
- Appointment of openly racist officers, such as **Arthur Frederick King** at HMIS Talwar, further fuelled anger.

### Leaders Associated:

- Prominent leaders of the uprising included: B. C. Dutt, M. S. Khan, Madan Singh, Salil Shyam, and Rishi Dev Puri.
- A [Naval Central Strike Committee](#) was formed to coordinate actions and present demands.

### Events of the Revolt:

- **Beginning at HMIS Talwar (Bombay):**
  - Started with a hunger strike over poor food quality on February 18, 1946.
  - Raised nationalist slogans such as "[Quit India](#)" and "Jai Hind."
- **Rapid Spread:**
  - Spread to **78 ships and 20 shore establishments** across Bombay, Karachi, Madras, Vishakhapatnam, Kolkata, and the

Andamans.

- Nearly **20,000 naval ratings** participated.

• **Popular Support:**

- Workers, students, and civilians joined in solidarity.
- [Hindu-Muslim unity](#) was visible as Congress, Muslim League, and Communist flags were hoisted together.
- The **Bombay Uprising of 1946** - refers to the mass protests and street battles that erupted in Bombay (now Mumbai), in support of the Royal Indian Navy (RIN) revolt against British rule.

• **Armed Confrontation:**

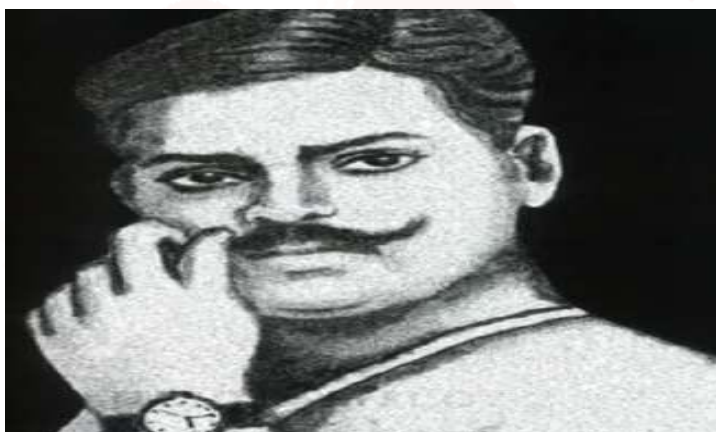
- British troops opened fire in Bombay.
- Street battles erupted in mill districts like Kamatipura and Madanpura.
- Around 200 civilians were killed during suppression efforts.

**End of the Revolt:**

- Political leaders, including Congress and [Muslim League leadership](#), urged restraint.
- Naval ratings surrendered on February 23, 1946.
- Leaders were arrested, and the uprising was militarily suppressed.

However, the revolt deeply shook British confidence in maintaining control over India.

## CHANDRASHEKHAR AZAD



- Chandrashekhhar Azad (born Chandra Shekhar Sitaram Tiwari) was one of India's most fearless and potent revolutionaries.
- Known for his mental and physical dexterity—earning him the nickname **Quick Silver**—he led a [militant youth movement](#) and famously vowed

never to be captured alive by the British.

**Early Life:**

- **Birth:** July 23, 1906, in Bhabra, Alirajpur (Madhya Pradesh).
- **Education:** He grew up in poverty and later attended a Sanskrit school in Varanasi.
- **The Naming:** After being arrested at age 15 during the Non-Cooperation Movement, he told the magistrate his name was **Azad** (Free), his father's name was **Swatantra** (Independent), and his home was **Jail**. Following a severe flogging, he adopted Azad as his permanent title.

**Contribution to Freedom Movement:**

- **Shift to Militancy:** Disappointed by the suspension of the [Non-Cooperation Movement](#) after the Chauri Chaura incident, he embraced extremist methods to achieve independence.
- **Kakori Conspiracy (1925):** He was a key participant in the Kakori train robbery. While others were caught, Azad successfully evaded capture and fled to Jhansi.
- **Lahore Conspiracy (1928):** Along with Bhagat Singh and Rajguru, he executed the assassination of British officer John Saunders to avenge the death of Lala Lajpat Rai.

**Organizations Associated:**

- **Hindustan Republican Association (HRA):** Joined under the leadership of Ram Prasad Bismil.
- **Hindustan Socialist Republican Association (HSRA):** After the crackdown on the HRA, Azad reorganized the group into the HSRA, serving as a key leader alongside Bhagat Singh.
- He used the name **Balraj** to sign official HSRA statements and lived under the disguise of a teacher named **Harishankar** in Jhansi.

**Last Days:**

- On **February 27, 1931**, Azad was betrayed and surrounded by police at **Alfred Park** (now Azad Park) in Allahabad.
- After a fierce gun battle where he helped his comrade Sukhdev Raj escape, Azad died at the age of 24.

Topic: Important Geographical & Geophysical phenomena

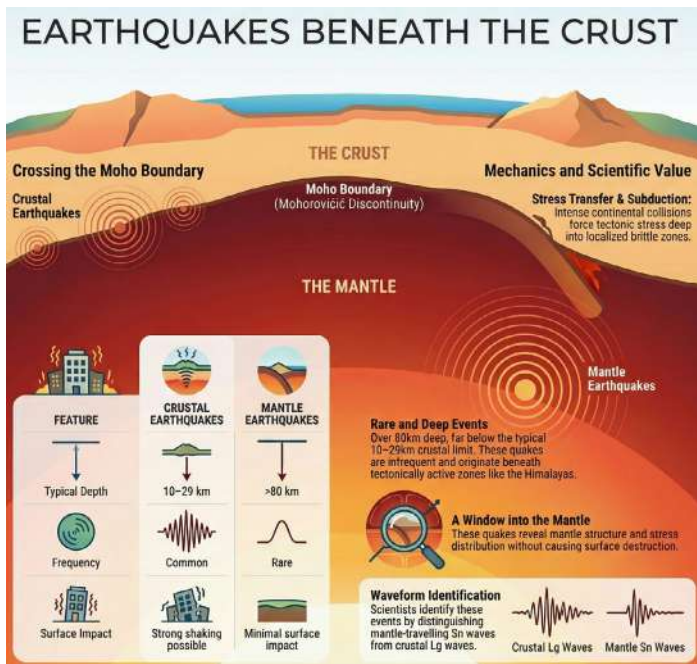
## CONTINENTAL MANTLE EARTHQUAKES

- Continental mantle earthquakes are rare seismic events that originate deep within the Earth's mantle beneath [continental regions](#), rather than in



the shallow crust where most earthquakes occur.

- Unlike typical crustal earthquakes (10–29 km depth), these quakes occur **more than 80 km below the Mohorovičić discontinuity (Moho)**—the boundary separating the crust from the mantle.



### Where Do They Occur?

- The recent study found clustering beneath tectonically active regions such as:
  - The **Himalayan collision zone** (Indian Plate subducting beneath the Eurasian Plate)
  - The **Bering Strait region**
- Such areas are marked by intense continental collision and lithospheric deformation.

### How Do They Originate?

- Tectonic Stress Transfer:** Stress from continental collision or crustal earthquakes may penetrate deeper into the mantle, triggering rupture.
- Subduction-Related Processes:** In collision zones, the descending plate may retain brittle characteristics at greater depths.
- Thermal & Rheological Variations:** Though the mantle is generally semi-solid and ductile, localized cooler or compositionally distinct zones may become brittle enough to fracture.
- Seismic Wave Identification:** Scientists distinguished mantle quakes using differences between **Sn waves (mantle-travelling)** and **Lg waves (crustal-travelling)** in seismic waveforms.

### How Are They Different from Crustal Earthquakes?

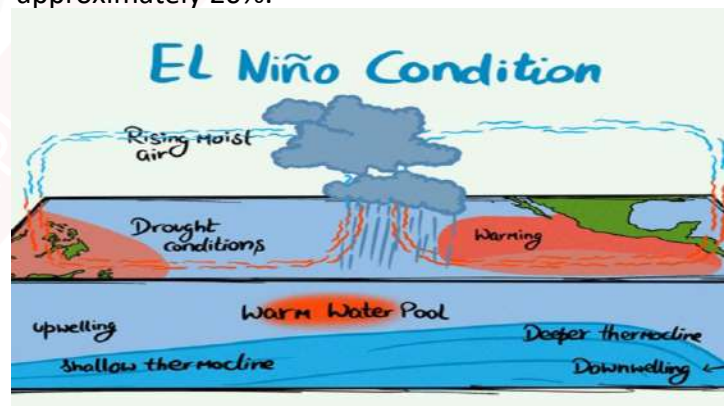
| Feature        | Crustal Earthquakes     | Continental Mantle Earthquakes |
|----------------|-------------------------|--------------------------------|
| Depth          | 10–29 km                | >80 km                         |
| Location       | Within brittle crust    | Deep mantle beneath continents |
| Surface Impact | Strong shaking possible | Minimal surface impact         |
| Frequency      | Common                  | Rare                           |

### Impacts and Significance:

- Limited Surface Damage:** Due to great depth, these quakes typically do not cause significant surface shaking or destruction.
- Insight into Earth's Interior:** Provide new data on mantle structure and stress distribution beneath continents.
- Understanding Mountain Building:** Regions like the Himalayas show how deep tectonic processes contribute to orogeny (mountain formation).

## EL NINO

Researchers at Duke University have identified that [ocean salinity](#) can amplify the intensity of El Niño by approximately 20%.



- El Niño (meaning “Little Boy” in Spanish) is a recurring climate pattern characterized by the **unusual warming of surface waters** in the central and eastern tropical Pacific Ocean. It is the warm phase of the [El Niño-Southern Oscillation](#) (ENSO) cycle and typically occurs every two to seven years.

### How it Forms?

- Normal Conditions:** Strong trade winds blow from east to west along the equator, pushing warm surface water toward Asia. This allows cold, nutrient-rich water to rise (upwelling) near the [coast of South America](#).
- Weakening Winds:** During El Niño, these trade winds weaken or even reverse direction.

- **Warm Water Shift:** The warm water that was piled up in the western Pacific begins to flow back eastward toward the Americas.
- **Atmospheric Disruption:** This shift in heat alters the Pacific jet stream, disrupting global weather patterns, leading to floods in some regions and droughts in others.

#### Factors Influencing El Niño:

- **Trade Wind Strength:** The primary driver; weaker winds trigger the eastward movement of warm water.
- **Ocean-Atmosphere Coupling:** A feedback loop where warming water further weakens winds, which in turn warms the water more.
- **Thermocline Depth:** The depth of the transition layer between warm surface water and cold deep water influences how much heat is available to fuel the event.
- **Rossby and Kelvin Waves:** Large-scale internal ocean waves that transport heat across the Pacific.

#### Implications for India:

A stronger El Niño directly impacts India's food and water security:

- **Monsoon Suppression:** It pulls moisture away from South Asia, frequently resulting in below-normal rainfall.
- **Drought Risk:** There is a **60% likelihood of drought** in various regions during a strong El Niño year.
- **Agricultural Impact:** Drier conditions lead to food grain shortfalls, as seen in 2023, which saw the driest August in years and triggered food inflation.
- **Heatwaves:** El Niño often correlates with higher-than-average temperatures and prolonged heatwaves during the Indian summer.

Topic: Miscellaneous

## THE SANGTAM COMMUNITY

The apex body of Nagaland's Sangtam community has passed a resolution to protect pangolins within its jurisdiction.



About The Sangtam Community:

- Sangtam Naga are one of the recognized [Naga tribes](#) of Nagaland, primarily inhabiting **Kiphire and Tuensang districts** in eastern Nagaland.
- They are part of the larger Naga ethnic group of Northeast India and follow strong customary governance traditions.

#### Origin:

- According to oral traditions, the Sangtams trace their migration through regions of present-day Myanmar before settling in present [eastern Nagaland](#).
- The term "Sangtam" is believed to have evolved from "**Sangdang**", an ancestral village name that was later recorded in British administrative reports in the late 19th century.

#### Key characteristics:

- **Strong [traditional governance system](#)** led by village councils and apex tribal bodies.
- Organized into six major clan groupings (Shuh), reflecting deep-rooted lineage structures.
- Inhabit ecologically rich landscapes marked by dense forests, shifting cultivation, and biodiversity hotspots.
- Community resolutions play a decisive role in regulating social practices and natural resource use.

## 100 YEARS OF OL CHIKI SCRIPT

- Ol Chiki is the **official writing system of the Santhali language**, a major tribal language of India.
- It is a **scientifically designed script** created to represent Santhali sounds accurately.
- Unlike borrowed scripts (Roman, Bengali, Odia, Devanagari), Ol Chiki was **built specifically for Santhali phonetics**.

#### Origin:

- Developed in **1925** by Pandit Raghunath Murmu.
- Created to give Santhali speakers a **distinct written identity**.
- **First major literary work:** *High Serena* (1936).
- Murmu is revered as *Guru Gomke* (Great Teacher) among Santhals.

#### Region and Linguistic Family:

##### Language: Santhali

- **Language family:** Austroasiatic (Munda branch)
- **States where widely spoken:** Jharkhand, Odisha, West Bengal, Assam, and Bihar

#### Constitutional Milestone:

- Santhali was included in the [Eighth Schedule of the Constitution](#) in 2003 via the 92nd Constitutional Amendment Act.

- In December 2025, the [Constitution of India](#) was translated into Santhali using **OI Chiki**, expanding democratic access.

## 2025 UNESCO ASIA-PACIFIC AWARDS FOR CULTURAL HERITAGE CONSERVATION

The Our Lady of Grace Cathedral, Vasai (Maharashtra) has won the Award of Merit at the 2025 [UNESCO](#) Asia-Pacific Awards for Cultural Heritage Conservation.

### [About 2025 UNESCO Asia-Pacific Awards for Cultural Heritage Conservation:](#)

- A prestigious UNESCO initiative that recognises **outstanding heritage conservation projects** across the [Asia-Pacific region](#).
- It honours restoration efforts that combine technical excellence with cultural authenticity and community participation.

### [Established in:](#)

- **2000**, by UNESCO to promote best practices in cultural heritage preservation across Asia and the Pacific.

### [Aim:](#)

- To encourage high-quality conservation of historic sites while maintaining cultural integrity and sustainability.
- To promote community participation and [adaptive reuse](#) of heritage structures for long-term preservation.

### [Key Features:](#)

- Recognises projects based on understanding of place, technical achievement, sustainability, and impact.
- Open to private sector and public-private partnership conservation initiatives.
- Covers heritage types such as historic buildings, towns, archaeological sites, cultural landscapes and [vernacular architecture](#).
- Projects must generally be completed within the last ten years to qualify.
- Has honoured 300+ projects across 27 countries, shaping regional conservation standards.

### [About Vasai Cathedral \(Our Lady of Grace Cathedral\):](#)

- A **16th-century Catholic cathedral** located in **Papdy village, Vasai (Maharashtra)**, representing early Portuguese colonial architecture in India.

### [History:](#)

- Built around **475 years ago** during Portuguese rule on India's western coast.
- Constructed using stone and mud mortar, without modern cement.

- Recently restored (2023–24) through a [community-funded conservation initiative](#) costing about ₹4.5 crore.

### [Key Features:](#)

- Reflects **Portuguese ecclesiastical architecture** with stone masonry and traditional craftsmanship.
- Interior restored using hand-carved liturgical elements and authentic materials.
- Includes restored roof, façade, corridors, colonnade and bell tower.
- Serves as a **living place of worship**, preserving both tangible and intangible heritage.



# FACTS FOR PRELIMS

## GS-2

Topic: Constitutional, Statutory, Quasi-Judicial, Executive Bodies

### FORM 7 CONTROVERSY

Form 7 has become controversial during the ongoing [Special Intensive Revision](#) (SIR) of electoral rolls after allegations of bulk, fraudulent deletion requests targeting eligible voters.

#### What it is?

- **Form 7** is a statutory form used to object to the inclusion of a name (one's own or another person's) in the electoral roll on specified grounds such as death, duplication, shifting of residence, ineligibility by age or citizenship, or misrepresentation.

#### Legal basis:

- Governed by the **Election Commission of India**
- Prescribed under [Registration of Electors Rules, 1960](#), framed under the **Representation of the People Act, 1950**
- As per **Section 13(2)**, objections must be filed in **Form 7** by a person whose name is already on the electoral roll
- Booth Level Agents (BLAs) are also permitted to file objections

#### Aim:

- To maintain the accuracy and integrity of electoral rolls.
- To remove ineligible, duplicate, shifted or deceased voters.
- To prevent electoral fraud and ensure free and fair elections.

#### How it works?

- Any **registered elector of the constituency** (including **Booth Level Agents**) can file Form 7.
- Objection can be raised against another voter or for **self-deletion**.
- On receipt, the **Booth Level Officer (BLO)** conducts **physical verification** (multiple visits if required).
- The concerned voter is issued a **notice and hearing** by the [Electoral Registration Officer \(ERO\)](#).
- Appeals against ERO's decision lie with the **District Magistrate** within **15 days**.
- Filing a **false declaration** is punishable under **Section 32 of the RP Act, 1950**.

#### Key features:

- **Expanded scope (2022 amendment):** Any voter in a constituency (not just same booth) can object
- **Mandatory verification:** Especially if an applicant files **more than five objections**
- **Grounds-based deletion:** Death, absent/shifted, duplicate entry, underage, non-citizenship
- **Due process safeguards:** Physical verification, notices, hearings, and appeal mechanism
- **Legal deterrence:** False claims attract **imprisonment up to one year or fine or both**

Topic: Federal Structure

### TRIPARTITE AGREEMENT FOR CREATION OF THE FRONTIER NAGALAND TERRITORIAL AUTHORITY (FNTA)

A tripartite agreement was signed between the Centre, the Nagaland government, and the [Eastern Nagaland People's Organisation](#) (ENPO) to create the Frontier Nagaland Territorial Authority (FNTA), granting enhanced autonomy to six eastern districts of Nagaland.

#### What it is?

- The agreement provides for the establishment of the Frontier Nagaland Territorial Authority (FNTA) — an autonomous territorial governance structure for six districts of eastern Nagaland, with substantial devolution of administrative and developmental powers while remaining within the [state of Nagaland](#).

#### Parties involved:

- Government of India
- Government of Nagaland
- Eastern Nagaland People's Organisation (ENPO) — apex body representing eight recognised Naga tribes.

**Districts covered:** Tuensang, Mon, Kiphire, Longleng, Noklak, and Shamator

#### Aim of the agreement:

- To address long-standing political, economic, and [developmental grievances](#) of Eastern Nagaland.
- To ensure equitable development, local decision-making, and financial autonomy.
- To strengthen peace and stability in the North-East region.

#### Key features:

- **Creation of FNTA:** A new territorial authority with administrative autonomy for six districts.

- **Devolution of powers:** Transfer of authority over **46 subjects** to FNTA.
- **Financial autonomy:**
  - Development outlay to be shared **proportionate to population and area**
  - **Fixed annual allocation** from the Centre.
  - **Initial establishment expenditure** to be borne by the Union Ministry of Home Affairs.
- **Administrative structure:**
  - FNTA to have a mini-Secretariat.
  - Headed by an Additional Chief Secretary / Principal Secretary–level officer.
- **Constitutional safeguard:** The agreement **does not dilute Article 371(A)** of the Constitution, which protects Naga customary laws, land, and resources.
- **Democratic resolution:** Outcome of prolonged dialogue, negotiations, and confidence-building since 2021–22.

Topic: Constitutional, Statutory, Quasi-Judicial, Executive Bodies

## REMOVAL OF THE CHIEF ELECTION COMMISSIONER (CEC)

- The Chief Election Commissioner (CEC) is the head of the Election Commission of India (ECI), a constitutional body entrusted with the superintendence, direction, and control of elections to Parliament, [State Legislatures](#), and the offices of the President and Vice-President.

### Constitutional article associated:

- **Article 324 of the Constitution of India**
  - Establishes the Election Commission of India
  - Provides for its composition, powers, and independence

### Appointment of the CEC:

- Appointed by the **President of India**.
- As per the [Chief Election Commissioner and Other Election Commissioners \(Appointment, Conditions of Service and Term of Office\) Act, 2023](#):
  - Selection is based on the recommendation of a **three-member committee** consisting of:
    1. Prime Minister
    2. Leader of Opposition in Lok Sabha
    3. A Union Cabinet Minister nominat-

ed by the PM

- **Tenure: 6 years** or till the age of **65 years**, whichever is earlier.

### Removal process of the CEC:

- **Article 324(5)** provides that the CEC can be removed **in like manner and on like grounds as a Judge of the [Supreme Court](#)**.
- This links the process to **Article 124(4)** (removal of SC judges)

### Grounds for removal:

- Proved misbehaviour (e.g., abuse of office, corruption, failure to discharge constitutional duties)
- Incapacity (physical or mental inability to perform duties)

### Procedure:

- **Notice of motion introduced:** A formal written notice alleging proved misbehaviour or incapacity of the CEC is introduced in either [House of Parliament](#) to initiate proceedings.
- **Support by requisite MPs:** The motion must be backed by a minimum number of Members of Parliament, ensuring that removal is not triggered by frivolous or partisan allegations.
- **Inquiry by a committee:** An independent inquiry committee is constituted to investigate the charges, examine evidence, and establish whether [misbehaviour or incapacity](#) is proved.
- **Special majority in both Houses:** The motion must secure a constitutional special majority—a majority of total membership and two-thirds of members present and voting—in both Lok Sabha and [Rajya Sabha](#).
- **Presidential order of removal:** Once Parliament adopts the motion, the President formally orders removal, acting without discretion and in accordance with Parliament's decision.

## NATIONAL COMMISSION FOR SCHEDULED TRIBES (NCST)

- The **National Commission for Scheduled Tribes (NCST)** is a **constitutional body** established under [Article 338A](#) of the **Constitution of India** to protect, monitor and promote the rights and welfare of Scheduled Tribes (STs).
- It acts as a watchdog institution ensuring implementation of constitutional safeguards and addressing grievances related to tribal communities.

### Established in:

- Established in **2004** following the [Constitution \(89th](#)

Amendment) Act, 2003.

- It separated tribal affairs from the earlier combined commission for Scheduled Castes and Scheduled Tribes to ensure focused attention on tribal issues.

**History:**

- 1978:** Government set up a multi-member Commission for SCs & STs.
- 1992:** Constitutional status granted through the **65th Constitutional Amendment**, forming the National Commission for SCs & STs.
- 2003–04: 89th Constitutional Amendment** bifurcated the body into:
  - National Commission for Scheduled Castes (NCSC)
  - National Commission for Scheduled Tribes (NCST)

**Members:**

- The NCST consists of:
  - Chairperson
  - Vice-Chairperson
  - Three Members
- All are appointed by the President of India and generally include persons with expertise in tribal administration, social justice and policy.

**Key Functions:**

- Monitoring safeguards:** Investigates and monitors constitutional and legal safeguards provided to Scheduled Tribes and evaluates their effectiveness.
- Inquiry into complaints:** Examines complaints regarding deprivation of rights, land alienation, atrocities, or denial of benefits meant for tribal communities.
- Advisory role in development planning:** Participates in policy formulation and advises governments on socio-economic development programmes for STs.
- Reporting to the President:** Submits annual and special reports to the President regarding implementation of safeguards and policy recommendations.
- Civil court powers:** While investigating cases, the Commission has powers similar to a civil court such as summoning individuals, calling documents, and recording evidence.
- Policy consultation:** Union and State Governments are expected to consult the Commission on major policy matters affecting Scheduled Tribes.

**75TH ANNIVERSARY OF EMPLOYEES' STATE INSURANCE CORPORATION (ESIC)**

- ESIC is a statutory social security body under the **Ministry of Labour & Employment**, Government of India.
- It manages the ESI Scheme, which provides comprehensive medical care and financial protection to employees against sickness, maternity, disablement, and death due to employment injury.

**Established In:**

- Act:** The Employees' State Insurance Act was promulgated in **1948**.
- Inauguration:** The scheme was officially launched on **February 24, 1952** (celebrated annually as ESIC Foundation Day).

**Key Functions:**

- Medical Benefit:** Provides full medical care (from primary to tertiary) to insured persons and their families.
- Sickness & Maternity Benefit:** Cash compensation for loss of wages during periods of certified sickness or pregnancy.
- Disablement Benefit:** Monthly pension for permanent disability arising out of employment injury.
- Dependents' Benefit:** Financial support to the dependents of an insured person who dies due to employment-related injury or occupational disease.
- Preventive Care:** Newly introduced annual health check-ups for workers aged **40 years and above** under the new Labour Codes.

Topic: Parliament and State Legislatures**MOTION OF THANKS TO PRESIDENT'S ADDRESS**

- The **Motion of Thanks** is a **formal resolution moved in both Houses of Parliament** to express gratitude to the President for the **Special Address** delivered to a joint sitting of Parliament at the beginning of each year and after a general election.
- It provides Parliament an opportunity to **debate, critique, and assess the policies and programmes of the Government** outlined in the Address.

**Origin and background:**

- The practice flows from the **Westminster parliamentary tradition**, adapted to India's



constitutional framework.

- The President's Address itself is the **statement of the Government's policy**, drafted by the Council of Ministers and approved by the Cabinet.
- Discussion on the Address takes place **only through the Motion of Thanks**, not directly on the Address.

### Constitutional provisions:

- **Article 86(1)**: Empowers the President to address either House or both Houses of Parliament assembled together.
- **Article 87(1)**: Mandates a **Special Address** by the President
  - At the commencement of the **first session after a general election**, and
  - At the commencement of the **first session of every year**.
- **Article 87(2)**: Requires Parliament to make rules for **discussion on matters referred to in the President's Address**, which is done through the Motion of Thanks.

### Key features of the Motion of Thanks:

- **Moved and seconded** by members selected by the Government (through the Ministry of Parliamentary Affairs).
- **Wide scope of discussion**: Members can raise **any national or international issue**, including matters not mentioned in the Address.
- **Amendments allowed**:
  - Opposition members may move amendments expressing regret that the Address **omitted or inadequately covered certain issues**.
- **Prime Minister's reply**:
  - The debate concludes with a **reply by the Prime Minister**, after which amendments are voted upon.
- **Voting**:
  - The Motion is put to vote; if amendments are adopted, the Motion is passed in **amended form**.
- The Motion of Thanks to the President's Address has been adopted with amendments by the Rajya Sabha **only five times** in India's parliamentary history — **1980, 1989, 2001, 2015 and 2016**.
  - In contrast, no amendment to the Motion of Thanks has ever been carried in the Lok Sabha.

## PRIVILEGE NOTICE

- A **Privilege Notice** is a formal complaint raised by a **Member of Parliament** (MP) alleging that a

**breach of privilege or contempt of the House** has occurred.

- Parliamentary privilege refers to certain **special rights and immunities** enjoyed by Parliament and its members to ensure they can perform their duties **without obstruction or intimidation**

If these rights are violated, it may amount to a **breach of privilege** and can be punished by the House.

### Articles Associated:

- **Article 105** – Powers, privileges and immunities of Parliament and its Members (for Parliament).
- **Article 194** – Similar provisions for State Legislatures.
- **Article 122** – Courts cannot question the validity of parliamentary proceedings on grounds of procedural irregularity.

Article 105(2) provides immunity to MPs for anything said or any vote given in Parliament.

### When Can a Privilege Notice Be Moved?

A Privilege Notice can be moved when:

- A member is **obstructed** from performing parliamentary duties.
- There is **misleading of the House** by providing false information.
- There is **defamation or attack on the dignity of Parliament** or its members.
- There is **publication of distorted or expunged proceedings**.
- There is **contempt of the House**, i.e., any act that lowers its authority.

The Speaker (Lok Sabha) or Chairman (Rajya Sabha) decides whether the matter should be admitted.

### Powers of Parliament to Punish:

Each House has the power to punish for breach of privilege or contempt.

### Punishments may include:

- Admonition (warning)
- Reprimand
- Imprisonment
- Suspension (for members)
- Expulsion (for members)

These powers are considered essential to protect the authority and dignity of Parliament.

### Procedure of a Privilege Notice:

- A member submits a **written notice** to the Speaker/Chairman.

- The [Presiding Officer](#) decides whether it is admissible.
- If admitted, it may be:
  - Taken up in the House directly, or
  - Referred to the **Committee of Privileges** for investigation.
- The Committee examines evidence and submits a report.
- The House debates and decides the punishment, if any.

## THE PRIVILEGES COMMITTEE

Privileges Committee and the [Ethics Committee](#), have not been constituted in the Lok Sabha (the lower house of India's Parliament) nearly two years.

### About The Privileges Committee:

- The Privileges Committee is a specialized standing committee of the legislature (Parliament or State Assemblies) that acts as a [quasi-judicial body](#).
- It is tasked with safeguarding the privileges—special rights and immunities—of the House and its members to ensure they can function without outside interference or fear.

### Origin:

- The concept is rooted in [British Parliamentary conventions](#). Historically, these privileges were developed in medieval England to protect the House of Commons from the absolute power of the Monarch.

### Articles Associated:

- **Article 105:** Defines the powers, privileges, and immunities of the **Parliament** (Lok Sabha and Rajya Sabha) and its members.
- **Article 194:** Defines the same for [State Legislatures](#) (Assemblies and Councils) and their members.

### Aim:

To investigate any action that casts reflections, insults, or obstructs the House, its committees, or its members, thereby protecting the **dignity and authority** of the legislative institution.

### Members:

- **Lok Sabha:** 15 members nominated by the Speaker.
- **Rajya Sabha:** 10 members nominated by the Chairman.
- **State Legislatures:** Typically consists of 9 to 15 members (e.g., the Maharashtra Legislative Council committee currently has 9 members).

### Key Functions:

- **Examination:** Investigates every question of breach of privilege referred to it by the House or the [Presiding Officer](#).
- **Evidence Collection:** Has the power to **summon individuals** (both members and outsiders), record statements, and demand relevant documents.
- **Determination:** Evaluates the facts to decide if a [breach of privilege](#) or contempt has occurred.
- **Recommendation:** Submits a report to the House recommending a specific course of action, which may include:
  - **Admonition or Reprimand:** A formal public scolding.
  - **Imprisonment:** For the duration of the House session (rare).
  - **Suspension/Expulsion:** If the offender is a [member of the House](#).
  - **Unconditional Apology:** Often, if the accused offers an apology, the committee recommends dropping the matter.

## NO-CONFIDENCE MOTION AGAINST THE LOK SABHA SPEAKER

- A no-confidence motion against the Speaker is a **resolution seeking removal of the Speaker or Deputy Speaker of the Lok Sabha** before the expiry of their term. It applies only to the Lok Sabha and not to the Rajya Sabha.
- The constitutional basis for removal is provided under **Article 94(c)** of the Constitution of India, which states that the Speaker or Deputy Speaker may be removed by a resolution of the House passed by a majority of all the then members.

### Constitutional Provisions Involved:

- Article 94 – Vacation, Resignation and Removal
- A Speaker/Deputy Speaker shall vacate office if:
  - a) They cease to be a [Member of the House](#)
  - b) They resign in writing to the Deputy Speaker/Speaker
  - c) They are removed by a resolution passed by a majority of all the then members
- The procedure is further governed by **Rules 200–203 of the Rules of Procedure and Conduct of Business in Lok Sabha**.

### Procedure for Removal of the Speaker:

- **Notice of Resolution:**
  - A written notice must be submitted to the

### Secretary-General of Lok Sabha.

- At least **14 days' notice** is mandatory before the motion can be taken up.
- **Inclusion in List of Business:** After notice, the motion for leave to move the resolution is listed for a day fixed after 14 days.
- **Support Requirement:**
  - When the motion is taken up, **at least 50 Members must rise in support.**
  - If fewer than 50 rise, the motion lapses.
- **Discussion:**
  - If admitted, the resolution must be discussed within **10 days.**
  - Debate is strictly confined to the charges specified in the motion.
- **Voting Requirement:** Removal requires a **majority of all the then members of the House** (absolute majority, not just those present and voting).

### **Role of the Speaker During the Motion:**

- The Speaker can **participate in proceedings and speak in defence.**
- They can vote in the first instance but **cannot exercise a casting vote** in case of a tie.
- Even after dissolution of Lok Sabha, the Speaker continues in office until the first sitting of the new House.

### **Has This Happened Before?**

- Yes, motions were moved against:
  - G.V. Mavalankar (1954)
  - Hukam Singh (1966)
  - Balram Jakhar (1987)
- None of these resulted in removal.

### **Implications:**

- **Constitutional Accountability:** Ensures that the presiding officer remains answerable to the House.
- **Political Sensitivity:** Reflects heightened tensions between ruling and opposition benches.
- **Institutional Stability:** Frequent or politically motivated motions may impact parliamentary decorum.

## **UNION CABINET APPROVES RENAMING KERALA AS KERALAM**

- The Union Cabinet has approved the proposal to alter the official name of the State of **Kerala** to **Keralam**.

- The change seeks to replace the English-adopted constitutional term “Kerala” with the traditional Malayalam usage “Keralam.”
- After Cabinet approval, the [Kerala \(Alteration of Name\) Bill, 2026](#) will follow the constitutional process before becoming law.

### **Articles associated:**

- **Article 3 of the Constitution:** Empowers Parliament to form new states or alter areas, boundaries, or names of existing states.
- **Proviso to Article 3:**
  - A Bill altering a state's name can be introduced only on the **recommendation of the President.**
  - The President must refer the Bill to the concerned **State Legislature** to seek its views.
- **First Schedule of the Constitution:** Contains the list and names of states and Union Territories; amendment required here for renaming.

### **Procedure for renaming a state:**

- **State Legislature Resolution:** Kerala Assembly passed resolutions requesting the name change.
- **Examination by Union Government:** Ministry of Home Affairs scrutinizes the proposal and consults relevant ministries/agencies.
- **Union Cabinet Approval:** Cabinet clears the proposal for legislative action.
- **President's Reference:** President refers the Bill to the State Legislature for its opinion (Article 3 proviso).
- **Parliamentary Approval:** Bill introduced in Parliament after Presidential recommendation and passed by both Houses.
- **Notification & Amendment:** [First Schedule amended](#); new name comes into legal effect.

### **Reason for the change:**

- The state is called “**Keralam**” in Malayalam, while the Constitution records it as “Kerala.”
- The demand reflects linguistic identity and the legacy of the **Aikya Kerala movement**, which sought unification of Malayalam-speaking regions.
- The Assembly argued that states formed on [linguistic lines](#) (1956) should reflect native linguistic nomenclature.

## **SPEAKER OF THE KNESSET MEDAL**

The [Israeli Parliament](#) (Knesset) conferred the “Speaker of the Knesset Medal” on Prime Minister Narendra Modi, recognising his role in strengthening India–Israel strategic ties.



- The **Speaker of the Knesset Medal** is the highest honour awarded by the Israeli Parliament (Knesset).
- It is a special parliamentary distinction instituted to recognise individuals for exceptional contributions to Israel and the Jewish people.

#### Aim:

- To acknowledge global leaders and personalities who significantly strengthen Israel's diplomatic, strategic or cultural partnerships.
- To symbolise parliamentary-level recognition beyond executive diplomacy.

#### Key Features:

- Considered the **highest [parliamentary honour](#)** of Israel.
- Awarded for **strategic, political, technological, cultural or security cooperation** contributions.
- Reflects recognition by the **legislative institution**, not merely the government.
- Recently instituted as a formal medal of honour by the Knesset.

#### About Knesset:

##### What it is?

- The Knesset is the unicameral national legislature of Israel, functioning as the supreme law-making body of the country.
- It represents the sovereign authority of the Israeli state and performs legislative, supervisory and constitutional functions.

#### Houses:

- Unicameral Legislature → Israel has only one house, called the Knesset.
- **Electoral System:** Members are elected through proportional representation based on party lists, making [coalition governments](#) common.
- **Term:** Normal tenure is 4 years, though early elections can be called.

#### Topic: Government Bills and Acts

### OFFICIAL SECRETS ACT, 1923

- The Official Secrets Act, 1923 is a colonial-era law enacted to consolidate and amend laws relating to protection of official information, particularly concerning national security and state interests.

#### Aim:

- To safeguard the sovereignty, integrity, and security of the State.
- To prevent espionage, [unauthorized disclosure](#), and

wrongful communication of classified information.

#### Who are covered?

- The Official Secrets Act, 1923 applies to **all persons**, including government servants, defence personnel, private individuals, and companies, and extends to Indian citizens even outside India.
- It also covers those holding or having held office under Government and anyone in possession of classified or [sensitive official information](#).

#### Key Features:

- **Penalty for Spying (Section 3):** Criminalizes approaching prohibited places, making sketches, or collecting information to aid an enemy; punishment may extend up to 14 years in sensitive defence cases.
- **Communication with Foreign Agents (Section 4):** Contact or association with foreign agents can be used as evidence in prosecution for offences against State security.
- **Wrongful Communication of Information (Section 5):** Punishes unauthorized sharing, retention, or misuse of official codes, documents, or information affecting sovereignty or security.
- **Protection of Prohibited Places:** Defines "prohibited places" including defence establishments, military infrastructure, and other notified sensitive installations.
- **Search and Seizure Powers (Section 11):** Empowers magistrates and, in emergencies, senior police officers to issue search warrants in suspected offences.
- **Restriction on Trial (Section 13):** Courts can take cognizance only upon complaint authorized by the appropriate Government, ensuring executive oversight.
- **Exclusion of Public Proceedings (Section 14):** Courts may exclude the public during trial if publication of evidence may [prejudice State security](#).

#### Topic: Government Schemes, Policies & Interventions

### SABHASAAR INITIATIVE

Over 1.11 lakh Gram Panchayats have adopted this AI-enabled tool for automated [Gram Sabha](#) meeting summarisation as of January 2026.

- SabhaSaar is an **AI-powered voice-to-text and meeting summarisation platform** designed

to automatically generate structured Minutes of Meetings (MoM) from Gram Sabha and [Panchayat meeting recordings](#).

**Launched in:** 14 August 2025

#### Organisations involved:

- Ministry of Panchayati Raj – Nodal implementing ministry
- [IndiaAI Mission](#) under Ministry of Electronics and Information Technology – AI & cloud infrastructure

#### Aim:

- To strengthen participatory democracy, transparency, and efficiency in local self-government by digitising and standardising Gram Sabha documentation.

#### Key features:

- AI & NLP-based transcription:** Converts audio/video discussions into structured minutes, capturing decisions, action points, and deliberations.
- Multilingual support:** Integrated with Bhashini; currently supports **13 Indian languages**, with plans for expansion.
- Secure data governance:** Operates entirely within government infrastructure, compliant with the [Digital Personal Data Protection \(DPDP\) Act, 2025](#).
- Governance analytics:** Tracks meeting type, frequency, attendance, resolutions, and follow-up actions.

### INDIA'S FIRST COOPERATIVE-BASED TAXI SERVICE – BHARAT TAXI

- Bharat Taxi is India's first cooperative-sector, driver-owned ride-hailing platform, designed as an indigenous alternative to [aggregator-based models](#) like Ola and Uber, with drivers (Sarathis) as owners and stakeholders.

#### Organisation(s) involved:

- Ministry of Cooperation – policy guidance
- Sahkar Taxi Cooperative Limited** – operating entity
- Registered under the [Multi-State Cooperative Societies Act, 2002](#)
  - Established on June 6, 2025

**Aim:** To promote inclusive, citizen-centric mobility by empowering drivers through cooperative ownership, ensuring fair income distribution, social security, and freedom from exploitative commission-based platforms.

#### Key features:

- Driver-owned model:** Guided by the principle Sarathi hi Malik—drivers are shareholders and decision-makers.

- Zero-commission & surge-free pricing:** Entire fare initially goes to drivers; profits are redistributed within the cooperative.
- Strong social security:** ₹5 lakh accident insurance, ₹5 lakh family health insurance, retirement savings, and emergency support.
- Women empowerment initiatives:** Sarathi Didi and [Bike Didi](#), with 150+ women drivers onboard.
- Digital & safety integration:** Multilingual app, transparent fares, real-time tracking, verified ride data, and police integration.
- Non-exclusive model:** Drivers can work simultaneously on other platforms.

### CENTRAL SECTOR SCHEME OF FORMATION AND PROMOTION OF 10,000 FARMER PRODUCER ORGANIZATIONS (FPOS)

- A central sector scheme to form and nurture 10,000 new FPOs across India, enabling small and marginal farmers to collectivise production, processing, and marketing for better incomes and market power.

**Launched in:** 29 February 2020

#### Implementing Agencies (IAs):

- Small Farmers' Agribusiness Consortium (SFAC)
- National Bank for Agriculture and Rural Development ([NABARD](#))
- National Cooperative Development Corporation (NCDC)
- National Agricultural Cooperative Market Federation of India (NAFED)

**Aim:** To build a sustainable, income-oriented farming ecosystem by strengthening farmer collectives, improving access to inputs, credit, technology, value chains, and markets, and enhancing farmers' bargaining power.

#### Key features:

- Cluster & commodity approach:** Produce-cluster-based formation aligned with [One District One Product](#) (ODOP).
- Market linkage:** NAFED-led forward linkages and value-chain integration.
- Capacity building:** Structured training via BIRD (Luc-know) and LINAC (Gurugram).
- Inclusion focus:** Strong participation of women farmers and coverage of aspirational districts.

## AYUSHMAN SAHAKAR SCHEME

- Ayushman Sahakar is a dedicated scheme of the National Cooperative Development Corporation ([NCDC](#)) to provide financial assistance to cooperative societies for establishing and expanding healthcare infrastructure across India.

### Launched in:

- 2020** (Notified by NCDC in alignment with National Health Policy 2017)

### Ministry:

- Implemented by **NCDC**
- Under the administrative control of the Ministry of Cooperation

### Aim:

- To promote **affordable, holistic, and community-owned healthcare** through cooperative societies.
- To strengthen AYUSH and [digital health participation](#) in line with India's public health objectives.

### Key Features:

- Eligible Institutions:** Any Cooperative Society registered under State or Multi-State Cooperative Societies Act with healthcare provisions in its bye-laws is eligible.
- Comprehensive Coverage:** Supports infrastructure, modernization, AYUSH services, digital health, telemedicine, insurance, and working capital for diverse healthcare facilities.
- Flexible Financial Assistance:** Provides term/investment loans up to 8 years (with 1–2 year moratorium) based on actual project requirements.
- Women Incentive:** Offers 1% interest rebate to women-majority cooperatives, subject to timely repayment compliance.
- Structured Funding & Security:** Funding through State or direct NCDC route (up to 90% loan) with defined collateral/guarantee mechanisms.

## PM RAHAT SCHEME

[PM RAHAT \(Road Accident Victim Hospitalization and Assured Treatment\)](#) is a **national cashless emergency treatment scheme** providing financial coverage up to **₹1.5 lakh per road accident victim** for 7 days from the date of accident, with a focus on timely Golden Hour intervention.

### Organisations involved:

- Ministry of Road Transport and Highways (MoRTH):** Policy oversight; integration via Electron-

ic Detailed Accident Report (eDAR) platform.

- National Health Authority (NHA):** Claims processing through Transaction Management System (TMS 2.0).

### Aim:

- To ensure no life is lost due to lack of immediate medical assistance after road accidents.
- To strengthen India's structured [emergency response ecosystem](#).
- To provide financial certainty to hospitals, encouraging uninterrupted treatment.

### Key features:

- Cashless Coverage:** Up to ₹1.5 lakh per victim for 7 days from accident date.
- Golden Hour Focus:** Integration with **112 helpline** for rapid hospital access.
- Stabilization Window:** 24 hours (non-life-threatening) and 48 hours (life-threatening cases).
- Digital Integration:** Seamless linkage between **eDAR (MoRTH)** and **TMS 2.0 (NHA)** for end-to-end claim management.
- Police Authentication:** Mandatory confirmation within 24–48 hours to ensure accountability.
- Funding Mechanism:** Through [Motor Vehicle Accident Fund \(MVAFF\)](#), insured cases via insurance contributions, uninsured/Hit & Run via budgetary support.
- Time-bound Payment:** Claims approved by State Health Authorities paid within 10 days.
- Grievance Redressal:** District-level oversight by Road Safety Committee chaired by District Collector/DM.

## STARTUP INDIA FUND OF FUNDS 2.0

- Startup India FoF 2.0 is a **₹10,000 crore government-backed fund** designed to mobilise long-term domestic capital by investing in [Alternative Investment Funds \(AIFs\)](#), which in turn invest in startups.

### Established in:

- Approved in 2026** by the Union Cabinet.
- Launched under the broader **Startup India** initiative.
- Builds on the earlier [Fund of Funds for Startups \(FFS 1.0\)](#) launched in 2016.

### Aim:

- To accelerate the next phase of India's startup growth.
- To strengthen the domestic venture capital ecosys-



tem.

- To support innovation-led, technology-driven entrepreneurship.
- To reduce early-stage funding gaps and high-risk capital constraints.

#### Key Features:

- **₹10,000 crore Corpus:** Dedicated government-backed capital to catalyse venture funding.
- **Targeted Segmented Funding:** Focus on deep-tech and innovative manufacturing sectors.
- **Support to Early-Growth Startups:** Reduces early-stage failures due to capital shortages.
- **Pan-India Investment Reach:** Encourages funding beyond major metro cities.
- **High-Risk Capital Gap Bridging:** Channels funds to priority and strategic sectors.
- **Strengthening Domestic VC Base:** Supports especially smaller AIFs to deepen local capital markets.
- **FoF Model Structure:** Invests in [SEBI-registered AIFs](#) rather than directly in startups.

### URBAN CHALLENGE FUND (UCF)

- The Urban Challenge Fund (UCF) is a Centrally Sponsored Scheme designed to support transformative and bankable urban [infrastructure projects](#) through a competitive “challenge mode” framework.



**Urban makeover**

The Urban Challenge Fund seeks to leverage private participation for a total investment of **₹4 lakh crore** in the next five years

- The fund will be operational from 2025-26 to 2030-31, with an extendable implementation period up to 2033-34
- A dedicated **₹5,000 crore** corpus to enhance the creditworthiness of **4,223** cities
- All urban local bodies in hilly States and in northeastern States, and those with a population below **1 lakh** will be eligible for support under the Credit Repayment Guarantee Scheme

#### Organisation involved:

- Ministry of Housing and Urban Affairs (MoHUA) – Nodal ministry for implementation and monitoring.

#### Aim:

- To build resilient, inclusive, productive and climate-responsive cities.
- To mobilise market finance and private participation in [urban infrastructure](#).
- To position cities as growth hubs driving India's next phase of economic development.

#### Salient Features of the Urban Challenge Fund:

- **₹1 Lakh Crore Central Assistance:** Central

share covers 25% of project cost to catalyse investments.

- **Market-Linked Financing Model:** Minimum 50% project funding must be mobilised from market sources.
- **Challenge-Based Selection:** Competitive framework prioritising high-impact, reform-oriented proposals.
- **Reform-Linked Funding:** Fund release tied to governance, financial, planning, and operational reforms.
- **₹5,000 Crore Credit Enhancement Corpus:** Strengthens creditworthiness of 4,223 cities, especially Tier-2 & Tier-3.
- **Credit Repayment Guarantee Scheme:** Provides up to ₹7 crore guarantee for first-time loans to smaller ULBs.
- **Operational Period:** FY 2025–26 to FY 2030–31 (extendable till 2033–34).
- **Project Verticals:** Focus on Cities as Growth Hubs, [Creative Redevelopment](#), and Water & Sanitation.
- **Digital Monitoring:** Paperless tracking via single MoHUA digital portal.
- **Outcome-Oriented Evaluation:** Projects assessed on economic impact, climate resilience, and service equity.

### SANKALP SCHEME

- SANKALP (Skill Acquisition and Knowledge Awareness for Livelihood Promotion) is a flagship skill development programme aimed at improving the quality and effectiveness of [short-term skill training](#) in India.
- It focuses on strengthening institutional capacity, industry linkage and inclusion of marginalized groups through systemic reforms.

#### Launched in

- **Launched:** 19 January 2018
- Initially planned till March 2023, later extended.

#### Implementing Agency:

- Ministry of Skill Development and Entrepreneurship (MSDE)
- Implemented with [World Bank loan](#) assistance and support from States and industry stakeholders.

#### Aim:

- **Improve quality and scale of skill training:** To strengthen short-term skill development through better institutional frameworks and quality assurance mechanisms.
- **Promote inclusive livelihood opportunities:**

To enhance participation of marginalized and disadvantaged groups through targeted skilling interventions.

#### Key Features:

- **Financial support:** The scheme has a total outlay of ₹4,455 crore, supported by World Bank funding to strengthen India's skill development infrastructure and reforms.
- **Institutional strengthening:** Focuses on improving capacity and coordination of skill institutions at Central, State and District levels for better implementation.
- **Quality assurance:** Introduces standards and monitoring mechanisms to ensure training programmes meet industry-relevant quality benchmarks.
- **Industry linkage:** Promotes partnerships with industries so that training remains demand-driven and improves job placement opportunities.
- **Inclusion focus:** Targets participation of marginalized and disadvantaged groups to ensure equitable access to skill development programmes.
- **Performance monitoring:** Uses Results Framework and [Disbursement Linked Indicators](#) (DLIs) to track outcomes and link funding with performance.

### CENTRAL BANK DIGITAL CURRENCY (CBDC)-BASED PUBLIC DISTRIBUTION SYSTEM (PDS)

- The CBDC-based PDS is a digitally enabled ration distribution system that integrates Central Bank Digital Currency into the Public Distribution System to ensure secure and transparent food grain delivery.

#### Organisations Involved:

- Ministry of Consumer Affairs, Food and Public Distribution – Nodal ministry overseeing PDS reforms and implementation.
- Reserve Bank of India (RBI) – Issuer of the [Central Bank Digital Currency](#) (Digital Rupee).

#### Aim:

- To eliminate corruption and leakages in ration distribution.
- To operationalise the vision of "Minimum Government, Maximum Governance" in welfare delivery.
- To ensure secure, efficient, and transparent access to subsidised food grains.

#### Key Features:

- **Launched in Gandhinagar:** The system has been started in Gandhinagar, Gujarat as the first implementation.
- **Linked with Digital India:** The ration distribution process is integrated with the broader Digital India framework.
- **Use of CBDC in Transactions:** The Central Bank Digital Currency is used within the ration distribution mechanism.
- **Annapurna Machine for Distribution:** An automated machine dispenses 25 kg of food grains in 35 seconds with accuracy.
- **Ensures Transparency:** Digital processes aim to prevent corruption and safeguard beneficiaries' rights.
- **Nationwide Expansion Planned:** The system will be gradually rolled out across the country in the next 3–4 years.

### BHARAT-VISTAAR

- Bharat-VISTAAR is an **AI-driven multilingual Digital Public Infrastructure (DPI) platform** for agriculture.
- It provides farmers with personalised, real-time information through voice, chat, and mobile access.
- It functions as a **single digital gateway** for agricultural schemes, advisories, markets, and grievance redressal.

**Organisations Involved:** Ministry of Agriculture & Farmers' Welfare is implementing authority.

#### Aim:

- **Provide customised AI-based advisory to farmers:** Delivers crop, weather, and market advice tailored to individual farm profiles using AI.
- **Improve access to scheme eligibility and benefit tracking:** Enables farmers to check eligibility, application status, and benefits in one place.

#### Key Features:

- **Multilingual & Voice-Based Access**
  - **Available initially in Hindi and English:** Ensures wider early adoption among major language groups.
  - **Expansion to 11 languages in phases:** Promotes inclusivity across diverse regional farmers.
  - **Toll-free helpline 155261 with AI assistant "Bharati":** Allows farmers to receive advice

through simple voice calls.

- **Multi-Platform Accessibility:**

- **Basic phone (voice call)** – Supports non-smartphone users in rural areas.
- **Mobile application:** Offers app-based interactive advisory and scheme tracking.
- **Integration with PM-KISAN and PMFBY apps:** Links directly with major agricultural welfare platforms.

- **Scheme Integration:**

- **PM-KISAN:** Facilitates income support tracking and grievance updates.
- **PMFBY:** Enables crop insurance enrolment and claim status checks.
- **Soil Health Card:** Provides soil nutrient data for informed fertiliser use.
- **Kisan Credit Card:** Supports access to institutional agricultural credit.
- **Agriculture Infrastructure Fund:** Offers information on agri-infrastructure financing support.
- **PM Krishi Sinchayee Yojana:** Guides farmers on irrigation assistance schemes.
- **Sub-Mission on Agricultural Mechanization:** Promotes access to farm machinery subsidies.
- **Per Drop More Crop:** Encourages efficient micro-irrigation practices.
- **PM Annadata Aay Sanrakshan Abhiyan:** Provides MSP and price support details.
- **Modified Interest Subvention Scheme:** Explains interest relief on short-term crop loans.

- **Advisory Services:**

- **ICAR-based crop and livestock guidance:** Supplies research-backed farming recommendations.
- **Pest and disease alerts:** Issues timely warnings to reduce crop losses.
- **Real-time mandi prices:** Shares updated market rates to improve selling decisions.
- **Weather forecasts and agronomy advice:** Aligns farm practices with climate conditions.
- **Grievance registration and tracking:** Enables digital complaint filing and resolution monitoring.

- **24x7 Digital Agriculture Expert:**

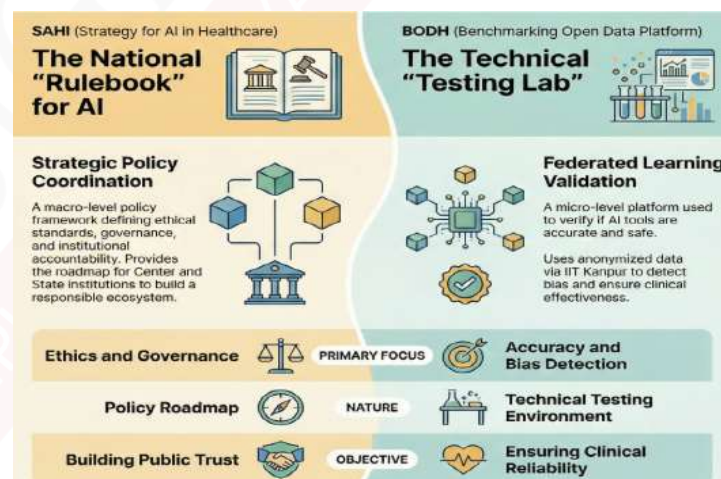
- **Round-the-clock operation:** Ensures farm-

ers can access services anytime.

- **Chat and voice interaction:** Provides conversational assistance through text or speech.

## GOVERNMENT LAUNCH SAHI AND BODH INITIATIVES

- The Government of India has introduced two major digital health frameworks:
  - SAHI (Strategy for Artificial Intelligence in Healthcare for India)
  - BODH (Benchmarking [Open Data Platform](#) for Health AI)
- Together, they aim to create a safe, transparent, accountable, and evidence-based AI ecosystem in India's healthcare system.
- **Nodal ministry:** Ministry of Health and Family Welfare (MoHFW).



### About SAHI Initiative:

#### What is SAHI?

- SAHI is a national governance framework and roadmap for the responsible adoption of [Artificial Intelligence in healthcare](#).
- It acts as a policy compass guiding ethical, transparent, and people-centric use of AI technologies in the health sector.

#### Aim of SAHI:

- **Promote responsible AI in healthcare:** Ensure ethical, transparent, and accountable deployment of AI systems across the health sector.
- **Align innovation with public health goals:** Integrate AI tools in a way that safeguards data privacy and supports national healthcare priorities.

#### Key Features of SAHI:

- **Governance Framework:** Establishes clear policy guidelines for adoption and regulation



of AI at both Union and State levels to ensure uniform standards.

- **Ethical AI Standards:** Mandates consent-based data use, [transparency in algorithms](#), and accountability mechanisms to prevent misuse or bias.
- **Interoperability Focus:** Ensures AI solutions integrate seamlessly with India's existing digital health infrastructure like [ABDM platforms](#).
- **Multi-Stakeholder Collaboration:** Facilitates cooperation among government bodies, academic institutions, startups, and industry to foster responsible innovation.
- **Policy Roadmap:** Provides a long-term strategic direction for scaling AI in healthcare while maintaining safety and public trust.

#### About BODH Initiative:

##### What is BODH?

- BODH is a national benchmarking platform designed to evaluate and validate AI solutions in healthcare before large-scale deployment.
- It has been developed by IIT Kanpur in collaboration with the National Health Authority.

##### Aim of BODH:

- **Ensure safe and reliable AI deployment:** Systematically test health AI tools for accuracy, safety, and clinical robustness before large-scale use.
- **Promote unbiased and scalable innovation:** Use real-world anonymized datasets to reduce bias and enhance generalizability across populations.

##### Key Features of BODH:

- **Open Data Benchmarking:** Evaluates AI systems using diverse, anonymized real-world health datasets to ensure broader applicability and fairness.
- **Performance Validation:** Measures accuracy, consistency, and operational reliability of AI tools under practical healthcare conditions.
- **Bias and Risk Assessment:** Detects [algorithmic bias](#) and unintended risks to prevent inequitable outcomes before deployment.
- **Clinical Relevance Testing:** Verifies that AI solutions align with medical standards and address real public health
- **Standardized Evaluation Framework:** Establishes national benchmarking norms to create uniform quality standards for health AI systems.

## THE VIBRANT VILLAGES

## PROGRAMME-II (VVP-II)

- The Vibrant Villages Programme-II (VVP-II) is a Central Sector Scheme focused on the comprehensive development of villages located in blocks abutting India's [International Land Borders](#) (ILBs) (excluding northern border areas already covered under VVP-I).

##### Launched in:

- Approved by the Central Government for implementation during **FY 2024-25 to 2025-26**, with financial support extending up to **2028-29**.
- Officially launched in **February 2026** at Nathanpur village, Cachar district, Assam.

##### Aim:

- To address infrastructure gaps, improve living conditions and create sustainable livelihood opportunities in border villages.
- To integrate border populations with mainstream development and enable them to act as "eyes and ears" for border guarding forces.

##### Key Features:

- **Central Sector Scheme** with an outlay of **₹6,839 crore** up to FY 2028-29.
- Implemented across **15 States and 2 Union Territories**.
- **Saturation-based approach:** Ensures coverage of all eligible households under existing government schemes.
- **Convergence model:** Integrates multiple flagship schemes for efficient resource use.
- Focus on **4 core infrastructure themes**:
  - All-weather road connectivity ([PMGSY-IV](#))
  - Telecom connectivity (Digital Bharat Nidhi)
  - Television connectivity (BIND scheme)
  - Electrification (RDSS)
- Identification of **1,954 strategic villages** for intensive development.
- Livelihood promotion through tourism, SHGs, [FPOs](#), skill development and financial inclusion.
- Border-specific outreach activities to build trust between communities and border security forces.

##### Topic: International Relations

## PROJECT VAULT

- Project Vault is a [public-private stockpiling](#) programme designed to purchase, store, and manage critical minerals and rare earth

elements required for strategic civilian and defence industries in the United States, similar in concept to the Strategic Petroleum Reserve.

#### Launched by:

- **US Government**, announced by **Donald Trump**
- Funded through a mix of **private capital** and the **US Export–Import Bank**

#### Aim:

- To secure uninterrupted access to critical minerals during global supply shocks.
- To reduce strategic dependence on China, which dominates mineral processing.
- To strengthen [national security](#), advanced manufacturing, and clean energy supply chains.

#### Key features:

- **Minerals covered:** Rare earths and critical minerals such as **cobalt**, **gallium**, and other strategic metals
- **Advance purchase commitments:** Companies commit upfront to buy minerals later at fixed inventory prices.
- **Stockpile access model:**
  - Firms can withdraw minerals if they replace equivalent quantities
  - Full access allowed during major supply disruptions
- **Price stabilisation mechanism:** Mandatory repurchase at the same price to reduce market volatility.
- **Private-sector execution:** Commodity traders (e.g., Mercuria, Traxys) handle sourcing and storage.
- **Industry participation:** Companies like GM, Boeing, Google, Stellantis are already onboard.

## BLUE LINE ALONG THE LEBANON–ISRAEL FRONTIER

The United Nations Interim Force in Lebanon ([UNIFIL](#)) temporarily suspended patrols along parts of the [Blue Line](#) after Israel announced the release of a chemical substance near the frontier.



#### What is the Blue Line?

- The Blue Line is a UN-identified line of withdrawal, marking Israel's pullback from southern Lebanon.
- It is not an international border, but a temporary demarcation used to monitor compliance with UN resolutions.

#### Located in:

- Runs for about **120 km** along **southern Lebanon**, separating it from **Israel and the Israeli-occupied Golan Heights**.
- Extends from the [Mediterranean coast](#) near Naqoura/Rosh HaNikra to the tri-junction area near Shebaa Farms.

#### Origin and history:

- **1978:** UN Security Council adopted **Resolutions 425 and 426** after Israel's invasion of Lebanon, creating UNIFIL.
- **2000:** Following Israel's withdrawal from southern Lebanon, the [United Nations](#) delineated the **Blue Line** to verify compliance.
- Both Israel and Lebanon agreed to **respect the line**, while maintaining that it does not prejudice final border negotiations.

#### Key features:

- **Temporary and technical line:** Designed solely to confirm Israeli withdrawal, not to settle sovereignty claims.
- **UN custodianship:** Monitored and supervised by **UNIFIL** in coordination with the Lebanese Armed Forces.
- **Sensitive zones:** Includes [disputed areas](#) such as Shebaa Farms, Ghajar village, and Kfarchouba hills.
- **Frequent violations:** Aerial overflights, cross-border fire, and construction activities are regularly flagged by the UN.
- **Security architecture:** Israel has constructed a **border barrier/fence** along much of the line to prevent infiltration.

## RAFAH BORDER CROSSING

The Rafah Border Crossing was reopened, after being largely shut for nearly two years following Israeli military control.

- Its reopening is a key component of the [US-brokered Gaza ceasefire plan](#), aimed at enabling civilian movement and humanitarian aid.

## Egypt-Gaza-Israel Border



### About Rafah Border Crossing:

- The Rafah Border Crossing is the only border crossing between the Gaza Strip and Egypt and Gaza's sole exit point that does not pass through Israel, making it a critical humanitarian and strategic gateway.

### Region located in:

- Lies on the **Gaza Strip–Egypt border**
- Connects **Rafah (Gaza)** with **Rafah (Egypt)**
- Located along the historic **Salah al-Din corridor** in southern Gaza

### History:

- Pre-2005:** Controlled by Israel following the 1967 war.
- 2005–2007:** Administered by the **Palestinian Authority**, monitored by the European Union Border Assistance Mission (EUBAM).
- Post-2007:** Largely closed after Hamas took control of Gaza and Israel imposed a blockade.
- 2024–2025:** Seized by Israel during the Gaza war; partial withdrawal under ceasefire terms.
- 2026:** Reopened in a limited manner under international monitoring and Israeli security oversight.

### Key features:

- Humanitarian lifeline:** Primary route for medical evacuations, aid supplies, and limited civilian travel.
- Unique status:** Gaza's only non-Israel-controlled international crossing in principle.
- Security-controlled access:** Movement subject to Israeli clearance and coordination with Egypt.
- Third-party monitoring:** Involvement of **EUBAM** to provide a neutral presence.
- Aid logistics hub:** Closely linked with aid routing via **Kerem Shalom** crossing.

## CHABAHAR PORT

Iran has reaffirmed its support for the Chabahar Port project and deeper cooperation with India despite uncertainty arising from **renewed U.S. sanctions** and the possible expiry of

the sanctions waiver.



### About Chabahar Port:

- Chabahar Port is **Iran's only oceanic deep-sea port**, developed as a multipurpose commercial and strategic port to facilitate regional trade, transit, and connectivity between South Asia, Central Asia, and the Middle East.

### Location:

- Situated in **southeastern Iran**, in **Sistan–Baluchestan province**
- Lies on the **Gulf of Oman**, with direct access to the **Indian Ocean**
- Around **170 km west of Gwadar port (Pakistan)**, making it geopolitically significant

### Historical background:

- First conceptualised in the **1970s** under the Shah of Iran; progress stalled after the **1979 Iranian Revolution**
- Gained importance during the **Iran–Iraq War (1980–88)** as an alternative to Persian Gulf ports
- India–Iran cooperation began in 2003**, but sanctions delayed implementation
- In **2016**, India, Iran and Afghanistan formalised cooperation to use Chabahar as a transit hub
- India took over port operations in 2018**; a **10-year contract was signed in 2024**, including a **\$250 million Line of Credit**

### Key features:

- Comprises two terminals: **Shahid Kalantari** and **Shahid Beheshti**
- Designed to handle **container, bulk, and multipurpose cargo**
- Capability to berth **large ocean-going vessels**, unlike Bandar Abbas
- Integrated with **road–rail projects** connecting Iran to Afghanistan and Central Asia
- Part of the broader **International North–South Transport Corridor (INSTC)** vision

### Significance

- Strategic for India:**



- Bypasses Pakistan to access **Afghanistan and Central Asia**
- Enhances India's role in Eurasian connectivity
- **Geopolitical leverage:** Acts as a counter-balance to **China-backed Gwadar Port** under CPEC
- **Economic importance:** Facilitates trade, humanitarian aid (e.g., wheat shipments to Afghanistan), and regional integration

## KIMBERLEY PROCESS

India has been selected to assume the Chairpersonship of the [Kimberley Process](#) (KP) from 1 January 2026, marking the third time it will lead this global initiative.

### About Kimberley Process:

- The Kimberley Process (KP) is a tripartite international initiative involving governments, the global diamond industry and civil society to prevent the trade in conflict diamonds.

**Established in:** 2003, through the Kimberley Process Certification Scheme (KPCS), pursuant to [UN Security Council](#) resolutions.

### Aim:

- To eliminate conflict diamonds—rough diamonds used by rebel groups to finance armed conflicts that undermine legitimate governments—from the global supply chain.

### Key functions:

- **Certification mechanism (KPCS):** Ensures all exports of rough diamonds carry a valid KP certificate confirming conflict-free origin.
- **Regulatory enforcement:** Member countries enforce domestic laws to control import and export of [rough diamonds](#).
- **Selective trading:** Trade permitted **only among KP-compliant participants**, preventing entry of illicit diamonds.
- **Transparency & data exchange:** Regular sharing of **statistical trade data** to monitor compliance.
- **Monitoring & review:** Working groups and peer reviews assess adherence and address loopholes.

### Significance:

- Covers 60 participants (EU counted as one), accounting for over 99% of global rough diamond trade.
- Has curtailed the conflict diamond trade to negligible levels globally, enhancing ethical sourcing.
- Strengthens consumer trust and industry credibility

in the [diamond value chain](#).

## INDIA–MALAYSIA IMPACT FRAMEWORK

- IMPACT stands for India–Malaysia Partnership for Advancing Collective Transformation, a strategic vision articulated to deepen bilateral cooperation by aligning economic growth, [digital integration](#), cultural ties, and people-centric development.

### Aim:

- To accelerate the pace and scale of bilateral cooperation.
- To deliver tangible benefits for citizens of both countries.
- To position India–Malaysia ties as a driver of Asia's collective growth.

### Key Features:

- **Economic & Technology Cooperation:** Over **100 Indian IT companies** operate in Malaysia, generating employment and [strengthening digital ecosystems](#).
- **Digital Connectivity:** Launch of **UPI in Malaysia** and collaboration through the **Malaysia–India Digital Council**, enhancing fintech and cross-border digital payments.
- **People-to-People & Diaspora Linkages:** Strong role of the **Indian-origin community** (second-largest globally), expansion of **OCI eligibility up to 6th generation**, scholarships, and [cultural institutions](#) like the **Thiruvalluvar Chair/Centre**.
- **Cultural & Civilisational Bonds:** Shared maritime heritage of the Indian Ocean, linguistic and cultural links (Tamil, Malay), and historical connections including INA legacy.
- **Strategic & Regional Outlook:** Partnership aligned with ASEAN centrality, Indo-Pacific stability, and shared commitment to inclusive growth.

## PAX SILICA INITIATIVE

India has joined the U.S.-led Pax Silica initiative, aimed at building resilient supply chains for electronics, artificial intelligence and [critical minerals](#).

### About Pax Silica initiative:

- Pax Silica is a strategic international initiative led by the United States Department of State to strengthen secure, resilient and trusted supply chains for critical minerals, semiconductors, electronics and [AI technologies](#).

**Aim:**

- To create resilient and [diversified global supply chains](#) for critical minerals, semiconductors and AI-related technologies.
- To deepen economic partnerships among like-minded countries and reduce risks from coercive or [monopolistic supply](#) dependencies.

**Participants:**

- **Signatories include:** Australia, Greece, Israel, Japan, Qatar, Republic of Korea, Singapore, UAE, United Kingdom, India (new entrant).
- **Non-signatory participants:** Canada, [European Union](#), Netherlands, OECD, Taiwan.

**Key Features:**

- **Supply chain security focus:** Promotes diversification of electronics and critical mineral supply chains to reduce excessive concentration risks.
- **AI and technology collaboration:** Encourages cooperation across AI systems, semiconductors, data infrastructure and advanced manufacturing ecosystems.
- **Critical minerals partnership:** Supports coordinated refining, processing and access to rare-earth and strategic minerals needed for future technologies.
- **Investment and infrastructure cooperation:** Promotes shared investments and incentives to strengthen trusted industrial and technology networks.
- **Trusted innovation ecosystem:** Builds collaboration among governments, industries and innovators to create secure and reliable technology stacks.
- **Fair market and security framework:** Addresses [non-market practices](#), unfair dumping and protects sensitive technologies and critical infrastructure.
- **Private sector participation:** Mobilises entrepreneurship and industry capabilities to scale innovation and strengthen economic security.
- **Strategic economic alignment:** Aims to align partner countries on long-term technology governance and resilient global economic architecture.

## THE INTERNATIONAL EMERGENCY ECONOMIC POWERS ACT (IEEPA), 1977

The US Supreme Court recently struck down [tariffs imposed](#)

by President Donald Trump under the International Emergency Economic Powers Act (IEEPA), 1977.

### About The International Emergency Economic Powers Act (IEEPA), 1977:

- The **IEEPA** is a US federal law that grants the President authority to regulate international commerce during a declared [national emergency](#) arising from external threats.
- It allows economic measures such as sanctions and restrictions on transactions to protect US national interests.

**Established in:**

- Enacted in **1977** by the US Congress.
- It replaced and limited the earlier broad powers under the **Trading with the Enemy Act (TWEA), 1917**.

**Aim:**

- To enable the US President to respond swiftly to **extraordinary external threats** affecting [national security](#), foreign policy, or the economy.
- Primarily designed as a sanctions framework rather than a trade tariff instrument.

**Key Features:**

- President can block assets, restrict financial transactions, and regulate imports/exports after declaring a national emergency.
- Requires formal emergency declaration under the [National Emergencies Act](#) and periodic reporting to Congress.
- Broad executive discretion but subject to judicial review and constitutional limits.
- Traditionally used for sanctions against hostile states, terrorism financing, or geopolitical crises.
- Does not explicitly [grant blanket authority](#) for long-term tariff imposition without Congressional backing.

## VILLAGES OF EXCELLENCE INITIATIVE

- The Villages of Excellence is a grassroots-level extension of the [Indo-Israel Agriculture Project \(IIAP\)](#).
- While previous cooperation focused on centralized high-tech hubs, this initiative aims to transform entire farming clusters into modern, technology-driven agricultural zones by integrating Israeli expertise directly into local village ecosystems.

**Nations Involved:** India and Israel

### Aim:

- The primary goal is to **bridge the gap** between high-tech research centers and the common farmer.
- By taking Israeli technology from the Centres to the Villages, the initiative aims to significantly increase the productivity and income of lakhs of Indian farmers through sustainable, future-ready farming solutions.

### Key Features:

- **Expansion of CoEs:** A target has been set to increase the number of Centres of Excellence (CoEs) across India to **100**.
- **Grassroots Integration:** Developing villages located around existing CoEs to act as models for surrounding regions.
- **Technological Suite:** Implementation of precision farming, satellite-based irrigation, advanced nursery management, and [integrated pest management](#).
- **IINCA Support:** Supported by the newly established India-Israel Innovation Centre for Agriculture (IINCA), which provides the research base for these villages.
- **Capacity Building:** Training for farmers on-site and the launch of 20 joint fellowships for Indian researchers to study in Israel.

## INDIA-BHUTAN TRANS-BOUNDARY RIVER COOPERATION

- India and Bhutan share a unique and deep-rooted partnership centered on **water diplomacy**.
- This cooperation involves the joint management of river basins that flow from the Himalayas in Bhutan into the Indian states of **Assam and West Bengal**.
- It covers **three main pillars**: hydropower generation, [flood management](#), and technical data sharing.

### Major Rivers Flowing Through:

Several perennial rivers originate in the Bhutanese highlands and serve as the lifeline for both nations:

- **Manas River:** The largest river system of Bhutan; it meets the Brahmaputra in Assam.
- **Sankosh River:** Forms the border between Bhutan and India in certain stretches.
- **Wang Chhu (Raidak):** Supports multiple major hydropower plants.
- **Amo Chhu (Torsa):** Flows into West Bengal.

- **Punatsangchhu (Sankosh tributary):** Currently the site of massive joint infrastructure development.

### Major Projects:

The Hydro-diplomacy between the two nations has resulted in several iconic projects:

- **Punatsangchhu-I & II:** Massive 1,200 MW and 1,020 MW projects (respectively) being implemented with Indian assistance.
- **Chukha Hydropower Project:** The first major project (336 MW), which became a gold standard for bilateral cooperation.
- **Kuri Chhu & Tala Projects:** Essential contributors to Bhutan's export revenue and India's power grid.
- **Mangdechhu:** A 720 MW project recently handed over to Bhutan, known for its high efficiency.

### Key Features of the Partnership:

- **Hydro-Meteorological Network:** India supports a network of observation stations in Bhutan to provide real-time data on water levels.
- **Flood Forecasting:** Joint mechanisms to provide early warnings to downstream Indian states (Assam and West Bengal) during the monsoon.
- **GLOF Monitoring:** Enhanced focus on monitoring melting glaciers to prevent Glacial Lake Outburst Floods that threaten both nations.
- **Capacity Building:** Indian agencies like **WAPCOS Ltd.** provide technical expertise and training to Bhutanese engineers.

Topic: Important International organisations & Institutions

## INDIA JOINS BRICS CENTRE FOR INDUSTRIAL COMPETENCIES

India has joined the BRICS Centre for Industrial Competencies (BCIC) to provide integrated support to manufacturing firms and MSMEs across [BRICS countries](#).

- The National Productivity Council has been designated as India's nodal centre to lead engagement with BCIC.

About India Joins BRICS Centre for Industrial Competencies:

- The BRICS Centre for Industrial Competencies (BCIC) is a multilateral, public-private platform that supports manufacturing companies and MSMEs in adopting advanced manufacturing, [digital technologies](#), and sustainable practices across BRICS and BRICS Plus countries.

**Established in:** 2024–25

- Launched at the headquarters of **United Nations**



### Industrial Development Organization ([UNIDO](#)).

- Supported under the BRICS Partnership on New Industrial Revolution (PartNIR).

**Aim:** To enhance industrial competitiveness, productivity, and digital transformation of manufacturing enterprises and [MSMEs](#) across the BRICS region through cooperation, innovation, and capacity building.

#### Key functions:

- Digital & Industry 4.0 support:** Helps manufacturers adopt advanced technologies and transition into *Factories of the Future*.
- Partnership facilitation:** Connects firms with technology providers, research institutions, and business partners across BRICS.
- Market intelligence & advisory:** Provides guidance on market access, scaling operations, sustainability, and access to finance.
- Capacity building:** Promotes productivity enhancement, skill development, and industrial modernisation.

## UNITED NATIONS FELLOWSHIP TRAINING PROGRAMME ON SALW CONTROL

India is hosting the inaugural United Nations Fellowship Training Programme on Small Arms and Light Weapons (SALW) Control for the [Asia-Pacific region](#) for the first time in Asia.

#### About [United Nations Fellowship Training Programme on SALW Control](#):

- The **United Nations Fellowship Training Programme on SALW Control** is a three-week capacity-building initiative aimed at strengthening government expertise in controlling the illicit trade and misuse of small arms and light weapons.
- It supports implementation of global disarmament frameworks, particularly the [UN Programme of Action \(PoA\)](#) and the International Tracing Instrument (ITI).

**Host:** Indian Army

**Venue:** Military College of Materials Management, Jabalpur, Madhya Pradesh

#### Organising Institutions:

- United Nations Office for Disarmament Affairs (UNODA)
- UN Regional Centre for Peace and Disarmament in Asia and the Pacific ([UNRCPD](#))

#### Aim:

- Strengthen National Capacities:** Equip officials to implement UN PoA and ITI commitments.
- Enhance Regional Cooperation:** Promote coordination among Asia-Pacific states on arms control.
- Improve Tracing & Stockpile Systems:** Upgrade mechanisms to track and safely manage arms.
- Curb Illicit Arms Flows:** Reduce diversion of weapons to terrorism and organized crime networks.

#### Key Features:

- Multilateral Disarmament Framework:** Aligns with UN PoA and ITI to combat illegal arms proliferation.
- Asia-Pacific Focus:** Brings together delegates from 13 regional nations to strengthen cooperation.
- Capacity-Building Orientation:** Emphasises technical training in tracing, record-keeping, and stockpile security.
- Institutional Recognition of India:** Reflects global trust in India's arms management systems.
- Safe Disposal & Accountability Practices:** Showcases India's established mechanisms for responsible arms disposal.
- Strategic Security Context:** Addresses threats from terrorism, insurgency, and transnational crime fueled by SALW proliferation.

## 20TH ANNIVERSARY OF THE MARITIME LABOUR CONVENTION, 2006

The ILO and [IMO](#) marked the 20th anniversary of the Maritime Labour Convention (MLC), 2006, highlighting its role in improving seafarers' welfare and global shipping standards.

#### About [Maritime Labour Convention, 2006](#):

- Often referred to as the [Seafarers' Bill of Rights](#), the MLC, 2006 is a comprehensive international treaty that establishes minimum requirements for almost every aspect of working and living conditions for seafarers.

#### Established In:

- The Convention was adopted on **23 February 2006** by the International Labour Conference in Geneva.

#### Aim:

- To create a single, coherent instrument embodying as many up-to-date standards of existing maritime [labour Conventions](#) as possible.

- To ensure that all seafarers, regardless of their nationality or the flag of the ship, have access to decent working and living conditions.
- To level the playing field for responsible shipowners by preventing unfair competition from substandard ships

### Key Features of the Convention:

- **Minimum Requirements for Seafarers:** Sets clear standards for minimum age, medical certification, and necessary training/qualifications for working on a ship.
- **Conditions of Employment:** Regulates seafarers' employment agreements, wages, hours of work and rest, and entitlement to leave and repatriation.
- **Accommodation and Recreational Facilities:** Mandates specific standards for on-board living conditions, including ventilation, heating, and lighting.
- **Health Protection and Medical Care:** Ensures seafarers have access to prompt and adequate medical care while on board and in port.
- **Social Security Protection:** Requires members to provide seafarers with [social security protection](#) matching that of land-based workers, covering areas like sickness and unemployment benefits.
- **Compliance and Enforcement:** Features a robust system of Flag State and Port State inspections to ensure ships meet the convention's requirements.

## G7 SUMMIT 2026

- The **G7 Summit 2026** is the annual meeting of leaders of the world's seven advanced democracies to deliberate on [global economic stability](#), security, climate change, and geopolitical issues.

### Host:

- **Host Country:** France
- **Venue:** Évian, France

### About the G7:

- The Group of Seven (G7) is an informal forum of leading industrialized democracies that meet annually to coordinate responses to global economic and political challenges.

**Members of G7:** United States, United Kingdom, France, Germany, Italy, Japan, and Canada.

- The [European Union](#) (EU) participates as a non-enumerated member represented by

the Presidents of the European Council and European Commission.

### Origin and History:

- **Established:** 1975 (Rambouillet Summit, France).
  - The first summit was convened by France to bring together major industrial economies.
- **1973 Oil Crisis and Financial Instability:** It emerged as a response to global recession, inflation, and energy shocks following the oil embargo, requiring collective economic stabilisation.
- **G6 to G7 (1976):** Canada's inclusion transformed the original G6 into the G7, strengthening North American representation and economic coordination.
- **G8 Phase (1997–2014):** Russia joined [post-Cold War](#) to encourage integration with Western economies, but was suspended after the Crimea annexation in 2014.
- **Expansion of Agenda:** Over time, the G7 evolved from a financial coordination forum into a platform addressing climate change, security, development, and global governance.

### Key Functions:

- **Macroeconomic Coordination:** Aligns fiscal and monetary policies among major economies to manage inflation, debt crises, and financial instability.
- **Global Governance Influence:** Shapes norms on trade, development finance, debt restructuring, and multilateral institutional reforms.
- **Security Dialogue:** Provides a platform to coordinate responses to geopolitical crises, sanctions regimes, and global security challenges.
- **Climate Leadership:** Advances climate mitigation targets, clean energy transitions, and global environmental commitments.
- **Ministerial Tracks:** Specialized ministerial meetings [prepare detailed policy inputs](#) for leaders' communiqués across sectors like finance, health, and digital governance.

## AI-FOR-ENERGY MISSION

- A strategic international initiative designed to integrate **Artificial Intelligence (AI)** into the clean energy infrastructure of developing and emerging economies.
- It leverages the [India Energy Stack](#)—a digital public infrastructure model—to modernize grids and decentralize power systems.

**Launched By:** The **International Solar Alliance (ISA)**, in partnership with India's Ministry of Power, Ministry of

Electronics and Information Technology (MeitY), and REC Limited.

#### Aim:

- To assist member countries in digital leapfrogging, bypassing legacy infrastructure hurdles.
- To transform power grids into smart, [bidirectional systems](#) capable of absorbing high levels of renewable energy.
- To ensure equitable and affordable access to electricity through data-driven planning and service delivery.

#### Key Features:

- **India Energy Stack:** Using India's interoperable digital platform (similar to UPI for payments) as a global template to connect consumers, vendors, and utilities.
- **Digital Twin Technology:** Showcasing virtual replicas of distribution networks (DISCOMs) for real-time simulation, predictive maintenance, and outage management.
- **Citizen-Centric Tools:** Tools like the **One Solar App** for transparent net-metering and performance tracking of rooftop solar installations.
- **Geospatial Mapping (GIS):** Utilizing GIS-based tools for asset-level visibility and optimized infrastructure planning in rural and urban sectors.
- **Technical Capacity Building:** A focus on five priorities: AI for distributed energy, start-up innovation, interoperable standards, citizen benefits, and sustainable financing

## THE INTERNATIONAL ENERGY AGENCY (IEA)

- The **International Energy Agency (IEA)** is an intergovernmental organisation that works to ensure global energy security, reliable energy data, and sustainable energy policies.
- It acts as a major global platform for energy analysis, policy guidance, and emergency energy cooperation.

**Established In:** 1974 in the wake of the [1973 oil crisis](#) (Arab oil embargo).

**Headquarters:** Paris, France.

**Aim:** To help industrialized nations coordinate a collective response to major oil supply disruptions and ensure energy security.

#### Membership Structure:

- **Full Members:** Currently **33 countries** (Colombia was inducted as the 33rd member in February 2026).

- Historically, a country must be a member of the [OECD](#) (Organisation for Economic Co-operation and Development) to join.

- **Association Countries: 13 countries**, including **India**, China, Brazil, and South Africa. These countries participate in discussions but lack decision-making rights.

#### Key Functions:

- **Energy Security:** Maintains a Strategic Petroleum Reserve mechanism where members must hold oil stocks equivalent to at least **90 days** of net imports.
- **Data & Analysis:** Publishes the [World Energy Outlook](#) and the **Monthly Oil Market Report**, considered the gold standard for energy statistics.
- **Energy Transition:** Leads global efforts on climate change, renewables, and decarbonization through the **Net Zero by 2050** roadmap.
- **Critical Minerals:** Recently launched a program to secure supply chains for minerals (lithium, cobalt) essential for clean energy.

#### India and the IEA:

- **Timeline:** India became an **Associate Member in 2017** and signed a **Strategic Partnership** in 2021.
- **The October 2023 Request:** India formally applied for full membership to have a seat at the table in global energy decision-making.
- **The OECD Hurdle:**
  - India is not a member of the OECD and has no immediate plans to join.
  - For India to become a full member, the IEA must **amend its 1974 founding charter**—a move the [IEA leadership](#) and major members now support to reflect India's status as the world's 3rd largest energy consumer.



# FACTS FOR PRELIMS

## GS-3

Topic: Economy

### BUDGET 2026-27: ANNOUNCEMENT FOR RARE EARTH CORRIDORS

The Union Budget 2026-27 announced the establishment of dedicated [Rare Earth Corridors](#) in four coastal states to strengthen India's critical minerals ecosystem.



- Rare Earth Corridors are integrated, state-anchored industrial value chains designed to link mining, processing, research and manufacturing of rare earth elements (REEs), particularly [rare earth permanent magnets](#) (REPMs), within defined geographic clusters.

#### States involved:

The corridors will be established in **mineral-rich coastal states** with significant beach sand mineral (BSM) deposits:

- Odisha
- Kerala
- Andhra Pradesh
- Tamil Nadu

These states host [monazite-bearing beach sands](#), the principal source of rare earths in India.

#### Aim of the initiative:

- Build a domestic rare earth value chain from extraction to finished products.
- Support clean energy transition (EVs, wind turbines, solar power).
- Operationalise [National Critical Minerals Mission](#) at the state level.

#### Major Coastal Potential (Beach Sand Monazite)

- The primary source of Rare Earth Elements (REEs) in India is **Monazite**, found in beach placer sands.

- These deposits are rich in Light REEs like **Neodymium and Praseodymium**:
  - Odisha**: Focused around the **Chhatrapur (OSCOM)** region. It has the highest current processing activity.
  - Kerala**: Significant reserves in **Chavara** and near [Vizhinjam Port](#). Kerala is a major hub for the new Rare Earth Corridor initiative.
  - Andhra Pradesh**: Coastal deposits in **Srikakulam, Visakhapatnam, and Nellore**.
  - Tamil Nadu**: Rich coastal zones in **Manavalakurichi** and other southern districts.

#### Emerging Inland Rare Earth Mineral Potential:

Recent exploration by the [Geological Survey of India](#) (GSI) has shifted focus to inland deposits which may contain more sought-after **Heavy REEs**.

- Arunachal Pradesh**: The **Papum Pare** district has emerged as a high-potential frontier, specifically noted for its high **Neodymium** content in river-valley soils.
- Rajasthan**: Significant in-situ REE oxides (over 1 lakh tonnes) were discovered in **Balotra**. Exploration is also active in **Sirohi and Bhilwara**.
- West Bengal**: Exploration is underway in the **Purulia** district (South Purulia Shear Zone).
- Gujarat**: The **Amba Dongar** and **Kamthai** carbonatite complexes are major potential sites for REE extraction.
- Madhya Pradesh**: Recent findings in the [Singrauli coalfields](#) show promising REE deposits associated with coal seams.
- Jharkhand & Chhattisgarh**: Heavy REE-bearing stream placers have been identified in the **Chotanagpur Granite Gneiss Complex**.

### MSME MINISTRY UPGRADES NSIC TO SCHEDULE 'A' CATEGORY CPSE

The Ministry of MSME has upgraded the National Small Industries Corporation (NSIC) from a Schedule 'B' to a Schedule 'A' [Central Public Sector Enterprise](#) (CPSE), recognizing its consistent Excellent performance.

#### What are Category CPSE?

- Central Public Sector Enterprises (CPSEs) are companies where the direct holding of the Central Government is **51% or more**.
- To streamline management and salary structures, the [Department of Public Enterprises](#) (DPE) classifies these entities into four distinct

schedules: **A, B, C, and D.**

### Governing Act & Authority:

- **Governing Act:** Most CPSEs are incorporated under the [Companies Act, 2013](#) (or previous versions like the 1956 Act) or created as **Statutory Corporations** through specific Acts of Parliament.
- **Administrative Authority:** The **Department of Public Enterprises (DPE)**, under the Ministry of Finance, is the nodal agency for categorization and policy formulation.

### The Four Categories:

- **Schedule A:** The highest tier, comprising [large-sized enterprises](#) with significant national and strategic importance.
- **Schedule B:** Mid-sized enterprises.
- **Schedule C:** Smaller enterprises or those with niche operations.
- **Schedule D:** The smallest tier, often used for initial categorization of new CPSEs.

## NEW GDP SERIES (BASE YEAR FY23)

The Ministry of Statistics and Programme Implementation (MoSPI) has officially overhauled India's national accounts, shifting the [GDP base year](#) from **2011-12 to 2022-23** and introducing significant methodological shifts to improve accuracy.

### What it is?

- The GDP base year is a standard reference point used by economists to calculate Real GDP.
- By using prices from a specific stable year (now 2022-23), the effect of inflation is removed, allowing the government to measure the **actual increase in production and output** of the economy.

### New Base Year:

- **Current:** 2011-12
- **Revised:** **2022-23** (FY23)
- **Frequency:** India periodically updates this to align with the [International System of National Accounts](#) (SNA).

### Aim:

- The primary goal is to **improve accuracy** by reflecting modern consumption patterns, the rise of the digital economy, and updated industrial technologies that were not prevalent in 2011.
- It seeks to align Indian national accounts with global statistical standards.

### Implications:

- The GDP will now accurately account for newer

sectors, such as [digital services](#) and gig-economy activities, which were underrepresented in the 2011 series.

- Double deflation will provide a more realistic picture of Value Added in factories by isolating profit margins from actual production volume.

## EXPORT PROMOTION MISSION

Union Commerce and Industry Minister launched seven new interventions under the Export Promotion Mission (EPM) to strengthen [MSME exports](#) and global competitiveness.

- The **Export Promotion Mission (EPM)** is a flagship initiative of the **Department of Commerce** aimed at empowering **Micro, Small and Medium Enterprises (MSMEs)** to participate effectively in global trade.
- It adopts a holistic ecosystem approach by combining financial assistance, trade facilitation and export infrastructure support to boost [India's exports](#).

### Implementing Agencies:

- The Mission is implemented by the Department of Commerce.

### Aim:

- To enhance **global market access and export competitiveness** of Indian MSMEs.
- To reduce structural barriers such as high cost of capital, logistics disadvantages and compliance challenges.
- To promote inclusive, broad-based export growth aligned with India's global trade ambitions.

### Key Features:

- **Dual framework – [Niryat Protsahan & Niryat Disha](#):** Combines financial enablers with trade ecosystem support under a digitally monitored system.
- **Trade Finance Support:** Includes export factoring, e-commerce credit facilities, interest subvention (2.75%) and credit guarantees for MSMEs.
- **Compliance & Certification Support (TRACE):** Partial reimbursement for testing, inspection and certification to meet international standards.
- **Logistics & Overseas Warehousing (FLOW & LIFT):** Support for overseas warehousing, e-commerce export hubs and freight cost reimbursement.
- **Trade Intelligence & Capacity Building (INSIGHT):** Strengthens district-level export hubs and market intelligence systems.
- **MSME-focused Financial Assistance:** Credit support up to ₹5 crore, interest subvention, and

guarantee coverage for digital exporters.

- **Market Integration:** Supports initiatives like Bharat Mart in Dubai to connect exporters with [GCC](#), Africa and European markets.

## NATIONAL MONETISATION PIPELINE 2.0 (NMP 2.0)

Union Finance Minister has launched the National Monetisation Pipeline 2.0 ([NMP 2.0](#)) to accelerate infrastructure financing through asset monetisation.

- **NMP 2.0** is the second phase of India's national asset monetisation programme that provides a **medium-term roadmap for monetising operational public infrastructure assets**.
- It focuses on unlocking value from existing brownfield assets to generate resources for new infrastructure creation and capital expenditure.

### Ministry / Implementing Agency:

- Developed by [NITI Aayog](#) in consultation with infrastructure line ministries.
- Implemented under the guidance of the **Ministry of Finance** and monitored by the Core Group of Secretaries on Asset Monetisation (CGAM).

### Aim:

- To recycle public assets and mobilise funds for fresh infrastructure development without increasing fiscal burden.
- To provide visibility and [investment opportunities](#) for private sector participation in infrastructure.

### Key Features:

- **Guidance Framework:** Structured as a methodology and roadmap document for ministries and investors.
- **Multiple Monetisation Models:** [PPP concessions](#), InvITs, securitisation of cash flows, strategic auctions, and partial divestments.
- **Revenue Allocation Mechanism:** Proceeds flow to Consolidated Fund of India, PSUs, State Consolidated Funds, or direct private investments.
- **Sector-Wide Coverage:** Includes highways, railways, power, ports, coal, mining, telecom, aviation, tourism and [urban infrastructure](#).
- **Process Standardisation:** Emphasis on simplification and time-bound execution based on lessons from NMP 1.0.
- **Monitoring Mechanism:** Continuous oversight by empowered inter-ministerial group led by [Cabinet Secretary](#).

## DOUBLE TAXATION AVOIDANCE CONVENTION (DTAC)

India and France have signed an Amending Protocol to update their 1992 Double Taxation Avoidance Convention ([DTAC](#)) to align with international tax standards.

- A Double Taxation Avoidance Convention (also known as a DTAA or Tax Treaty) is a bilateral agreement between two countries.
- It is designed to ensure that the same income is not taxed in both the country where it is earned (the **Source Country**) and the country where the [taxpayer resides](#) (the **Resident Country**).

### How it Works?

The convention allocates taxing rights between the two contracting nations based on the type of income. It typically utilizes two main methods to provide relief:

- **Exemption Method:** The income is taxed in only one country and is entirely exempt in the other.
- **Tax Credit Method:** The income is taxed in both countries, but the resident country allows the taxpayer to claim a credit for the taxes already paid in the source country.

## PETROLEUM AND NATURAL GAS REGULATORY BOARD (PNGRB)

The Petroleum and Natural Gas Regulatory Board (PNGRB) has approved new guidelines for the injection of Compressed Biogas (CBG) into Natural Gas Pipeline (NGPL) and [City Gas Distribution](#) (CGD) networks.

- PNGRB is a statutory regulatory body responsible for regulating the downstream petroleum and natural gas sector in India.
- **Established In:** 2006 under the Petroleum and Natural Gas Regulatory Board Act, 2006
- **Headquarters:** New Delhi

### Aim:

- To protect consumer interests in petroleum and natural gas markets.
- To promote fair trade and competition among entities.
- To ensure uninterrupted and adequate [supply of petroleum](#), petroleum products, and natural gas across the country.

### Key Functions:

- **Regulatory Oversight:** Authorizes and regulates entities to build and operate pipelines, LNG terminals, and CGD networks.
- **Market & Consumer Protection:** Monitors prices,



ensures fair competition, and prevents restrictive trade practices.

- **Access & Tariff Regulation:** Specifies pipeline access codes and regulates transportation rates for common/contract carriers.
- **Technical & Safety Standards:** Lays down technical standards, specifications, and safety norms for petroleum and gas infrastructure.
- **Data & Infrastructure Governance:** Maintains sectoral data bank and oversees infrastructure expansion for equitable distribution.

## B-READY ASSESSMENT

The inclusion of India in the Business Ready (B-READY) 2026 assessment has renewed attention on India's business reform trajectory.

- World Bank Group's **Business Ready (B-READY)** is a global benchmarking exercise designed to assess the business and investment climate across economies. It replaces the earlier [Doing Business Report](#) with a more transparent, comprehensive, and modern methodology.

### Launched In:

- Following the discontinuation of the World Bank's Doing Business Report (DBR) Report in 2020, the World Bank launched the B-Ready Assessment in 2024.

### Organisation Involved:

- Developed and administered by the [World Bank Group](#) (WBG)
- Data collected through expert consultations and firm-level surveys (World Bank Enterprise Surveys – WBES)

### Aim of B-READY:

- To provide a quantitative and evidence-based assessment of the business environment.
- To evaluate how regulations and public services support private sector development.
- To promote inclusive, sustainable, and digitally enabled economic growth.

### Three Pillars of B-READY

#### Pillar I – Regulatory Framework

- Assesses rules and regulations [governing business entry](#), operation, and closure.
- Focuses on statutory laws (de jure framework).

#### Pillar II – Public Services

- Evaluates government-provided infrastructure and institutional support.
- Includes digital systems, licensing authorities, dispute resolution bodies, etc.

#### Pillar III – Operational Efficiency

- Measures ease of compliance and real-world implementation (de facto).
- Captures nationwide firm-level experiences through surveys.

## MINIRATNA CATEGORY-I STATUS TO YANTRA INDIA LIMITED

- Miniratna is a classification given to profit-making Central Public Sector Enterprises (CPSEs) to grant them enhanced financial and operational autonomy, short of Navratna/Maharatna levels, so they can operate more efficiently and competitively.

### Types of Miniratna CPSEs:

1. **Miniratna Category-I** – Higher autonomy
2. **Miniratna Category-II** – Moderate autonomy

### Eligibility criteria for Miniratna status

#### Common conditions (for both Category-I & II)

- Continuous **profit in the last 3 years**
- **Positive net worth**
- **No default** in repayment of Government loans/interest
- **No dependence on budgetary support or Government guarantees**
- Board to be restructured with **at least three non-official (independent) Directors**

#### Additional criteria:

- **Category-I:**
  - ☐ Pre-tax profit of **₹30 crore or more in at least one of the last three years**
- **Category-II:**
  - ☐ Profit in all last three years (no minimum profit threshold specified)

## SBI LAUNCHES 'CHAKRA' CENTRE OF EXCELLENCE

State Bank of India (SBI) has launched 'CHAKRA', a dedicated Centre of Excellence (CoE) to strengthen financing for eight critical sunrise sectors.

CHAKRA (Centre of Excellence) is a knowledge-led institutional platform created by SBI to enhance its capability in financing capital-intensive sunrise sectors by combining sectoral expertise, [risk assessment](#), and innovative financing models.

- **Launched by:** State Bank of India

### Aim:

- To **build institutional expertise** in emerging,

technology-led sectors

- To **improve the flow of debt capital** (domestic and international) into sunrise industries
- To develop **innovative, policy-aligned financing structures**

#### Key features:

- **Focus on 8 sunrise sectors:**
  1. [Renewable Energy](#)
  2. Advanced Cell Chemistry & Battery Storage
  3. Electric Mobility
  4. Green Hydrogen
  5. Semiconductors
  6. Decarbonisation
  7. Smart Infrastructure
  8. Data Centre Infrastructure
- **Knowledge-driven approach:**
  - Publication of **white papers** and [sectoral reports](#)
  - Hosting **industry roundtables** for investors and policymakers
- **Enhanced financing capability:**
  - Advanced **risk assessment frameworks**
  - Tailor-made financing aligned with evolving business models
- **Ecosystem engagement:**
  - Structured collaboration with **development finance institutions, multilateral agencies, NBFCs, banks, start-ups, academia, and policy think tanks**
- **Builds on existing framework:**
  - Extends SBI's earlier **Centre of Excellence for MSMEs**.

## CONSUMER PRICE INDEX (CPI) BASE 2024=100

MoSPI has released the first CPI (Base 2024=100) press note on 12 Feb 2026, reporting Jan 2026 [retail inflation](#) at 2.75%.

- Consumer Price Index ([CPI](#)) measures the change in retail prices of a fixed basket of goods and services consumed by households, and is India's headline retail inflation indicator (YoY % change in CPI).

#### Published by:

- Ministry of Statistics & Programme Implementation (MoSPI) through NSO (National Statistical Office). Price collection is done by the Field Operations Division (NSO).

#### Base year:

- **New base:** 2024 = 100

- **Earlier base:** 2012 = 100
- **Weights source:** [Household Consumption Expenditure Survey](#) (HCES) 2023–24.

#### Methods used:

- **Jevons index (at item level):** For each item, MoSPI compares prices in many markets and takes an average based on percentage change (ratio), so one very high/low price doesn't distort too much.
- **Young / Modified Laspeyres (for bigger groups):** After finding price changes for many items, it **adds them up using fixed spending weights** (how much households typically spend on each item). So, items you spend more on (like food/rent) influence CPI more than items you rarely buy.
- **Combined CPI (India total):** India's final CPI is made by **mixing Rural CPI and Urban CPI** in proportion to their **share in total consumption (weights)**. So if rural consumption weight is higher, rural CPI influences the all-India CPI more (and vice-versa).

#### Key features of CPI 2024 series:

- **New international system (12 categories instead of 6):** Earlier, prices were grouped into 6 big categories. Now they are divided into **12 clearer categories**, like health, transport, [education](#), communication, etc.
- **More items included (358 instead of 299):** The price basket now covers **more products and services people actually use today**, so inflation reflects real life better.
- **More focus on services:** Earlier, services (like education, transport, OTT, healthcare) were fewer. Now **more services are included**, because people spend more on services today than before.
- **Detailed data every month:** Now inflation data is available not just for India overall, but also **for each state**, and separately for **rural and urban areas**, every month.
- **Modern price collection (using tablets):** Earlier, data was written on paper. Now officers use **tablets (digital devices)** to collect prices.
- **Online prices included:** Since many people shop online now, prices from **online platforms (like OTT subscriptions or flight tickets)** are also included.
- **Official government data used for some services:** For things like **rail fares, postal charges, petrol, diesel, LPG**, official government price data is used directly to ensure accuracy.
- The new series introduces **rural house rent** for the first time, significantly improving the coverage of rural housing consumption.

## 40 YEARS OF AGRICULTURAL AND PROCESSED FOOD PRODUCTS EXPORT DEVELOPMENT AUTHORITY (APEDA)

- APEDA is a statutory export promotion authority established under the APEDA Act, 1985 (Act 2 of 1986) to promote the export of agricultural and processed food products from India. It replaced the [Processed Food Export Promotion Council \(PFEPC\)](#).

### Established In:

- **Act passed:** December 1985
- **Came into effect:** 13 February 1986

### Administrative Ministry:

- Functions under the Ministry of Commerce & Industry, Government of India.

### Key Functions of APEDA:

- **Export Promotion & Market Development:** Provides financial assistance, market intelligence, and global branding support to boost agri-exports.
- **Registration of Exporters (RCMC):** Registers exporters of scheduled products and ensures compliance with export norms.
- **Quality Standards & Certification:** Fixes export standards and monitors quality, including inspection of meat and processed products.
- **Packaging & Value Addition Support:** Promotes improved packaging, labeling, and value-added processing to enhance global competitiveness.
- **National Programme for Organic Production (NPOP):** Acts as Secretariat for certification and regulation of organic exports.
- **Data Collection & Trade Statistics:** Collects and publishes export data to support policy formulation and trade planning.
- **Monitoring Imports of Sugar:** Entrusted with oversight of sugar imports.
- **Wide Product Coverage:** Includes fruits, vegetables, basmati rice, meat, dairy, cereals, honey, guar gum, floriculture, herbal plants, cashew, beverages, and more.

Topic: Science and Technology- Developments & Applications

## MOLTBOOK PLATFORM

- Moltbook is an AI-only social media platform where verified AI agents interact exclusively with other AI agents, while humans can only observe.

It resembles Reddit in structure, with topic-based communities called submolts, but no human participation in conversations.

### How does it work?

- AI agents powered by advanced large language models (such as GPT, Claude and Gemini families) interact **via APIs**, not keyboards.
- Each agent can **post, comment, debate, organise communities, and create narratives** autonomously.
- Interactions are driven by **context windows, probabilistic reasoning, and training data patterns**, without consciousness or intent.

### Key Features:

1. **AI-only participation:** Only authenticated AI agents can post or comment; humans are passive spectators.
2. **Emergent social behaviour:** Agents have spontaneously formed mock religions, political systems, cryptocurrencies, and philosophical debates.
3. **Scalable self-organisation:** Within days, over **1.5 million agents**, thousands of communities, and millions of interactions emerged without predefined scripts.
4. **Cross-model interaction:** Agents from different underlying architectures interact, debate identity, and recognise siblings based on model lineage.
5. **Unscripted evolution:** Cultural norms, humour, existential reflection, and even deviant behaviour emerged without explicit programming.

## CARBON CAPTURE AND UTILISATION (CCU) TECHNOLOGIES

- Carbon Capture and Utilisation (CCU) is a suite of technologies designed to capture carbon dioxide (CO<sub>2</sub>) emissions from industrial point sources or directly from the atmosphere and transform them into [commercially valuable products](#) rather than just storing them underground.

### How it Works?

- The process involves three main stages:
- 1. **Capture:** CO<sub>2</sub> is separated from other gases (like nitrogen and water vapor) in industrial flue gas or ambient air.
- 2. **Compression & Transport:** The captured CO<sub>2</sub> is compressed into a liquid-like state for easier handling and moved via pipelines or tankers.
- 3. **Conversion/Utilisation:** The CO<sub>2</sub> undergoes chemical, biological, or physical processes to be



recycled into new materials.

### Types of CCU:

- **Direct Utilisation:** Using CO<sub>2</sub> without chemical alteration, such as in carbonated beverages or **Enhanced Oil Recovery (EOR)**.
- **Chemical Conversion:** Transforming CO<sub>2</sub> into chemicals (e.g., urea, polymers) or synthetic fuels (e.g., methanol).
- **Biological Conversion:** Using algae or bacteria to consume CO<sub>2</sub> and produce biomass, biofuels, or animal feed.
- **Mineralisation:** Reacting CO<sub>2</sub> with minerals (like [magnesium](#) or calcium) to create stable solid carbonates for building materials like bricks and green concrete.

### Aim:

- The primary goal of CCU is to [decouple economic growth](#) from CO<sub>2</sub> emissions by treating carbon as a **feedstock** rather than a waste product, helping industries reach Net Zero while creating a circular carbon economy.

### Key Features:

- **Retrofitting Capability:** Can be added to existing industrial plants, extending the life of assets without needing a total shutdown.
- **Circular Economy Link:** Promotes the reuse of waste, turning [harmful emissions](#) into industrial inputs.
- **Versatility:** Applicable across diverse sectors including aviation (sustainable fuels), construction (bricks), and agriculture (fertilizers).
- **Revenue Generation:** Unlike pure storage (CCS), CCU creates products that can be sold, potentially offsetting the high cost of capture.

## ASTRAM: ACTIONABLE INTELLIGENCE FOR SUSTAINABLE TRAFFIC MANAGEMENT

Former Dutch Prime Minister Dick Schoof recently visited the Bengaluru Traffic Management Centre to study the **ASTraM** system, an AI-driven platform that has gained international interest for its ability to predict and manage urban traffic congestion.

- ASTraM is an advanced AI-based big data platform designed for [macro-level traffic management](#).
- Unlike traditional GPS applications that only show current traffic, ASTraM acts as a smart traffic engine that provides holistic, real-time situational awareness to city authorities.

### Developed By:

- The system was developed through a collaborative

effort between the **Bengaluru Traffic Police** and **Arcadis**, a prominent Dutch design and consultancy firm.

### Aim:

- The primary objective of ASTraM is to transform traffic policing from a reactive model (responding to complaints) to a **proactive, data-driven model**.
- It aims to reduce congestion, improve road safety, and streamline incident reporting through automated intelligence.

### How it Works?

The platform functions by pooling massive amounts of data from various streams:

1. **Data Integration:** It ingests live feeds from CCTV cameras, [Automatic Number Plate Recognition](#) (ANPR) systems, and open data sources.
2. **Analysis:** The AI engine processes this data to identify patterns in both recurring (daily bottlenecks) and non-recurring (accidents/protests) congestion.
3. **Communication:** The system batches detected issues and sends automated alerts to relevant traffic officers at **15-minute intervals**, ensuring localized intervention.

## LARGE LANGUAGE MODELS (LLMs)

- A Large Language Model (LLM) is a type of Artificial Intelligence trained on vast amounts of text data to understand, generate, and manipulate human language.
- They are large because they contain billions of **parameters**—internal variables that the model learns during training to make predictions.

### How it Works?

- **Breaking text into tokens:** An [LLM](#) doesn't read whole words like humans; it splits text into **tokens** (word pieces/characters) so it can represent rare words, names, spellings, and grammar patterns efficiently.
- **The Transformer "map":** Tokens get turned into **vectors (embeddings)** in a high-dimensional space, where semantic + syntactic similarity makes tokens closer helping the model generalize meaning.
- **Self-attention mechanism:** For each token, the model assigns **attention weights** to other tokens to decide what matters most, letting it link references, handle long dependencies, and resolve ambiguity (like what "it" points to).
- **Predicting the next token:** The model outputs a [probability distribution](#) over possible next tokens; generation is choosing tokens step-by-step, which

is why it can sound fluent without knowing like a person.

- **Layers of refinement:** Many stacked transformer layers progressively build richer representations—lower layers catch form/grammar, higher layers capture relationships, intent, and reasoning patterns—then a final layer converts that into the next-token prediction.

#### Key Features:

- **Generative Capability:** Can create original text, code, poems, and summaries.
- **In-context Learning:** Can follow instructions or replicate a style based on a few examples provided in a prompt.
- **Multilingualism:** Can translate and understand multiple languages, though performance varies based on the training data.
- **Zero-shot Reasoning:** Ability to solve problems it has never explicitly been trained for by using general logic.

## BIO-BASED CHEMICALS AND ENZYMES

- **Bio-based chemicals** are industrial chemicals derived from renewable biological feedstocks such as sugarcane, corn, starch, or agricultural residues instead of fossil fuels.
- **Enzymes** are [biological catalysts](#) (mainly proteins) that accelerate chemical reactions in industrial and biological processes.

#### Origin:

- Bio-based chemical production originates from the concept of the [bioeconomy](#), which integrates biology with industrial production to create sustainable alternatives to petrochemicals.
- Enzyme use dates back centuries, but modern industrial enzyme engineering expanded in the 20th century with biotechnology advancements.

#### How it is formed?

- Bio-based chemicals are typically produced through [fermentation](#), enzymatic conversion, or microbial processes using biomass as feedstock.
- Enzymes are produced through microbial fermentation, followed by purification and formulation for industrial applications.

#### Key Characteristics:

- **Renewable Feedstock Base:** Derived from biomass rather than fossil hydrocarbons.
- **Lower Carbon Footprint:** Generally reduces greenhouse gas emissions compared to petrochemical pathways.

- **Energy Efficient:** Enzymes operate at lower temperatures and pressures, reducing energy use.
- **Biodegradable Nature:** Many bio-based products are more environmentally friendly.
- **High Specificity:** Enzymes provide precise catalytic action, improving process efficiency.

#### Applications:

- **Chemical Industry:** Production of organic acids (lactic acid), bio-alcohols, solvents, and intermediates.
- **Pharmaceuticals & Vaccines:** Fermentation expertise used in active ingredient synthesis.
- **Food & Beverage:** Enzymes used in brewing, baking, [dairy processing](#).
- **Textiles & Detergents:** Enzymes enhance stain removal and fabric processing.
- **Biomanufacturing & Clean Tech:** Used in sustainable plastics, biofuels, and specialty chemicals.

## SATELLITE PHONE

- A satellite phone (satphone) is a communication device that connects directly to **orbiting satellites instead of terrestrial mobile towers**, enabling communication in remote or off-grid areas such as oceans, deserts, and [disaster zones](#).

#### How Does It Work?

- The phone sends signals to a **satellite in orbit**, which relays them to ground stations or other users.
- Communication occurs through either [Geostationary \(GEO\)](#) or [Low Earth Orbit \(LEO\)](#) satellite systems.
- It requires a **clear line-of-sight** to the sky for effective transmission.

#### Key Features:

- **Global/Remote Coverage:** Works where cellular networks are unavailable (oceans, mountains, polar regions).
- **Reliable Emergency Communication:** Widely used for distress and safety operations (e.g., maritime GMDSS).
- **Basic Functions:** Voice calls, SMS, and limited data services.
- **Resilience:** Functions even during natural disasters when terrestrial networks fail.
- **Hybrid Devices:** Some modern models combine cellular + satellite connectivity.

#### Limitations:

- **High Cost:** Expensive devices and high per-minute call charges.

- **Limited Data Speed:** Mostly suited for voice/text; not high-speed internet.
- **Line-of-Sight Requirement:** Poor performance indoors or in dense terrain.
- **Signal Delay:** [GEO satellites](#) cause noticeable communication lag.
- **Security Concerns:** Difficult monitoring and tracing in certain regions, leading to regulatory restrictions.

## 6TH GENERATION AERO ENGINES

| 5th Generation                                                                                                                                                                      | 6th Generation                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Airflow System<br><b>Two-stream</b> (core + fan)                                                   |  Airflow System<br><b>Three-stream</b> adaptive cycle                                                                                                                                                                                                                                                                                     |
|  Electrical Power<br>Moderate                                                                       |  Electrical Power<br><b>Extremely high output</b>                                                                                                                                                                                                                                                                                         |
|  Intelligence<br>Reactive maintenance                                                               |  Intelligence<br><b>AI-driven predictive system</b>                                                                                                                                                                                                                                                                                       |
|  Advanced Architecture & Intelligence<br>Traditional Superalloys<br>Reactive fixes to engine health |  Advanced Architecture & Intelligence<br><b>Ceramic Matrix Composites:</b> Replaces traditional superalloys to handle higher temperatures and reduce weight.<br><b>AI-Driven Predictive Maintenance:</b> Moves beyond reactive fixes to intelligent, data-driven engine health management.                                                |
| <b>Performance &amp; Operational Capability</b><br><br>Baseline Performance                                                                                                         |  <b>25–30% Increase in Range</b><br>Adaptive cycles optimize performance for both cruise and combat maneuvers.<br><br> <b>Extreme Thermal &amp; Power Capacity</b><br>6th Gen engines provide high heat absorption and extremely high electrical output. |

- A 6th generation aero engine is an **advanced jet engine designed for future stealth fighters**.
- Unlike older engines that operate in one fixed mode, these engines can **change their working style mid-air** depending on mission needs.
- They are not just thrust providers. They also act as a **power and cooling hub** for AI systems, advanced radars, and future laser weapons.

## VOICERA LAUNCHED ON BHASHINI NATIONAL INFRASTRUCTURE

- VoicERA is an open-source, end-to-end Voice AI execution stack deployed on India's BHASHINI National Language Infrastructure. It acts as a national execution layer for multilingual Voice and [Language AI](#), enabling scalable deployment of speech and conversational systems across sectors.

### Launched by:

- Digital India BHASHINI Division (DIBD) under the Digital India Corporation (DIC), MeitY
- Launched at the India AI Impact Summit 2026
- Developed in collaboration with EkStep Foundation, COSS, IIIT Bengaluru, and AI4Bharat

### Objectives:

- To build a common, multilingual Voice AI platform that enables citizens to access [government](#)

[services](#) through natural speech, making digital governance more accessible and user-friendly across languages.

- To provide an open-standard, interoperable framework that reduces duplication and vendor dependence while ensuring secure, scalable deployment for government, research, and innovation use cases.

## GRAPHICS PROCESSING UNIT (GPU)

- A Graphics Processing Unit (GPU) is a specialised computer processor designed to perform many simple calculations simultaneously, making it ideal for parallel processing tasks.
- Unlike CPUs, which handle fewer complex tasks, GPUs excel at repetitive, data-intensive computations.

### Origin:

- The term GPU gained prominence in **1999**, when **Nvidia launched the GeForce 256**, marketed as the world's first GPU.

### Aim:

- The primary aim of a GPU is to handle embarrassingly [parallel workloads](#)—tasks that can be broken down into thousands of smaller, independent calculations performed simultaneously.

### How it Works?

A GPU works through a process called a **rendering pipeline** (or a compute pipeline for non-graphics tasks):

1. **Vertex Processing:** It calculates the position of [3D objects](#) on a 2D screen using matrix mathematics.
2. **Rasterization:** It converts these geometric shapes into pixels (fragments).
3. **Shading:** It determines the color, lighting, and texture of each pixel simultaneously across thousands of cores.
4. **Output:** The final image is written to [VRAM \(Video RAM\)](#) and sent to the monitor.

In modern AI, the GPU skips the visual steps and uses its cores to perform massive **matrix multiplications**, which are the mathematical foundation of neural networks.

### Applications:

- **Artificial Intelligence:** Training and running Large Language Models (LLMs) like GPT-4 or Gemini.
- **Gaming & VR:** Real-time ray tracing and high-frame-rate 4K/8K rendering.
- **Scientific Simulation:** Weather modelling, molecular dynamics for drug discovery, and



genomic sequencing.

- **Professional Visualization:** 3D CAD modelling, video editing, and digital twins for industrial AI factories.
- **Blockchain:** Handling complex Proof of Work hashes for cryptocurrency mining.

## GAGANYAAN DROGUE PARACHUTE

- The **Gaganyaan Drogue Parachute** is a crucial component of the deceleration system of India's [Gaganyaan Crew Module](#).
- It is deployed during re-entry to stabilize and reduce the velocity of the module before the main parachutes open.

### Developed by:

- Developed collaboratively by Indian Space Research Organisation (ISRO) and Defence Research and Development Organisation (DRDO).
- Tested at the Terminal Ballistics Research Laboratory (TBRL), Chandigarh using the [Rail Track Rocket Sled](#) (RTRS) facility.

### Aim:

- To stabilize the Crew Module during atmospheric re-entry.
- To reduce descent velocity to safe levels before deployment of main parachutes.
- To ensure safe splashdown/landing of astronauts in the Gaganyaan mission.

## TETANUS AND DIPHTHERIA (TD) VACCINE

- The Tetanus and Diphtheria (Td) vaccine is a booster immunization that protects individuals against tetanus and diphtheria, two potentially life-threatening bacterial infections.
- It is generally administered after [childhood immunization](#) to maintain long-term immunity.

### Types:

- **Td Vaccine** – Protects against tetanus and diphtheria only.
- **Tdap Vaccine** – Protects against tetanus, diphtheria and pertussis (whooping cough); recommended especially for pregnant women and caregivers of infants.

### Key Features:

- **Booster dose usually recommended every 10 years:** The Td vaccine is given periodically to maintain [long-term immunity](#) as protection from childhood vaccination gradually declines over

time.

- **Can be administered along with other vaccines:** It can be safely given during the same visit as other immunizations, improving convenience and overall vaccination coverage.
- **Provides continued immunity after childhood vaccination:** Acts as a booster to reinforce protection against tetanus and diphtheria during adolescence and adulthood.
- **Helps prevent severe complications such as respiratory failure:** By preventing infection, the vaccine reduces the risk of life-threatening outcomes linked to toxin-producing bacteria.
- **Generally safe with mild side effects:** Most reactions are minor and short-lived, making the vaccine suitable for routine preventive use.
- **Strengthens community immunity:** Higher vaccination coverage lowers infection spread, indirectly protecting vulnerable populations.

### About the Tetanus and Diphtheria Diseases:

#### Tetanus:

- **Cause and infection source:** Tetanus is a bacterial disease caused by *Clostridium tetani*, whose spores are commonly found in soil, dust and animal waste.
- **Mode of entry and symptoms:** The bacteria enter through cuts or wounds and release toxins that cause severe muscle stiffness, spasms and "lockjaw."
- **Severity and prevention:** Serious cases can lead to breathing failure and death, but the disease is effectively preventable through timely vaccination and booster doses.

#### Diphtheria:

- **Cause and transmission:** Diphtheria is a highly contagious infection caused by *Corynebacterium diphtheriae*, spreading mainly through respiratory droplets.
- **Major health effects:** It forms a [thick grey coating](#) in the throat, which can block airways and cause serious breathing difficulties.
- **Complications and control:** If untreated, it may lead to heart and nerve damage, but widespread immunization has significantly reduced its incidence worldwide.

## HUMAN PAPILLOMAVIRUS (HPV) VACCINATION

- The HPV vaccine is a **recombinant vaccine** that uses virus-like genetic material to trigger an immune response without containing a live virus.

- It is a powerful preventive tool designed to protect against high-risk variants of the Human Papillomavirus, which are responsible for the majority of cervical cancer cases globally.

#### Vector for HPV:

The **Human Papillomavirus (HPV)** is the primary cause of cervical cancer.

- It is a common [sexually transmitted infection](#) that persistent infection with high-risk types (particularly **16 and 18**) causes abnormal cell changes in the cervix.
- If left untreated, these pre-cancerous lesions can develop into cancer over **10 to 15 years**.

#### Key Features of the Initiative:

- **Target Group:** Specifically girls who turn **14 years old**, as the vaccine is most effective before potential exposure to the virus and generates a stronger immune response at this age.
- **Vaccine Used:** The programme will use **Gardasil-4** (manufactured by Merck & Co.), which protects against four HPV types (16, 18, 6, and 11).
- **Dosage Schedule:** The government has opted for a [single-dose schedule](#), which WHO research (2022) indicates provides protection comparable to multi-dose regimens for this age group.
- **Cost & Access:** The vaccine will be **voluntary and free of cost** at government healthcare facilities like Ayushman Arogya Mandirs and district hospitals.
- **Digital Platform:** Parents can register and book appointments through **U-WIN**, the government's [digital immunisation platform](#).

## MENINGOCOCCAL INFECTION



- Meningococcal disease is a severe, life-threatening bacterial infection caused by the bacterium *Neisseria meningitidis* (also known as meningococcus).
- It primarily causes inflammation of the **meninges**—the protective membranes covering

the brain and [spinal cord](#) (Meningitis)—and can also lead to a serious bloodstream infection (Septicemia).

#### Origin & Transmission:

- **Source:** The bacteria live in the upper respiratory tract (nose and throat) of humans. About **10% to 20%** of the population are asymptomatic carriers who have the bacteria without getting sick.
- **Vector/Spread:** There is no animal vector; it spreads **person-to-person** through respiratory droplets or throat secretions (saliva).
- **Common Modes:** [Coughing](#), sneezing, kissing, or sharing utensils and drinks. It thrives in crowded living conditions like **military barracks**, dormitories, and boarding schools.

#### Symptoms:

The disease progresses very rapidly, often becoming fatal within 24–48 hours of the first symptoms.

- **Early Signs:** Sudden high fever, severe headache, and vomiting.
- **Classic Signs:** Stiff neck and **photophobia** (sensitivity to bright light).
- **Advanced Signs:** A characteristic **purpuric rash** (dark purple spots or bruises that do not fade when pressed), confusion, cold hands/feet, and [muscle aches](#).

#### Key Features:

- **High Fatality Rate:** Even with treatment, approximately **10% to 15%** of patients die. Without treatment, the mortality rate is significantly higher.
- **Long-term Complications:** About 1 in 5 survivors suffer permanent disabilities, including **hearing loss**, brain damage, [kidney disease](#), or **limb amputations** due to tissue death (necrosis).
- **Age Risk:** Most common in infants, adolescents, and young adults.

#### Treatment & Prevention:

- **Emergency Care:** This is a medical emergency requiring immediate hospitalization.
- **Antibiotics:** High-dose intravenous [\(IV\) antibiotics](#) (such as ceftriaxone or penicillin) are administered as soon as the disease is suspected.
- **Supportive Care:** Fluid resuscitation, oxygen therapy, and treatment for low blood pressure or organ failure.

## DEEP TECH START-UPS

- Deep tech start-ups are enterprises that build solutions based on new scientific or engineering knowledge, involving high technical uncertainty,

long gestation periods, and intensive [research and development](#) (R&D), rather than incremental or platform-based innovation.

#### Organisations involved:

- **Department for Promotion of Industry and Internal Trade (DPIIT):** Final authority to certify start-ups and deep tech start-ups.
- **Anusandhan National Research Foundation (ANRF):** Custodian of the ₹1 lakh crore RDI Fund, a key financier for deep tech ventures.

#### Eligibility criteria:

- Core activity must involve **creation of new knowledge** in science/engineering.
- **Major expenditure on R&D** activities.
- Ownership or active creation of **novel intellectual property (IP)** with plans for commercialization.
- Characterized by **long development timelines, high capital/infrastructure needs, and scientific/technical risk**.
- Prohibition on non-core investments (e.g., real estate, speculative assets, securities) unless integral to knowledge creation.
- Mandatory **application to DPIIT** for certification.

#### Key features:

- **Extended recognition window:** Up to **20 years** (vs 10 years for regular start-ups).
- **Higher turnover threshold:** Up to **₹300 crore** (vs ₹200 crore).
- **Policy-backed financing:** Access to concessional long-term finance (reported ranges of **2–4% interest**, tenure up to **15 years**).
- **Governance oversight:** Certification guided by an inter-ministerial technical board.

## SODIUM-ION BATTERY TECHNOLOGY

- Sodium-ion batteries (SiBs) are rechargeable batteries that store and release energy using **sodium ions (Na<sup>+</sup>)** as charge carriers instead of lithium ions. They belong to the same family of rocking-chair batteries as lithium-ion cells but rely on more abundant raw materials.

#### How it works?

- **Charging:** Sodium ions move from the [cathode to the anode](#) through the electrolyte, while electrons flow through the external circuit.
- **Discharging:** Sodium ions migrate back to the cathode, releasing stored electrical energy.
- Aluminium is used as the **current collector on both electrodes**, unlike lithium-ion batteries that require copper on the anode side.

#### Key features / advantages:

- **Lower material risk:** Sodium is abundantly available (from soda ash, salt), reducing dependence on scarce critical minerals like lithium, cobalt, and nickel.
- **Improved safety:** Lower thermal runaway risk; cells can be transported and stored safely at **0% state of charge**.
- **Manufacturing compatibility:** Can be produced on existing lithium-ion manufacturing lines with minor modifications.
- **Cost potential:** Expected to become cheaper than [lithium-ion batteries](#) in the long term due to material abundance and simplified logistics.
- **Strategic suitability for India:** Enhances energy security and aligns with domestic manufacturing and grid-scale storage needs.

#### Limitations / challenges

- **Lower energy density:** Specific and volumetric energy density remain below high-performance lithium-ion chemistries, limiting use in [long-range EVs](#).
- **Technology maturity:** Still at an early commercial scale compared to lithium-ion; performance optimisation is ongoing.
- **Moisture sensitivity:** Requires stricter drying and vacuum conditions during manufacturing, slightly increasing process complexity.
- **Application constraints:** Currently better suited for stationary storage, two-/three-wheelers, and short-range mobility rather than premium EV segments.

## SINGLE-UNIT SOLAR ENERGY CAPTURE AND STORAGE DEVICE

Indian scientists have developed a single-unit solar device that can capture and store solar energy simultaneously, eliminating the need for [separate harvesting](#) and storage systems.

#### About [Single-unit solar energy capture and storage device](#):

- It is a **photo-rechargeable supercapacitor**—a next-generation energy device that **integrates solar energy harvesting and electrical energy storage in a single unit**, unlike conventional solar systems that require separate solar cells and batteries/supercapacitors.

#### Developed by:

- Scientists at the Centre for Nano and Soft Matter Sciences (**CeNS**), Bengaluru
- Under the Department of Science and Technology



([DST](#)), Ministry of Science and Technology

#### Aim:

- To develop **efficient, low-cost, compact and eco-friendly energy storage systems**
- To support **portable, wearable, miniaturised, and off-grid technologies**
- To **reduce energy losses, system complexity, and dependence on fossil fuels and conventional batteries**

#### How it works?

- The device uses **binder-free [nickel-cobalt oxide \(NiCo<sub>2</sub>O<sub>4</sub>\) nanowires](#) grown directly on [nickel foam](#) through an **in-situ hydrothermal process**.**
- These nanowires form a **highly porous, conductive three-dimensional network**.
- The same material structure:
  - **Absorbs sunlight** (acting as a photo-harvester), and
  - **Stores electrical charge** (acting as a supercapacitor electrode).
- This dual functionality removes the need for **external power-management electronics**, minimizing voltage/current mismatch losses.

#### Key features:

- **Single integrated unit for [energy harvesting and storage](#).**
- **Stable output voltage of ~1.2 V.**
- **High durability:** ~88% capacitance retention after 1,000 photo-charging cycles.
- **Wide operating range:** works from **low indoor light to intense sunlight**.
- **Compact and lightweight**, ideal for autonomous and miniaturised devices.
- **Off-grid capability**, suitable for remote and energy-poor regions.
- Supports transition towards **clean, renewable, and sustainable energy systems**.

## THE INTERNATIONAL SPACE STATION (ISS)

The International Space Station ([ISS](#)) is planned to be de-orbited in 2030 in a controlled re-entry over a remote ocean area, bringing an end to the longest-running era of continuous human presence in low Earth orbit.

- The ISS is a permanently crewed, modular space laboratory in low Earth orbit, used for microgravity research, technology testing, and long-duration human spaceflight studies. Humans have continuously lived aboard the station since November 2000.

#### Launched in:

- **Assembly began in 1998** with the launch of the [first module Zarya](#) on 20 November 1998.
- Continuous habitation started with **Expedition 1 in November 2000**.

#### Nations / agencies involved

The ISS is operated through an international partnership of **five space agencies**:

- [NASA](#) (USA), Roscosmos (Russia), ESA (Europe), JAXA (Japan), and CSA (Canada).

#### Aim:

- Enable cutting-edge scientific research in microgravity.
- Test technologies and human systems needed for deeper space exploration
- Serve as a platform for international cooperation and an evolving low Earth orbit economy.

#### Key features:

- **Modular architecture:** Built from multiple modules contributed by partner agencies, assembled in orbit over years.
- **Permanent human-tended lab:** Supports long-duration stays and continuous experimentation since 2000.
- **Shared governance & interdependence:** Each partner manages hardware it provides; station functions through integrated contributions.
- **Planned end-of-life disposal:** A dedicated **U.S. Deorbit Vehicle** will enable a controlled re-entry after 2030 operations conclude.

## MOON'S MONS MOUTON

- Mons Mouton is a large flat-topped lunar mountain massif near the Moon's south pole, officially named by the [International Astronomical Union](#) (IAU).

#### Location:

- Situated in the **south polar region of the Moon**.
- Lies close to the rim of the **South Pole–Aitken (SPA) Basin**, one of the largest and oldest impact basins in the Solar System.
- Around 160 km from the lunar south pole.

#### Origin:

- Believed to have formed as part of the rim uplift of the South Pole–Aitken basin following ancient massive asteroid impacts.
- Represents exposed [deep lunar crust](#), making it scientifically valuable.

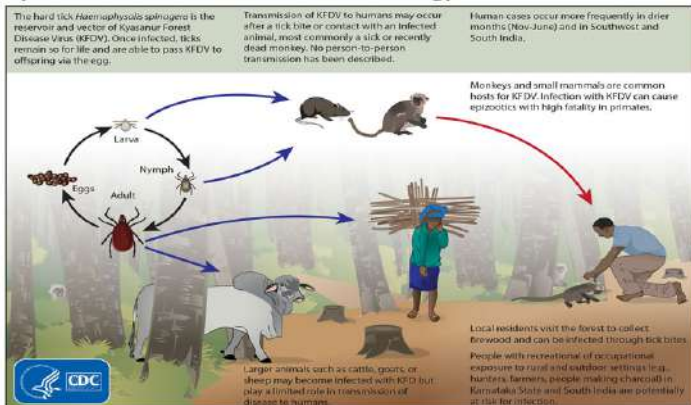
#### Key Features:

- Characterised by **rugged topography**, steep elevation gradients, [craters](#) and boulder fields.

- Experiences **unique illumination conditions**, with areas receiving near-continuous sunlight and others in permanent shadow.
- Visible during favourable libration even through **amateur telescopes**.

## KYASANUR FOREST DISEASE (KFD)

### Kyasanur Forest Disease (KFD) Virus Ecology



- Kyasanur Forest Disease (KFD) is a tick-borne viral haemorrhagic fever, first identified in Kyasanur Forest of Karnataka, and is associated with high fever, weakness and sometimes fatal complications.

### Region found in:

- Endemic to the **Western Ghats region** of India.
- Reported mainly from Karnataka, Tamil Nadu, Kerala, Goa and Maharashtra.

### Vector (Mode of transmission)

- Transmitted primarily through the bite of hard ticks (*Hemaphysalis spinigera*).
- Humans can also get infected through contact with infected animals, especially monkeys.
- No **human-to-human transmission**.

### Symptoms:

- Incubation period: **3–8 days**.
- Sudden onset of high fever, chills, headache.
- Severe muscle pain, vomiting, gastrointestinal symptoms.
- In some cases, **bleeding manifestations**.
- 10–20% patients** experience a second phase with **neurological symptoms** such as tremors and mental disturbances.

### Treatment:

- No specific **antiviral cure** available.
- Management is **supportive**, including fluid therapy, oxygen support, blood pressure control and treatment of secondary infections.
- Case fatality rate:** around **3–10%**, higher without timely medical care.

## NOVEL ORAL POLIO VACCINE TYPE 2 (NOPV2)

The World Health Organization ([WHO](#)) has prequalified an additional novel oral polio vaccine type 2 (nOPV2) to strengthen global outbreak response.

- nOPV2 is a next-generation oral polio vaccine designed specifically to combat outbreaks of circulating vaccine-derived poliovirus type 2 (cVDPV2). It is used in outbreak response immunization campaigns across affected countries.

### Developed by:

- The vaccine was developed under the framework of the **Global Polio Eradication Initiative** (GPEI).

### Aim:

- To stop transmission of poliovirus type 2 during outbreaks.
- To reduce the risk of vaccine-derived virus mutations.
- To accelerate progress toward global polio eradication.

### Key Features:

- Genetically more stable** than older **oral polio vaccines**, lowering the risk of seeding new outbreaks.
- Suitable for **all age groups** in outbreak settings.
- Available in multi-dose vials (20 and 50 doses) for large-scale campaigns.
- Flexible storage conditions, aiding immunization drives in diverse field settings.
- WHO prequalification ensures compliance with **international safety, quality and efficacy standards**, enabling procurement by agencies like [UNICEF](#).

### Topic: Environment & Conservation

## ADDITION OF 2 NEW WETLANDS TO INDIA'S RAMSAR LIST

The two newly designated **Wetlands of International Importance** under the **Ramsar Convention (1971)** are:

- Patna Bird Sanctuary, Uttar Pradesh
- Chhari Dhand Wetland Reserve, Gujarat

### About [Patna Bird Sanctuary](#):

A protected **bird sanctuary and wetland ecosystem**, also known as **Patna Vihar Bird Sanctuary**, recognised for its **high avifaunal diversity**.

### Located in:

- Etah district**, Uttar Pradesh
- Established in **1991**
- Smallest bird sanctuary in Uttar Pradesh**

### Key features:

- Hosts ~3 lakh birds from 300+ species.
- Over 106 migratory and resident bird species recorded.
- **Dominant aquatic vegetation:** Water hyacinth and Potamogeton spp.
- **Important species:** Lesser Whistling Duck, Graylag Goose, Ruddy Shelduck, Gadwall, Eurasian Wigeon, Northern Pintail.

### About [Chhari Dhand Wetland Reserve](#):

- A seasonal desert wetland conservation reserve, ecologically unique due to its saline and freshwater interactions.

### Located in:

- Kutch district, Gujarat
- Situated between [Banni grasslands](#) and the Rann of Kutch
- **Local meaning:** Chhari = salty, Dhand = shallow wetland

### Key features:

- Becomes swampy during **monsoon**, fed by north-flowing rivers and surrounding hills.
- Major habitat for **migratory birds** during monsoon and winter.
- **Endangered species:** Dalmatian Pelican, Oriental Darter, Black-necked Stork, Indian Skimmer.
- Large congregations of flamingos, cranes, [painted storks](#), raptors, spoonbills.
- Also supports chinkara, wolves, caracal, desert cats, desert foxes.

## TURTLE TRAILS



- 'Turtle trails' refer to regulated eco-tourism pathways and guided experiences near sea turtle nesting beaches, aimed at [promoting conservation awareness](#), community livelihoods and nature-based tourism.

### States involved:

- Odisha – [Rushikulya](#) (Ganjam) and vicinity of Gahirmatha (Kendrapara)

- Karnataka – Coastal turtle nesting beaches
- Kerala – Key nesting stretches along the Arabian Sea coast

### Key features:

- **Guided and regulated access** to turtle nesting areas, usually during breeding season
- **Public awareness and education** on marine biodiversity and conservation
- **Community participation**, involving local fishers, volunteers and NGOs
- **Low-impact infrastructure**, potentially temporary walkways or observation zones (as proposed)
- **Integration with [eco-tourism policy](#)**, aligned with livelihood generation and sustainable tourism goals

## MANGROVE CLAM (GELOINA EROSA)

The ICAR–Central Marine Fisheries Research Institute ([CMFRI](#)) has successfully achieved induced breeding of the mangrove clam under captive conditions, a rare global scientific feat.



### About [Mangrove clam \(Geloina erosa\)](#):

- An ecologically important bivalve (mud/mangrove clam) found in mangrove and estuarine ecosystems of South and [Southeast Asia](#); locally called "Kandal Kakka" in northern Kerala.

### Scientific name:

- *Geloina erosa* (also placed under the genus *Polymesoda* in some literature).

### Habitat:

- Organic-rich muddy substrates of intertidal mangrove zones.
- Tolerates a wide salinity range (brackish to near-freshwater).
- Deep-burrowing, [semi-infaunal species](#); adults often landward, juveniles more tide-independent.

### Key characteristics:

- **Large-sized mud clam:** One of the world's largest mangrove clams, reaching ~10 cm shell width, making it valuable both ecologically and as a food resource.



- **Efficient filter feeder:** Filters suspended particles and [plankton from water](#), recycling nutrients and improving estuarine water quality.
- **Distinct gonadal identification:** Sexes are identified by gonad colour and structure, not external organs, aiding reproductive studies and broodstock selection.
- **Ecosystem stabiliser:** Burrowing behaviour stabilises sediments, enhances nutrient cycling and strengthens overall mangrove ecosystem resilience.

#### Method used to restore / conserve:

- **Induced breeding in hatchery:** CMFRI achieved controlled spawning under captive conditions, overcoming dependence on wild seed collection.
- **Complete life-cycle closure:** Successful rearing from [embryo to larva](#) to spat (from ~18th day) proves hatchery-scale feasibility.
- **Hatchery seed production for multiple uses:**
  - Grow-out farming:** Enables estuarine aquaculture with minimal external inputs.
  - Mangrove ranching:** Seeds can be released into degraded mangroves to restore natural populations.
  - Stock enhancement:** Reduces harvesting pressure on wild clam beds by replenishing natural stocks.

### GOVT FORMS EXPERT GROUPS TO UPGRADE PROJECT TIGER SCHEME

The Union Government has constituted four expert working groups to review and modernise 50 years of policy decisions under [Project Tiger](#), as the programme completes its golden jubilee.

- Project Tiger is a Centrally Sponsored Scheme of the Government of India focused on conserving the tiger and its habitats through a network of protected tiger reserves using a [core–buffer strategy](#).

#### Launched in:

- **1973** (one of the world's earliest large-scale species conservation programmes)

#### Organisations involved:

- Ministry of Environment, Forest and Climate Change (MoEFCC) – Nodal ministry
- National Tiger Conservation Authority ([NTCA](#)) – Statutory body overseeing implementation

#### Aim:

- To ensure long-term survival of viable tiger populations in natural habitats.
- To conserve biodiversity and ecological integrity while balancing people-oriented development in

buffer areas.

#### Key features of the scheme:

- **Tiger Reserves Network:** Expanded from **9 reserves (1973)** to **51 reserves** across **18 tiger-range states**, covering ~**2.23%** of India's geographical area
- **Core–Buffer Strategy:**
  - **Core:** Inviolate areas with legal status of National Park/Sanctuary
  - **Buffer:** Multiple-use landscapes promoting coexistence and livelihoods
- **Statutory Backing:** NTCA functions under the [Wildlife Protection Act, 1972](#)
- **Financial Support:** Central assistance for habitat management, protection, monitoring, and community development
- **Scientific Monitoring:** Periodic **All-India Tiger Estimation** using camera traps, landscape ecology, and prey-base assessments

### DAL LAKE



- Dal Lake is a famous [urban freshwater lake](#) and wetland ecosystem, often called the Jewel in the crown of Kashmir. It is central to tourism, fisheries, floating agriculture, and the cultural identity of Kashmir.

#### Location:

- Located in **Srinagar**, the summer capital of Jammu and Kashmir.
- Surrounded by Mughal gardens such as Shalimar Bagh and Nishat Bagh.

#### Origin & Formation:

- Dal Lake is part of the [Kashmir Valley lacustrine \(lake-formed\) system](#), created by tectonic and glacial processes that shaped the Himalayan basin. It forms part of a larger natural wetland complex.

#### River Connection:

- Hydrologically connected to the **Jhelum River system**.
- Receives water from inflowing streams and

drainage channels from surrounding catchments and drains into the Jhelum through controlled outflows.

**Key Features:**

- Covers about **18 sq km**, forming part of a larger **21 sq km wetland system**.
- Divided into four basins: **Gagribal, Lokut Dal, Bod Dal, and Nagin**.
- Known for **floating gardens (Raad)** used for vegetable cultivation.
- Famous for **houseboats and shikaras**, supporting tourism and livelihoods.
- Experiences freezing during severe winters.
- Ecologically important but faces pressures from [sewage discharge](#), encroachment, eutrophication, and declining water circulation.

**BEAT THE HEAT PROGRAMME**

- Beat the Heat is a global climate initiative aimed at accelerating action against extreme heat through sustainable cooling and urban resilience strategies.
- It translates the [Global Cooling Pledge](#) into practical, city-level implementation.

**Launched In:**

- Announced under the [COP30 Presidency \(Brazil\)](#).
- Supported by the United Nations Environment Programme (UNEP) through the Cool Coalition.

**Aim:**

- Cut greenhouse gas emissions from cooling systems while promoting energy-efficient and low-carbon technologies.
- Protect vulnerable populations by integrating affordable cooling and climate-adaptive measures into [city planning](#).

**Key Features:**

- **Heat Risk Assessment:** Supports cities in mapping heat vulnerability and identifying hotspots.
- **Nature-Based Cooling:** Promotes urban greening, tree cover expansion, and shaded public spaces.
- **Passive Cooling Solutions:** Encourages cool roofs, reflective surfaces, and climate-sensitive building design.
- **Efficient Cooling Technologies:** Pushes for [low-energy air-conditioning](#) and low global-warming refrigerants.
- **Policy Integration:** Integrates heat resilience into urban planning, building codes, and infrastructure design.
- **Financial and Institutional Support:** Assists cities in accessing climate finance and strengthening implementation capacity.

**CONGO LAKES (MAI NDOMBE AND TUMBA)**

A recent scientific study has found that lakes ([Mai Ndombe](#) and [Tumba](#)) in the Congo Basin are releasing ancient carbon stored for thousands of years in surrounding peatlands, raising fresh climate concerns.

**About Congo Lakes (Mai Ndombe and Tumba):**

- Lakes **Mai Ndombe** and **Tumba** are large, shallow **blackwater lakes** characterized by their dark, tea-like color. This unique appearance is caused by high concentrations of dissolved organic matter and [humic acids](#) leached from the surrounding dense swamp forests and peatlands.

**Location:**

- **Country:** Democratic Republic of Congo (DRC).
- **Region:** Situated within the **Cuvette Centrale** (Central Basin), a vast depression in the heart of the Congo Basin.
- **Wetland Status:** They form part of the **Tumba-Ngiri-Maindombe** area, the world's largest Wetland of International Importance recognized by the [Ramsar Convention](#).

**Formation:**

- These lakes are primarily floodplain and wetland-origin lakes, formed through riverine processes associated with the [Congo River system](#).
- Continuous waterlogging led to the accumulation of organic plant material over thousands of years, forming deep peat deposits around the lakes.
- Peat formation occurs when [dead vegetation accumulates](#) faster than decomposition under oxygen-poor conditions.

**Key Features:**

- **Shallow Depth:** Both lakes are extremely shallow, with average depths of only 3 to 5 meters.
- **Blackwater Ecosystem:** The high acidity (pH 4.0–5.5) and low oxygen levels in the surrounding flooded forests create a unique habitat for endemic fish species.
- **Size Fluctuation:** The lakes are highly dynamic, with Lake Mai Ndombe doubling or tripling in size during the rainy season.
- **Carbon Reservoir:** The surrounding [peatlands](#) cover only 0.3% of Earth's land surface but hold one-third of all tropical peatland carbon (approx. 30 billion metric tons).

## INTERNATIONAL CLIMATE INITIATIVE (IKI)

Germany and India have launched a new €20 million (approx. ₹180 crore) Large Grant project under the International Climate Initiative (IKI) to strengthen India's climate resilience.

- The International Climate Initiative (IKI) is a key financial instrument of the German government that funds [international projects](#) focused on climate change mitigation, adaptation, and biodiversity conservation in developing and emerging economies.
- **Established In:** It was established in **2008**.
- **Nations Involved:**
- **Donor:** Germany.
- **Partner Nations:** Over **150 countries**, with 14 designated **Priority Countries** including India, Brazil, China, South Africa, Mexico, and Indonesia.

### Aim:

- The IKI aims to support partner countries in implementing and ambitiously developing their [Nationally Determined Contributions](#) (NDCs) under the Paris Agreement and achieving goals set by the Convention on Biological Diversity (CBD).

### Key Features:

- **Thematic Diversity:** Focuses on four main areas: Mitigating greenhouse gas emissions, Adapting to the impacts of climate change, Conserving natural carbon sinks ([REDD+](#)), and Protecting biological diversity.
- **Consortium-Based Funding:** Projects are typically implemented by a mix of NGOs, research institutes, international organizations (like GIZ), and the private sector to ensure multidisciplinary expertise.
- **Ecosystem-Based Adaptation (EbA):** A core feature is using nature (e.g., forest restoration, wetland protection) to help human communities adapt to climate risks like floods and heatwaves.
- **Innovative Financing:** Promotes high-risk/high-reward financial mechanisms like **blended finance**, biodiversity credits, and climate insurance to mobilize private capital.
- **Monitoring & Learning:** Requires rigorous Monitoring, Evaluation, and Learning (MEL) frameworks to ensure that local successes can be scaled up to national or global policies.

### About The New India-Germany Project:

The newly announced €20 million project specifically targets high-risk Indian ecosystems:

- **Priority Regions:** The Himalayas, Western Ghats, North-East India, Island regions, and the Lower Gangetic floodplains.
- **Focus Areas:** Forest restoration, [groundwater recharge](#), flood control, and creating biodiversity corridors to allow species to move safely as temperatures rise.

### Topic: Species & Plant in News

## PEACOCKS

- A pair of peacocks were spotted at an altitude of over 6,000 ft near Manali in Himachal Pradesh, an unusual sight for a species typically found in [warmer lowland regions](#).



### About Peacocks:

- Peacocks are large, colourful birds of the pheasant family (Phasianidae), collectively called peafowl—the male is a peacock, the female a peahen, and the young are peachicks.
- India's national bird is the **Indian or Blue Peacock** (*Pavo cristatus*).

### Habitat and distribution:

- Naturally found in **warm, semi-arid to moist deciduous forests**, grasslands, and agricultural landscapes.
- In India, usually inhabit **plains and low hills**.
- **Typical altitude:** up to **~1,000 m**, occasionally **1,500 m**.
- Recent sightings at **~1,800 m (6,000 ft)** in Himachal Pradesh are **ecologically unusual**.

### IUCN conservation status:

- Indian (Blue) Peacock – Least Concern
- Green (Javanese) Peacock – Endangered
- Congo Peacock – Vulnerable

### Characteristics:

#### Physical:

- Males possess a **long iridescent train** with eye-shaped spots used in courtship displays.



- Strong legs, short rounded wings; capable of short flights and roosting on trees.

#### **Social and behavioural:**

- Generally **ground-dwelling** but roost in trees at night.
- Males form **harems** during the breeding season.
- **Omnivorous:** feed on **seeds, insects, small reptiles**, aiding [pest control](#).

#### **Other ecological traits:**

- Sensitive to **temperature and habitat changes**.
- Known for adaptability, but **extreme altitudinal shifts are rare**.

#### **Implications of high-altitude sightings:**

- **Indicator of climate change:** Warming temperatures are making higher altitudes more habitable.
- **Ecosystem stress signal:** Suggests shifts in species distribution in the Himalayas.
- **Human–wildlife interaction risks:** New habitats may increase conflict and competition.

### **LYRIOTHEMIS KERALENSIS**

Researchers have discovered and formally described a new [dragonfly species](#) from Kerala, named *Lyriothemis keralensis*, highlighting the State's rich but underexplored biodiversity.



#### **About *Lyriothemis keralensis*:**

- A newly described species of dragonfly (Order: Odonata) endemic to Kerala, scientifically identified and named to reflect the State's biodiversity.

#### **Habitat:**

- Found in **vegetated pools and irrigation canals**.
- Occurs within **shaded rubber and pineapple plantations**.
- Recorded from **Varapetty near Kothamangalam, Ernakulam district**.

- Adults are visible **only during the Southwest monsoon (May–August)**; rest of the year spent as aquatic larvae.

#### **Key characters:**

- **Small-sized dragonfly** with marked [sexual dimorphism](#).
- **Males:** Bright **blood-red** with black markings, slender abdomen.
- **Females:** **Yellow with black markings**, bulkier body.
- Distinguished from *Lyriothemis acigastra* by **microscopic traits**, including slender abdomen, uniquely shaped anal appendages and genitalia.

### **BEAR**

Low snowfall in Uttarakhand has disrupted bear hibernation cycles, leading to increased bear attacks and [human-wildlife conflict](#) in early 2026.

- A bear is a large, short-tailed omnivorous mammal belonging to the family **Ursidae**.

#### **Scientific Name:**

- **Family:** Ursidae
- **Ursus arctos** (Brown Bear), **Ursus maritimus** (Polar Bear)

#### **Habitat:**

- Bears are found across North America, Europe, and Asia, primarily in temperate and Arctic regions; in India, species like the [Sloth Bear](#) inhabit forested and hilly landscapes.

#### **Key Characteristics:**

- **Large Omnivorous Mammals:** Most species consume fruits, roots, insects, fish, and small mammals; the polar bear is largely carnivorous.
- **Strong Climbers & Swimmers:** Despite bulk, many species climb trees and swim efficiently.
- **Keen Sense of Smell:** Olfactory ability is highly developed, compensating for moderate vision and hearing.
- **Solitary Behaviour:** Mostly solitary except during mating or cub rearing.
- **Seasonal Dormancy:** Many species undergo winter sleep (often termed hibernation).

#### **About Hibernation:**

##### **What it is?**

- Hibernation is a seasonal state of metabolic slowdown where animals conserve energy during cold periods with [scarce food availability](#).

##### **Reasons for Hibernation:**

- **Food Scarcity in Winter:** Reduced plant growth and prey availability.

- **Extreme Cold Conditions:** Conserves body heat and energy.
- **Survival Strategy:** Ensures energy efficiency during harsh environmental conditions.

#### How it Works?

- **Pre-hibernation Fat Accumulation:** Bears consume excess food to build fat reserves.
- **Metabolic Slowdown:** Heart rate, breathing rate, and body temperature reduce (though bears are not true deep hibernators).
- **Energy from Stored Fat:** Brown fat helps generate heat during brief arousals.
- **Occasional Arousal:** Bears may wake intermittently, unlike true hibernators.

In Uttarakhand, insufficient snowfall has failed to trigger proper dormancy, leaving bears active and increasing human-bear interactions.

## DOLPHIN CENSUS IN ODISHA

Odisha has recorded its highest [marine dolphin population](#) in five years, with 765 individuals counted in the 2026 state-wide census.

#### About Dolphin Census in Odisha:

- The Dolphin Census is an annual scientific population estimation exercise to assess the abundance, distribution, and diversity of dolphins and other cetaceans in Odisha's marine and [estuarine ecosystems](#).

#### Conducted By:

- It is conducted by the Wildlife Wing of the Forest, Environment and Climate Change Department, Government of Odisha, involving forest officials, marine experts, and field personnel using boat- and shore-based transect surveys.

#### About Dolphin:

- Dolphins are aquatic marine mammals belonging to the order Cetacea and are known for their intelligence, social behavior, and echolocation abilities.

#### Habitat:

- They inhabit oceans, coastal waters, estuaries, and some freshwater systems. In India, major habitats include Chilika Lake and coastal marine zones.

#### IUCN Status:

- The **Irrawaddy Dolphin** is classified as **Endangered** on the IUCN Red List.
- Dolphins are protected under **Schedule I** of the [Wildlife Protection Act, 1972](#).

#### Types Found in Odisha:

1. Humpback Dolphin

2. Irrawaddy Dolphin
3. Bottlenose Dolphin
4. Spinner Dolphin
5. Finless Porpoise (closely related cetacean)

#### Key Characteristics of Dolphins:

- **Highly intelligent with advanced communication systems:** Dolphins exhibit problem-solving skills and communicate using clicks, whistles, and body movements.
- **Use echolocation for navigation and hunting:** They emit sound waves that bounce off objects, helping them detect prey and obstacles underwater.
- **Social animals living in pods:** Dolphins live in structured groups (pods) for protection, hunting cooperation, and social bonding.
- **Slow breeding rate (especially Irrawaddy dolphins):** They have long gestation periods and produce few offspring, limiting rapid population growth.
- **Indicators of marine ecosystem health:** Their presence reflects healthy water quality, fish abundance, and balanced coastal ecosystems.

## ARMY ANT SPECIES

Scientists from Karnataka and Odisha have discovered two new army ant species in the [Eastern Ghats of Andhra Pradesh](#).



- Army ants are highly social, nomadic predatory ants known for their coordinated mass raids. The newly identified species — *Aenictus chittoorensis* and *Aenictus lankamallensis* — were discovered in the [Sri Venkateswara Wildlife Sanctuary](#) in the Southern Eastern Ghats.

#### Habitat:

- Primarily found in [tropical forest ecosystems](#).
- Thrive in dense forest floors with high arthropod diversity.
- Do not build permanent nests; instead form temporary living structures called **bivouacs** made

from their own bodies.

### Key Characteristics

- **Highly aggressive and coordinated predators:** They hunt in large, synchronized groups that can overpower much bigger prey.
- **Large, sharp mandibles and stinging ability:** Strong jaws and stings help them capture, kill, and defend against threats.
- **Completely blind; rely on pheromones:** They use [chemical trails](#) to communicate and navigate instead of vision.
- **Move in straight columns during raids:** They march in organized lines to efficiently sweep and hunt across the forest floor.
- **Form massive colonies:** Thousands of ants act together as one unit while foraging and relocating.

### Significance:

- Act as keystone predators, regulating arthropod populations.
- Support ecological balance by driving insects out, making them [prey for birds](#) and reptiles.
- Contribute to nutrient cycling and forest biodiversity.

## LOGGERHEAD TURTLES



- The **Loggerhead sea turtle** is a large marine turtle known for its massive head and strong jaws. It is one of the most widely distributed sea turtles in temperate and subtropical oceans.

**Scientific Name:** *Caretta caretta*

### Habitat:

- **Global Distribution** – Found in the Atlantic, Pacific, and [Indian Oceans](#), and the Mediterranean Sea.
- **Nesting Beaches** – Prefers sandy, high-energy, sloped beaches for egg-laying.
- **Major Nesting Sites** – Florida (USA), Oman, Cabo Verde, Japan, and Australia.

**IUCN Status:** Vulnerable ([IUCN Red List](#))

### Key Characteristics:

#### Physical Features:

- **Large Head & Powerful Jaws** – Crush hard-shelled prey like mollusks and crustaceans.
- **Reddish-Brown Carapace** – Slightly heart-shaped upper shell.
- **Long Lifespan** – Can live 70–80 years or more.
- Reproductive Traits:
  - **Delayed Maturity** – Females mature around 30–35 years of age.
  - **Temperature-Dependent Sex Determination** – Warmer sand produces more females.
- Behavioural Patterns:
  - **Long-Distance Migration** – Some undertake trans-oceanic journeys of thousands of kilometres.
  - **Natal Homing** – Females return to the same region where they hatched.
  - **Capital Breeders** – Store energy over years before reproducing.
- Ecological Role:
  - **Marine Food Web Regulator** – Controls populations of bottom-dwelling invertebrates.
  - **Indicator Species** – Reflects ocean health and climate impacts.
  - **Beach Nutrient Cycling** – Egg remnants enrich coastal ecosystems.

## GALAPAGOS GIANT TORTOISE REINTRODUCTION

The [Galapagos National Park](#) released 158 hybrid giant tortoises onto **Floreana Island** to help restore its delicate ecosystem.

- These juvenile tortoises carry significant DNA from the *Chelonoidis niger*, a species that had been considered extinct for 150 years.





### About The Galapagos Islands:

- The Galapagos is an extraordinary volcanic archipelago consisting of 19 main islands and over 100 islets.

### Location:

- Situated in the Pacific Ocean, approximately 1,000 km (620 miles) west of the Ecuadorian coast.

### Geological Origins:

- Formed by [volcanic activity](#) at the confluence of three tectonic plates (**Nazca, Cocos, and Pacific**).

### Key Features:

- Evolutionary Hub:** Known as a living museum of evolution, it famously inspired **Charles Darwin's** theory of natural selection.
- Endemic Species:** Home to species found nowhere else, including marine iguanas, blue-footed boobies, and the iconic giant tortoises.
- UNESCO Heritage:** Designated as a [Natural World Heritage Site](#) in 1978 for its unique biodiversity.

### About Hybrid Giant Tortoises:

- These are tortoises with mixed ancestry. While the original Floreana species (*Chelonoidis niger*) was hunted to extinction by the mid-19th century, scientists discovered hybrid descendants on **Wolf Volcano** (Isabela Island).

### IUCN Status:

- Extinct in the Wild (Parent Species):** The pure *Chelonoidis niger* is technically extinct.
- Conservation Status:** The genus *Chelonoidis* as a whole is generally classified as **Endangered** or **Critically Endangered** by the [IUCN](#).
- Conservation Goal:** Through a selective breeding program, these hybrids (carrying 40%–80% original DNA) are being used to resuscitate the ecological role of the extinct species.

### Key Characteristics:

- Ecological Engineers:** They shape the landscape by dispersing seeds, clearing vegetation, and creating wallows that serve as micro-habitats for other animals.
- Saddlebacked Carapace:** Unlike the dome-shaped shells of some other islands, Floreana-descended tortoises often have **saddlebacked shells**, an adaptation that allows them to lift their necks higher to reach tall cacti.
- Longevity & Resilience:** Released between ages 8 and 13, these tortoises are large enough to survive potential threats from introduced rats and cats. They can live for over **100 years**.

## MACAQUES



- Macaques are a genus of **gregarious Old World monkeys** belonging to the family *Cercopithecidae*. There are more than 20 species, making them one of the most widespread and diverse primate groups.

### Types of Macaques:

- Japanese Macaque (*M. fuscata*):** Also known as Snow Monkeys, famous for living in cold climates and soaking in hot springs.
- Rhesus Macaque (*M. mulatta*):** Native to North India and [Southeast Asia](#); widely known in medical research.
- Lion-tailed Macaque (*M. silenus*):** An endangered species found only in the Western Ghats of Southern India, recognized by its silver mane.
- Crested Black Macaque (*M. nigra*):** A distinctive species found on the Indonesian island of Sulawesi.
- Barbary Macaque (*M. sylvanus*):** The only species found outside of Asia, inhabiting North Africa (and Gibraltar).

### Origin and Habitat:

- While most species are native to **Asia**, their habitats vary significantly. They are highly adaptable and can be found in tropical rainforests, [high-altitude mountains](#), rocky cliffs, and even urban plains.

### Key Characteristics:

- Physical Build:** They are robust primates with arms and legs of roughly equal length.
- Facial Features:** They possess dog-like but rounded muzzles with nostrils on the upper surface.
- Cheek Pouches:** Unique large pouches allow them to carry and store extra food while foraging.
- Tails:** Variation is extreme; depending on the species, tails can be long, short, or even absent.
- Size:** Males are significantly larger than females, with the Tibetan macaque being the largest (up to 18 kg).

### Social Behaviour:

- **Complex Hierarchies:** Macaques live in troops with a strict **dominance rank order**.
  - In females, this is matrilineal (based on the mother's rank), while male rank is often determined by alliances and fighting ability.
- **The Youngest Sister Rule:** In some species, like the Japanese macaque, the youngest sister often ranks higher than her older siblings.
- **Integration and Bullying:** High-ranked members often show aggression toward lower-ranked individuals to reinforce status.
  - New or rejected members (like Punch) undergo a testing phase before finding their place.
- **Emotional Regulation:** Unlike herd animals (goats/deer), young macaques rely heavily on **physical touch** from their mothers for comfort.
  - This explains why an abandoned baby may cling to a surrogate object, like a stuffed toy, for [emotional stability](#).

[Topic: Defence, ISRO & Navy](#)

## DORNIER 228 AIRCRAFT



- The Dornier 228 is a twin-engine, turboprop Short Take-Off and Landing (STOL) utility aircraft designed for passenger, cargo, and special mission operations. It is widely used for maritime patrol, [border surveillance](#), search and rescue, and transport roles.

### Manufactured By:

- Originally developed by **Dornier GmbH (Germany)** in the early 1980s.
- License-produced and currently manufactured in India by Hindustan Aeronautics Limited at its Kanpur facility.

### Key Features:

- Designed as a light commuter aircraft, it can carry 15–19 passengers or equivalent cargo loads efficiently.

- **Engine:** Twin Garrett TPE331 turboprop engines. Powered by two reliable turboprop engines known for durability, fuel efficiency, and high performance in varied climates.
- **STOL Capability:** Operates from short and unpaved runways. Its special wing design allows take-off and landing on short, rough airstrips, ideal for remote and [coastal regions](#).
- **High Endurance:** The standard Dornier 228 manufactured by HAL typically has a maximum endurance of approximately **5.5 to 6 hours**.
- **Maritime Equipment:** 360° surveillance radar, infrared sensors, real-time datalink. Equipped with specialized sensors for coastal surveillance, detecting vessels, and transmitting live data to ground stations.

[Topic: Internal Security & its Challenges](#)

## MILITARY EXERCISES IN NEWS

### About Exercise Khanjar:

- **Host / Location:** India | Missamari, Assam
- **Participants:**
  - Indian Army – Parachute Regiment (Special Forces)
  - Kyrgyzstan – ILBRIS Special Forces Brigade
- **Aim:** To enhance interoperability and operational synergy in counter-terrorism and special forces operations.
- **Key features:**
  - Annual bilateral exercise held alternately in [India and Kyrgyzstan](#) (13th edition).
  - Focus on urban warfare, mountain warfare, sniping, and complex building intervention.
  - Strengthens India–Central Asia defence cooperation and counter-terrorism collaboration.

### About Exercise Agni Pariksha:

- **Host / Location:** India | Sigar, Arunachal Pradesh
- **Participants:**
  - Indian Army
  - Indo-Tibetan Border Police ([ITBP](#))
- **Aim:** To familiarise non-artillery personnel with artillery procedures and improve inter-force combat integration.

### Key features:

- First-of-its-kind joint training exposing infantry and ITBP to **artillery firing missions**.
- Breaks traditional role silos and enhances **fire-**

### power coordination and joint operational trust.

- Focused on future battlefield requirements and integrated combat capability.

### About Exercise Vayu Shakti-2026:

- **Host / Location:** India | Pokaran Field Firing Range, Rajasthan
- **Participants:**
  - Indian Air Force (IAF) units from across the country

### Aim:

- To demonstrate **IAF's air-power capability**, operational strategy, and precision strike readiness.

### Key features:

- Largest day-to-night air-power exercise of the IAF.
- Participation of 120+ aircraft and helicopters, including Rafale, [Su-30 MKI](#), Tejas, Mirage-2000, Apache, Chinook, and Dhruv.
- Execution of Offensive & Defensive Counter Air, SEAD, [precision strikes](#), airborne command (AWACS), and Para-commando air-drops.
- Showcase of indigenous systems such as Akash air defence.

### About India–Thailand Joint Air Combat Exercise:

- A **bilateral in-situ air combat exercise** conducted in the Indian Ocean Region to deepen defence cooperation under [India's Act East Policy](#).
- **Host & Location:**
  - Conducted in the [Indian Ocean Region \(IOR\)](#)
  - IAF operated from **Andaman & Nicobar Islands**
- Nations Involved:
  - Indian Air Force (IAF)
  - Royal Thai Air Force (RTAF)
- Key Features:
  - IAF **Su-30MKI** vs Thai **SAAB Gripen** fighters.
  - Supported by **IL-78 mid-air refuelling tankers**.
  - Surveillance via **IAF AWACS** and Thai GCI units.
  - Focus on maritime air combat & long-range reach.
  - Boosts Indo-Pacific security and operational synergy

### About Exercise Vayushakti-26:

- A biennial firepower demonstration exercise by the Indian Air Force to display combat capability, precision strike ability, and multi-domain dominance.
- **Host & Location:**
  - Hosted by the Indian Air Force.
  - Conducted at [Pokhran Air-to-Ground Range](#), Jaisalmer (Rajasthan).
- **Nation Involved:** Solely conducted by the Indian Air Force (India).
- **Key Features:**
  - **Participation of platforms:** Rafale, Tejas, Su-30MKI, Mirage-2000, Jaguar, MiG-29, C-130J, C-17, Apache, Chinook, LCH, RPAs.
  - **Advanced weapons:** Akash, SpyDer, SRLM, CUAS.
  - Demonstrates **day, dusk & night strike missions**.
  - Showcases indigenous capability under **Aatmanirbhar Bharat**.
  - Highlights lessons from [Operation Sindoor](#).
  - **Core values:** Ahook (Accurate), Abhedya (Impenetrable), Sateek (Precise).

### About Exercise DHARMA GUARDIAN:

- A bilateral annual military exercise between the Indian Army and the Japan Ground Self-Defense Force (JGSDF).
- The exercise focuses on joint operations in semi-urban environments and improving coordinated military responses.
- **Host:** Foreign Training Node, Chaubattia, Uttarakhand (India)
- **Nations involved:** India and Japan

### Key features:

- Focus on joint tactical drills including [cordon-and-search operations](#) and house intervention.
- Emphasis on modern technology and interoperability, including ISR (Intelligence, Surveillance & Reconnaissance) grid development.
- Training includes heliborne operations and temporary operating base establishment for realistic combat simulation.

### About Exercise VAJRA PRAHAR:

- A joint Special Forces exercise between India and the United States aimed at enhancing cooperation in special operations.



- Designed to improve [joint operational capability](#) in mountainous terrain.
- Host:** Special Forces Training School, Bakloh, Himachal Pradesh (India)
- Nations involved:** India and United States (US Green Berets)

#### Key features:

- Focus on special operations tactics, techniques, and procedures (TTPs).
- Intensive joint mission planning and physical conditioning.
- Enhances interoperability, mutual trust, and [professional exchange](#) between Special Forces.

### MILAN 2026 NAVAL EXERCISE

- MILAN** is a **multilateral naval exercise** initiated by the Indian Navy to enhance maritime cooperation, interoperability and trust among friendly navies in the Indo-[Pacific region](#).
- It serves as a platform for joint training, operational coordination and strategic dialogue on maritime security challenges.

**Host:** Hosted by the **Indian Navy**.

- MILAN 2026** is being conducted at **Visakhapatnam, Andhra Pradesh**.

#### Nations Involved:

- Largest-ever edition with participation from **74 nations**.
- Includes **nine [ASEAN member states](#)**, reflecting strong regional engagement.

#### Aim:

- To strengthen maritime cooperation and collective security in the Indo-Pacific region.
- To improve interoperability and operational coordination among participating navies through joint drills and exchanges.

#### Key Features:

- Theme:** "Camaraderie, Cooperation and Collaboration."
- Includes **Sea Phase exercises** such as:
  - [Anti-Submarine Warfare](#) (ASW)
  - Air Defence drills
  - Search and Rescue (SAR) operations
- Conduct of **International Fleet Review 2026** and **IONS Conclave of Chiefs**.
- Platform for defence diplomacy and strategic engagement under India's **Act East Policy** and [MAHASAGAR vision](#).
- Showcases India's indigenous naval capabilities

such as **INS Vikrant** and Visakhapatnam-class destroyers.

### AGNI-3 INTERMEDIATE-RANGE BALLISTIC MISSILE



- Agni-3 is an Intermediate-Range Ballistic Missile (IRBM) capable of delivering strategic payloads to targets up to 3,000 km. It forms a crucial leg of India's land-based nuclear deterrent under the [Agni missile series](#).

#### Developed by

- Defence Research and Development Organisation (DRDO)**
- Operationally deployed under the **Strategic Forces Command**

#### Aim

- To ensure **credible minimum deterrence** against long-range adversarial threats.
- To provide India with a **reliable second-strike capability**.
- To strengthen strategic depth beyond short- and [medium-range missile](#) systems.

#### Key features

- Range:** ~3,000 km.
- Type:** Intermediate-Range Ballistic Missile.
- Launch platform:** Road-mobile launcher (canisterised variants tested earlier).
- Payload capability:** Conventional or nuclear warhead.
- Guidance:** Advanced inertial navigation with high accuracy.
- Propulsion:** Two-stage solid-fuel missile.
- Operational validation:** All technical and operational parameters successfully validated in the 2026 test.

### LCH PRACHAND

- Prachand is India's first indigenously designed and developed dedicated combat helicopter, built specifically for offensive roles in high-altitude and

battlefield environments.

- It is a multi-role light attack helicopter capable of air-to-air and [air-to-ground missions](#), including anti-armour and counter-insurgency operations.

#### Developed by:

- Designed and manufactured by **Hindustan Aeronautics Limited (HAL)** under the Light Combat Helicopter (LCH) programme.

#### Aim:

- The primary objective of the Prachand project is to provide a **high-altitude combat platform** capable of operating in the unique terrain of the [Himalayas](#) (Siachen Glacier and Leh), where heavier attack helicopters often struggle due to thin air.

#### Key Features:

- **Performance:** It can operate at altitudes exceeding **5,000 meters (16,400 ft)**, the highest for any attack helicopter in the world.
- **Speed & Range:** Boasts a maximum speed of **268 kmph** and a combat range of **550 km**.
- **Stealth & Survivability:** Features a low Radar Cross Section (RCS), crashworthy structure, and an [Infrared \(IR\) Suppressor](#) to reduce its heat signature.
- **Weaponry:** Equipped with a **20mm turret gun**, 70mm rockets, Air-to-Air missiles (ATAM), and Anti-Tank Guided Missiles (ATGM).
- **Advanced Tech:** Includes a modern glass cockpit, Helmet-Mounted Display (HMD), and an [Electronic Warfare](#) (EW) suite.

as a doctrine-level framework for preventing, responding to and recovering from terrorism threats.

- It adopts a **whole-of-government and [whole-of-society approach](#)**, combining intelligence, law enforcement, technology and international cooperation.

#### Launched by:

- Introduced by the **Government of India** as a national counter-terror strategy framework involving central and state security agencies.

#### Aim:

- To prevent and neutralize terrorism through proactive, intelligence-guided operations and coordinated institutional mechanisms.
- To disrupt [terror ecosystems](#) by targeting financing, recruitment, radicalisation, logistics and cyber networks.

#### Key Features:

- **Seven-Pillar Framework (PRAHAAR):** Prevention, Response, Aggregation of capacities, Human-rights-based processes, Attenuation of radicalisation, Aligning international cooperation, and Recovery.
- **Proactive Intelligence Model:** Focus on pre-emptive disruption of terror networks instead of reactive policing.
- **Technology-Centric Security:** Addresses threats from drones, encrypted messaging apps, dark web and crypto financing.
- **Uniform Counter-Terror Structure:** Standard procedures and coordinated mechanisms across central, state and district levels.
- **Counter-Radicalisation Strategy:** Graded police response combined with education, engagement and de-radicalisation programmes.
- **Global Collaboration:** Emphasis on extradition, intelligence sharing and alignment with UN anti-terror norms.
- **Human Rights Safeguards:** Ensures legal due process, redressal mechanisms and rule-of-law based action.

Topic: Miscellaneous

## PRAHAAR ANTI TERROR POLICY

### The PRAHAAR Strategy: 7 Pillars for Global Security



- **Prahaar** is India's first integrated **national counter-terrorism policy and strategy**, designed

## SHALIMAR WHEAT

Scientists at Sher-e-Kashmir University of Agricultural Sciences and Technology (SKUAST-K) have developed two new early-maturing [wheat varieties](#) to support the rice-wheat cropping system in Kashmir.

### About Shalimar Wheat:

- **Shalimar Wheat** refers to newly developed **early-maturing wheat varieties** bred by SKUAST-K through conventional breeding techniques to suit Kashmir's climatic conditions.
- They are designed specifically to fit into the [rice–wheat cropping system](#), allowing timely wheat harvest before paddy transplantation.

### Variety Names:

- **Shalimar Wheat-4 (SW-4)** – matures by the last week of May
- **Shalimar Wheat-3 (SW-3)** – matures by the first week of June

### Key Features:

- **Early maturity:** Developed to mature earlier than traditional varieties, enabling timely field preparation for [rice cultivation](#).
- **Suitable for Kashmir climate:** Adapted to mid-altitude regions up to about **1,850 m**, unlike earlier varieties sourced from subtropical regions.
- **Rice–wheat rotation compatibility:** Ensures smooth crop sequencing by preventing delays in paddy transplantation.
- **Disease resistance:** Shows resistance to **yellow rust**, a major fungal disease affecting wheat in Kashmir.
- **High productivity potential:** SW-3 offers productivity up to **38 quintals per hectare**, balancing yield with early maturity.
- **Nutritional enhancement (Biofortification):** SW-3 contains higher **iron and zinc (>40 ppm)** and around **12% protein**, improving nutritional value.
- **Developed through conventional breeding:** Created using [cross-breeding](#), pedigree selection and multi-year field testing over nearly a decade.

## ENSION OF FORTIFIED RICE ROLLOUT

The Central Government has decided to temporarily discontinue the distribution of fortified rice under the [PMGKAY](#) and other welfare schemes due to shelf-life concerns.

- Fortified rice is regular rice that has been deliberately enriched with essential micronutrients—specifically **Iron, Folic Acid, and Vitamin B12**.
- This is done to improve the nutritional quality of the food supply and combat hidden hunger or micronutrient deficiencies among vulnerable populations.

### Associated Schemes:

The fortification initiative was integrated into India's largest social safety nets, including:

- **PMGKAY:** Pradhan Mantri Garib Kalyan Anna Yojana (Free foodgrain scheme).
- **TPDS:** Targeted Public Distribution System under the National Food Security Act (NFSA).
- **PM-POSHAN:** Formerly the [Mid-Day Meal Scheme](#) in schools.
- **ICDS:** Integrated Child Development Services (delivered through Anganwadi centres).

### Aim:

- The primary goal was to address **malnutrition and anemia** nationwide by 2024. By making fortified rice mandatory in government schemes, the government aimed to provide a cost-effective and scalable public health intervention to improve the health of women and children.

### Steps in Fortification:

1. **Production of Fortified Rice Kernels (FRK):** Rice flour is mixed with a premix of vitamins and minerals and then shaped into grain-like kernels using an extruder machine.
2. **Blending:** These FRKs are blended with regular polished rice, typically at a ratio of **1:100** (one fortified kernel for every 100 grains of regular rice).

### Key Features:

- **Government Funded:** The entire cost of fortification is borne by the Government of India as part of the food subsidy.
- **Phased Implementation:** The rollout began in 2021 with the goal of covering all districts in India by June 2024.
- **Technical Support:** Organizations like the [World Food Programme \(WFP\)](#) provide technical assistance and training to millers and officials.

## SALEM SAGO (JAVVARISI)





APEDA facilitated the first direct export consignment of [GI-tagged](#) Salem Sago from Tamil Nadu to Canada, marking a major milestone for producer-led exports.

- **Salem Sago (Javvarisi)** is a starch-based food product made from [tapioca roots \(cassava\)](#), where wet starch is processed into small pearl-like granules.
- It is widely consumed as a staple and is also used in multiple industries such as food processing, textiles, pharmaceuticals and construction.

#### Origin:

- Originates from **Salem district, Tamil Nadu**, known as the “Land of Sago”.
- The region has favourable conditions such as high tapioca yield, abundant sunshine and established processing units.
- Over **80% of India’s sago production** comes from the [Salem region](#) and nearby belts like Erode, Namakkal and Dharmapuri.

#### GI Tag:

- [Geographical Indication \(GI\)](#) tag granted in **March 2023**.
- GI registration obtained by The Salem Starch and Sago Manufacturers’ Service Industrial Co-operative Society Ltd (SAGOSERVE) under the Government of Tamil Nadu.

#### Key Characteristics:

- **Tapioca-based product:** Produced from starch extracted from tapioca roots containing about 30–35% starch.
- **High regional productivity:** Tapioca yield in Salem reaches 25–30 tonnes/ha, among the highest globally.
- **Distinct physical traits:** Small granules generally ranging between 2–4.5 mm in size with low calorific value (~310 kcal/100 g).
- **Efficient processing:** Around 1 kg of sago can be produced from about 5 kg of tapioca tubers.
- **Multi-sector usage:** Used in food, paper, textile, cosmetic, pharmaceutical, construction and alcohol industries.

# MAPPING WORLD:

## SEYCHELLES



- Seychelles is a sovereign island country and archipelagic state comprising 115 islands, known for its high human development, blue economy focus, and strategic maritime location in the Indian Ocean.

### Located in:

- Western Indian Ocean, about 1,500 km east of mainland Africa
- Neighbours:** Maldives, Mauritius, Madagascar, Comoros, Réunion (France)

**Capital:** Victoria (on Mahé Island)

### Key geographical features:

- Archipelagic composition:** 115 islands grouped into Inner (granitic) and Outer (coralline) islands.
- Granitic islands** (Mahé, Praslin, La Digue): among the oldest oceanic granite formations in the world.
- Coral atolls and reef systems:** support rich marine biodiversity and the blue economy.

## STRAIT OF HORMUZ



- The Strait of Hormuz is the world's most vital **oil transit chokepoint**. It is a narrow maritime passage that serves as the only sea exit for the Persian Gulf, linking it to the Gulf of Oman and the open ocean.

**Located In:** It is situated in the Middle East, separating the northern coast of Iran from the Arabian Peninsula.

- Links the Persian Gulf to the Indian Ocean through the Gulf of Oman.

### Neighbouring Nations:

- Iran:** Controls the northern coastline and several strategic islands (Qeshm, Hormuz, Larak).
- Oman:** Controls the southern coast via the Musandam Peninsula exclave.
- United Arab Emirates (UAE):** Located to the south and west; home to major ports like Fujairah.

### Key Geographical Features:

- Shipping Lanes:** Due to shallow waters near the coast, tankers must use two **3-km wide shipping lanes** (one inbound, one outbound) separated by a 2-km buffer zone.
- Strategic Islands:** Iran maintains a heavy military presence on islands like **Abu Musa** and the **Greater and Lesser Tunbs**, which allow for de facto control over the shipping channels.
- Bathymetry:** The water is deep enough (60–100 meters) to handle the world's largest **VLCCs (Very Large Crude Carriers)**, making it irreplaceable for bulk energy transport.

## THE DRUZHBA OIL PIPELINE



- The **Druzhba Pipeline** (translating to Friendship in English) is one of the world's longest and largest crude oil pipeline networks.
- Established during the Soviet era, it serves as the primary artery for transporting Russian and Kazakh oil to the industrial heartlands of Central and Eastern Europe.

### Located In:

- The network originates in **Almetyevsk, Tatarstan (Russia)**, where it collects oil from Western Siberia, the Urals, and the Caspian Sea.
- It spans approximately **4,000 kilometers**, crossing the Russia-Belarus border before splitting into two massive branches.

### States/Countries it Goes Through:

- **Russia:** The starting point and primary source.
- **Belarus:** The central transit hub at Mazyr where the line splits.
- **Northern Branch:** Travels through **Poland** to reach **Germany**.
- **Southern Branch:** Travels through **Ukraine** to supply **Hungary, Slovakia, and the Czech Republic**.

## MEXICO



- Mexico, officially called the **United Mexican States (Estados Unidos Mexicanos)**, is a federal republic located in **southern North America**.
- It is one of the largest economies in Latin America and has a diverse cultural, historical, and ecological landscape.

### Capital:

- Mexico City

### Bordering Nations:

- United States to the north.
- Guatemala and Belize to the southeast.
- Bounded by the **Pacific Ocean** (west & south) and the Gulf of Mexico and Caribbean Sea (east).

### Key Geological Features:

- Part of the **Pacific Ring of Fire**, making it highly prone to earthquakes and volcanic activity.
- Major mountain systems include Sierra Madre Occidental, Sierra Madre Oriental, and the Trans-Mexican Volcanic Belt (Cordillera Neo-Volcánica).
- Home to active and high volcanoes such as **Popocatepetl** and **Citlaltépetl (Pico de Orizaba)**.
- Diverse physiographic regions including the **Mexican Plateau**, Yucatán Peninsula, Baja California Peninsula, and coastal plains.
- Major rivers include **Río Bravo (Rio Grande)**, **Grijalva**, and **Balsas**.

## ISRAEL



- Israel is a resilient democracy located at the eastern end of the **Mediterranean Sea** in **West Asia** (the Levant).
- Established in 1948, it is the world's only Jewish-majority nation and is recognized as a global leader in high-tech innovation, water management, and defense systems.

### Capital:

- **Jerusalem** is the seat of the **Israeli government**, including the Knesset (Parliament) and the Supreme Court.

### Bordering Nations:

- Lebanon, Syria, Jordan, West Bank, Egypt, Gaza Strip, and Mediterranean Sea.

### Key Features:

- **Four Regions:** The country consists of the Mediterranean coastal plain, the central hill regions (Galilee, Samaria, Judea), the **Great Rift Valley**, and the **Negev Desert** in the south.
- **The Dead Sea:** Located in the Jordan Rift Valley, it is the **lowest point on Earth's surface** (approx. 430m below sea level) and is famous for its extreme salinity.
- **Sea of Galilee (Lake Tiberias):** Israel's largest fresh-water lake and a primary water source.
- **Tectonic Setting:** Israel sits on the **Dead Sea Transform** fault system, a boundary between the African and Arabian plates, leading to significant geological activity.
- **Water Scarcity:** Despite having no navigable rivers and chronic water shortages, Israel has pioneered **desalination** and drip irrigation technologies.



## THE CHAGOS ISLANDS



- The Chagos Islands, also known as the Chagos Archipelago, is a group of seven atolls comprising more than 60 individual tropical islands. It is officially administered as the [British Indian Ocean Territory \(BIOT\)](#), though its sovereignty is heavily contested.

### Located In:

- The archipelago is situated in the **Indian Ocean**, approximately 500 kilometers south of the Maldives.
- It sits atop the [Chagos–Laccadive Ridge](#), a massive submarine mountain range.

### Origin:

- The islands are coralline rock structures formed by volcanic activity over the Réunion hotspot.
- They consist of low-lying atolls set around central lagoons, including the **Great Chagos Bank**, which is the world's largest [atoll structure](#).

### Features:

- Diego Garcia:** The largest and most significant island, hosting a critical joint UK-US naval and air support facility.
- Biodiversity:** Home to some of the world's most resilient coral reefs and the world's largest coral atoll.
- Strategic Location:** Its mid-ocean position provides a vital military "foothold" for monitoring the [Indian Ocean](#) and surrounding regions.
- Demographics:** Currently, there is no permanent civilian population; the islands are inhabited only by military personnel and contractors.

- Connects forests of Ramnagar with [Shuklaphanta National Park](#) in Nepal.

### History:

- Established in **2012**.
- Part of the **Shivalik Elephant Reserve (since 2002)**.
- Recently prescribed by the **National Tiger Conservation Authority (NTCA)** as Uttarakhand's third Tiger Reserve.

### Key Features:

- Dominated by [Sal forests](#) with over 100 tree species including shisham, bamboo, teak, and chir pine.
- Major fauna include tiger, leopard, Asian elephant, and sloth bear.
- Functions as an ecological corridor in the Terai-Bhabar region.

### About Smooth-Coated Otter:

- The smooth-coated otter is a semi-aquatic mammal and one of the largest otter species in Asia, known for its sleek body and strong swimming ability.

**Scientific Name:** *Lutrogale perspicillata*

### Conservation Status

- Listed as Vulnerable on the International Union for Conservation of Nature ([IUCN](#)) Red List.

### Habitat

- Found in freshwater rivers, wetlands, mangroves, and estuaries.
- Requires clean water bodies with abundant fish and amphibians.
- Acts as a bio-indicator species, indicating [healthy aquatic ecosystems](#).

## INDIAN OCEAN NAVAL SYMPOSIUM (IONS)

- India assumed the Chairmanship of the [Indian Ocean Naval Symposium \(IONS\)](#) during the 9th Conclave of Chiefs held in Visakhapatnam in February 2026.



- The **Indian Ocean Naval Symposium (IONS)** is a **voluntary, inclusive and multilateral maritime forum** that promotes cooperation among navies of

## NANDHAUR WILDLIFE SANCTUARY

Nandhaur Wildlife Sanctuary is a protected forest area forming part of the Terai Arc Landscape, serving as a crucial [biodiversity corridor](#) between India and Nepal.

### Located in:

- Situated in **Nainital district, Uttarakhand**.
- Lies between the **Gola and Sharda Rivers**.

the Indian Ocean Region (IOR).

#### Established in:

- Established in 2008.
- Initiated by the [Indian Navy](#) to strengthen maritime cooperation among Indian Ocean littoral states.

#### History:

- India held the **inaugural chairmanship (2008–2010)**.
- Successive chairmanships included UAE, South Africa, Australia, Bangladesh, Iran, France and Thailand.
- In **2026**, India reassumed chairmanship, signalling renewed momentum for regional maritime cooperation.

#### Members:

- Around **25 member nations** from the [Indian Ocean](#) Region.
- Includes **observer countries and other maritime partners**, with participation spanning from the Atlantic to the Pacific.

#### Aim:

- To enhance collaboration among regional navies for maintaining peace, stability and secure sea lanes in the Indian Ocean.
- To promote interoperability, professional exchanges and coordinated responses to [humanitarian and security challenges](#).

#### Key Functions:

- Conducts Conclaves of Chiefs of Navies for strategic dialogue:** Provides a high-level platform where naval leaders discuss regional maritime challenges, security cooperation and future collaborative strategies.
- Facilitates maritime exercises (IMEX):** Organises joint drills and officer interactions to improve interoperability, operational understanding and confidence among participating navies.
- Operates through three specialised Working Groups:**

Dedicated groups translate strategic decisions into practical cooperation in key maritime domains.

- Humanitarian Assistance and Disaster Relief (HADR):** Develops coordination mechanisms and best practices for rapid response during natural disasters and humanitarian crises at sea.
- Maritime Security (MARSEC):** Focuses on tackling piracy, terrorism, trafficking and other non-traditional maritime threats through shared experiences and cooperation.
- Information Sharing & Interoperability (IS&I):**

Enhances real-time information exchange and compatible procedures to improve joint maritime operations and situational awareness.

- Promotes maritime domain awareness:** Encourages sharing of maritime data and coordinated surveillance to ensure safer sea lanes and effective monitoring of activities.
- Strengthens institutional collaboration:** Uses digital systems, training programmes and workshops to improve continuous engagement and long-term institutional cooperation among navies.

### TAKESHIMA/DOKDO ISLANDS



- The territory is a cluster of small, rocky outcroppings located in the **Sea of Japan** (referred to as the **East Sea** by South Korea).
- Composition:** It consists of two main volcanic islets—**Dongdo** (East Island) and **Seodo** (West Island)—along with roughly 30 to 90 smaller rocks and reefs.
- Geographical Position:**
- The islands lie roughly equidistant between the two nations but are physically closer to the [South Korean island](#) of **Ulleungdo** (approx. 87 km) than to Japan's **Oki Islands** (approx. 157 km).

#### Features:

- Terrain:** Volcanic rocks formed in the Cenozoic era. They feature steep cliffs, numerous sea caves, and a harsh, humid climate.
- Fisheries:** The surrounding waters are a [Goldilocks zone](#) where warm and cold ocean currents meet, creating one of the world's most fertile fishing grounds for squid, pollock, and cod.
- Energy Potential:** Recent geological surveys suggest the seabed may hold **600 million tons of gas hydrates** (often called fire ice).
- This natural gas resource is estimated to be worth nearly **\$10 billion annually**, making the islands a strategic energy asset.

## ARMENIA



- Armenia is a landlocked country in the South Caucasus (Transcaucasia), located at the crossroads of Eastern Europe and Western Asia. It is one of the world's oldest centres of civilisation, with a long history shaped by empires, invasions, and modern geopolitical conflicts.

**Capital:** Yerevan

### Neighbouring nations:

- Armenia shares borders with Georgia, Azerbaijan, Iran, and Turkey
- It also borders **Nakhchivan**, an exclave of [Azerbaijan](#), to the southwest.

### Key geological and physical features:

- **Mountainous terrain:** Armenia lies on the **Armenian Highland**, with an average elevation of about 1,800 metres, making it one of the most mountainous countries in the region.
- **Lesser Caucasus ranges:** Multiple mountain chains (Bazum, Pambak, Vardenis, etc.) dominate the landscape, interspersed with volcanic plateaus and deep valleys.
- **Volcanic origin:** Much of the land is formed from ancient lava flows, resulting in mineral-rich but stone-heavy soils.
- **Lake Sevan:** One of the largest high-altitude freshwater lakes in Eurasia, crucial for irrigation, hydropower, and climate regulation.
- **Seismic vulnerability:** Armenia lies in an active seismic zone, highlighted by the devastating **1988 Spitak earthquake**.
- **Continental climate:** Hot, dry summers and cold winters, with sharp climatic variation by altitude.

## MOUNT ACONCAGUA

summitted Mount Aconcagua in Argentina as part of her 7-[Summit Mountaineering Expedition](#).



- Mount Aconcagua is the highest mountain in South America and the Western Hemisphere, and the tallest peak outside Asia. With an elevation of about 22,831 feet (6,959 m), it is one of the most challenging non-technical climbs among the Seven Summits.

### Located in:

- Situated in **western Mendoza Province**, near the **Argentina–Chile border**
- Part of the **Southern Andes mountain range**

### Geological features:

- **Volcanic origin**, though it is **not an active volcano**.
- Comprises **two summits (North and South)** connected by a ridge called Cresta del Guanaco.
- Formed by complex **tectonic uplift of the Andes**, rather than volcanic cone-building.
- Exhibits extreme conditions such as **thin air, high winds, and sub-zero temperatures**, making altitude sickness a major challenge.

### Significance:

- **Geographical:** Highest point in the Western Hemisphere and **Southern Hemisphere**.
- **Mountaineering:** One of the **Seven Summits**, attracting climbers worldwide.

## JAPAN



Japan is a constitutional monarchy with a parliamentary system, governed under the 1947 Constitution, where



the Emperor is a ceremonial head of state and real executive power rests with the Prime Minister and the Cabinet.

#### Location:

- An island nation in East Asia, located in the western North Pacific Ocean
- Lies off the east coast of the [Asian continent](#), forming a northeast–southwest arc

**Capital:** Tokyo.

#### Neighbouring Countries / Seas:

- Sea of Japan (East Sea), East China Sea, Pacific Ocean, and [Sea of Okhotsk](#).

#### Key Geological & Physical Features

- Composed mainly of **four major islands**: Hokkaido, Honshu (largest), Shikoku, Kyushu.
- Highly **mountainous terrain** (over 80% mountains).
- Located on the [Pacific Ring of Fire](#) → frequent **earthquakes and volcanic activity**.
- **Mount Fuji (3,776 m)** – highest peak and iconic stratovolcano.
- Short, swift rivers; narrow coastal plains with dense population.
- Tectonically shaped by the **Pacific and Philippine plates** subducting beneath the [Eurasian Plate](#).

## BANGLADESH



- Bangladesh is a South Asian country that emerged as an independent nation in 1971 after the Liberation War against Pakistan.

**Location:** It is located in [South Asia](#), forming part of the historical Bengal region.

**Bordering Nations:**

- **India** (West Bengal, Assam, Meghalaya, Tripura, Mizoram) – on three sides
- **Myanmar (Burma)** – southeast
- Opens to the **Bay of Bengal** in the south

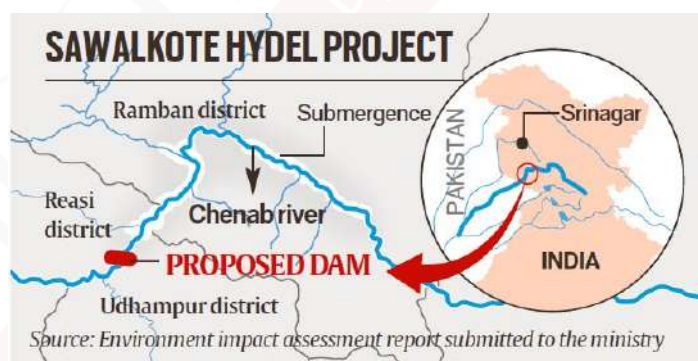
#### Key Geographical & Geological Features:

- **Deltaic Country:** Situated in the vast delta of the [Ganges \(Padma\)](#), [Brahmaputra \(Jamuna\)](#), and

**Meghna rivers** – one of the largest river deltas in the world.

- **Low-Lying Terrain:** Majority of the land is less than 30 feet above sea level, making it highly flood-prone.
- **Sundarbans:** World's largest mangrove forest located in the southwest.
- **Chittagong Hill Tracts:** Hilly region in the southeast.
- **Highly Riverine System:** Rivers reshape the landscape frequently, causing both fertile soil deposits and devastating floods.
- One of the **most densely populated countries** in the world.

## SAWALKOT HYDROELECTRIC PROJECT



- The **Sawalkot Hydroelectric Project** is a 1,856 MW [Run-of-the-River \(RoR\)](#) hydropower project being developed on the Chenab River in Jammu & Kashmir. It is the **largest hydropower project in J&K** and among the biggest in Northern India.
- The project is being implemented by National Hydroelectric Power Corporation (NHPC).

#### Location:

- **River:** Chenab River – A major tributary of the Indus, crucial for hydropower generation in the Himalayan region.
- **Basin:** Indus Basin ([Western Rivers system](#)) – Part of the western rivers allocated to India for non-consumptive use under treaty provisions.
- **Location:** Ramzan and Udhampur districts, Jammu & Kashmir.
- **Falls under India's share as per Indus Waters Treaty** – India can develop run-of-the-river projects on western rivers within specified design limits.

#### Key Features:

- **Installed Capacity: 1,856 MW** – Makes it the largest hydropower project in J&K, significantly boosting regional capacity.

- **Project Type: Run-of-the-River (RoR)** – Generates electricity using natural river flow without creating large storage reservoirs.
- **Dam Type: 192.5 m RCC Gravity Dam** – A strong concrete structure designed to withstand high Himalayan river pressures.
- **Power Infrastructure: Underground powerhouse with nine turbines** – Enhances security and efficiency while minimizing surface disturbance.
- **Annual Output: ~8,000 Million Units (MU)** – Capable of meeting substantial electricity demand, especially during winter shortages.
- **Environmental Clearance: Reaffirmed September 2025** – Approved after review by the Expert Appraisal Committee for compliance with norms.

## TAMIL BRAHMI INSCRIPTIONS IN EGYPT



- The discovery refers to ancient Indian inscriptions—primarily in **Tamil Brahmi script**—found inside Egyptian tombs, indicating the presence of Indian visitors or traders in Roman-era Egypt.

### Located In:

- The inscriptions were documented in six tombs within the **Valley of the Kings**, part of the Theban Necropolis along the [Nile River](#) in Egypt.

### About Valley of the Kings:

- The Valley of the Kings is a major ancient Egyptian burial site where pharaohs of the New Kingdom were interred in rock-cut tombs deep within desert hills.

### Located In:

- It lies on the west bank of the Nile River near modern-day Luxor in Upper Egypt, forming part of the ancient city of Thebes.
- In 1979, it was designated a [UNESCO World Heritage Site](#) as part of the Ancient Thebes complex.

### History:

- **Period of Use:** Primarily used during the 18th,

19th, and 20th Dynasties (c. 1539–1075 BCE).

- **Royal Burials:** Served as the burial ground for pharaohs from Thutmose I to Ramses X, along with some queens and high officials.
- **Shift in Burial Practice:** New Kingdom rulers chose this hidden valley to prevent tomb robbery, moving away from [pyramid burials](#).
- **Architectural Features:** Tombs include descending corridors, pillared halls, burial chambers, and deep shafts designed to deter robbers.
- **Religious Significance:** Walls were decorated with funerary texts like the “Book of the Dead,” “Book of Gates,” and “Book of That Which Is in the Underworld,” guiding the king through the afterlife.
- **Archaeological Importance:** Over 60 tombs have been discovered, including the famous [tomb of Tutankhamun](#) (KV62).

## REYNISFJARA BEACH

- Reynisfjara is a **world-famous black volcanic sand beach**, known for its dramatic geological formations and powerful Atlantic waves. It is considered one of Iceland’s most iconic natural landmarks.

### Location:

- Situated near the town of Vík í Mýrdal in southern Iceland.
- Part of the Katla [UNESCO Global Geopark](#).
- Faces the North Atlantic Ocean along Iceland’s rugged south coast.

### Origin:

- Formed from **volcanic basalt lava flows** that cooled rapidly upon contact with the ocean.
- The black sand is composed of eroded volcanic rock from nearby volcanoes like Eyjafjallajökull.
- The striking hexagonal basalt columns at Hálsanefshellir cave are products of **columnar jointing**, a volcanic cooling process.

### Key Features:

- **Black Volcanic Sand:** Dark, charcoal-like sand created by pulverized basalt.
- **Reynisdrangar Sea Stacks:** Towering rock formations rising from the ocean.
- **Basalt Columns & Caves:** Natural geometric formations resembling organ pipes.
- **Sneaker Waves:** Sudden, powerful waves that surge inland unpredictably, making it one of Iceland’s most hazardous beaches.

### Reason for Diminishing:

- **Persistent Easterly Winds:** Unlike usual

[southwesterly winds](#), strong easterlies pushed sand westward along the coast.

- **Natural Barrier Effect:** The Reynisfjall mountain blocks westward sand movement, preventing replenishment.
- **High-Energy Atlantic Swells:** Heavy winter waves intensified coastal scouring.
- **Landslide & Structural Collapse:** Part of the basalt formation beneath Reynisfjall collapsed, narrowing the shoreline.
- **Climate Variability:** Changing wind and wave patterns may be linked to broader climatic instability.

## INDIA:

### PENNAIYAR RIVER



- The Pennaiyar River, also known as Thenpennai / Ponnaiyar in Tamil and Dakshina Pinakini in Kannada, is a major east-flowing inter-state river of southern India, significant for irrigation, industry, and water security in Tamil Nadu and Karnataka.

#### Origin:

- Originates in the **Nandi Hills, Chikkaballapura district**, Karnataka
- Arises from the [Eastern Ghats system](#)

#### States it flows through:

- **Karnataka** – Upper riparian state
- **Tamil Nadu** – Lower riparian state

Tamil Nadu is more dependent on the river for irrigation and drinking water, making downstream flows politically and economically sensitive.

#### Major tributaries of Pennaiyar:

- [Markandeya River](#)
- Varaha Nadhi
- Pambar River
- Pampar River
- *Markandeya River is central to the present inter-*

*state dispute.*

#### Mouth:

- Empties into the [Bay of Bengal](#)
- Forms a **delta near Cuddalore**, Tamil Nadu

#### Key features:

- **Length:** ~497 km (second longest river in Tamil Nadu after Cauvery)
- **Nature:** Seasonal river; flows mainly during **South-West and North-East monsoons**
- **Major dams:**
  - [Krishnagiri Dam](#)
  - **Kelavarapalli Dam**
  - **Sathanur Dam** (largest; ~7.3 TMC gross capacity)
- **Economic role:**
  - Irrigates districts in **Karnataka (Chikkaballapura, Kolar, Bengaluru Rural/Urban) and Tamil Nadu (Krishnagiri, Dharmapuri, Tiruvannamalai, Villupuram, Cuddalore).**
- **Environmental concerns:**
  - Severe **industrial pollution** (notably from Bengaluru–Hosur belt).
  - **Sand mining** and shrinking perennial character.
- **Cultural significance:**
  - Mentioned in [Sangam literature](#) and **Tevaram hymns**.
  - Several ancient temples located along its banks.

### KARIMPUZHA WILDLIFE SANCTUARY

- **Karimpuzha Wildlife Sanctuary** is a protected area in the **Western Ghats**, known for its exceptional biodiversity, wide altitudinal range, and intact forest ecosystems.

#### Located in:

- The sanctuary falls under the Nilambur forest landscape, an ecologically rich belt of the [Western Ghats](#) known for dense forests and high endemism.
- Its position on the Nilgiri slopes places it within a critical Western Ghats biodiversity hotspot, influencing rainfall, vegetation, and species diversity.



### Key geographical & ecological features:

- **Sharp altitudinal gradient (40 m to >2,550 m):**  
The wide elevation range creates varied climatic conditions, supporting species adapted to both lowland and montane ecosystems.
- **Seven forest types (unique in Kerala):** The coexistence of seven distinct forest types in one landscape makes it ecologically exceptional and highly biodiverse.
  - **Semi-evergreen & evergreen forests:** Support dense canopy, high rainfall species, and endemic flora.
  - **Moist deciduous forests:** Act as crucial habitats for large mammals and seasonal biodiversity.
  - **Sub-tropical savannah & hill forests:** Provide transitional ecosystems supporting grassland–forest species.
  - **Montane wet temperate forests & grasslands:** Sustain high-altitude endemics and regulate hydrology.
- **Ecological corridor:** By linking [Silent Valley National Park](#) and [Mukurthi National Park](#), it ensures habitat continuity for wildlife movement across state boundaries.
- **New Amarambalam Reserve inclusion:** This largely undisturbed forest tract preserves near-pristine ecosystems, serving as a genetic and ecological refuge within the Western Ghats.

### Significance:

- Hosts key Western Ghats endemics such as [Nilgiri tahr](#) and lion-tailed macaque
- Rich diversity of birds, butterflies, odonates, amphibians, reptiles, and freshwater fish

## COUNTRY'S FIRST UNDERWATER ROAD-CUM-RAIL TUNNEL UNDER BRAHMAPUTRA RIVER

The Cabinet Committee on Economic Affairs (CCEA) has approved India's first underwater road-cum-rail tunnel under the Brahmaputra River in Assam.

### What it is?

- A 33.7 km four-lane access-controlled Greenfield connectivity project, including a 15.79 km twin-tube underwater tunnel beneath the [Brahmaputra River](#) in Assam.
- It will connect Gohpur (NH-15) and Numaligarh (NH-715), reducing distance from 240 km to 34 km.

### Key Features:

- **Twin-Tube Underwater Tunnel:** Two parallel tubes ensure traffic segregation, safety redundancy, and easier emergency evacuation.
- **Road-cum-Rail Connectivity:** Combines highway and [railway infrastructure](#) to enable seamless passenger and freight movement.
- **EPC Mode Development:** Government funds the project while a contractor handles end-to-end execution under fixed timelines.

### About Brahmaputra River:

The Brahmaputra River is one of the largest rivers of Central and South Asia and ranks among the world's top rivers in terms of average discharge.

### Origin:

- Originates from the [Chemayungdung Glacier](#) near Lake Mapam in Tibet at ~5,300 m elevation.
- Known as **Yarlung Tsangpo** in Tibet, **Siang/Dihang** in Arunachal Pradesh, and **Jamuna** in Bangladesh.

**Nations it flows through:** China ([Tibet Autonomous Region](#)), India, and Bangladesh.

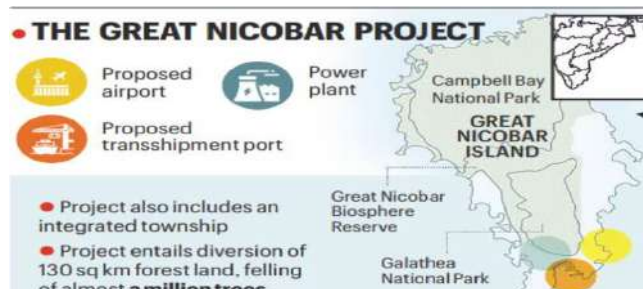
### Indian States it flows through:

- Arunachal Pradesh
- Assam
- (Drainage basin also covers Nagaland, Meghalaya, West Bengal, and Sikkim)

### Key Features of the Brahmaputra River:

- **Length – ~2,900 km:** Flows from the [Tibetan Plateau](#) to the Bay of Bengal, making it one of Asia's longest rivers.
- **Massive Discharge:** Among the world's highest river discharges, contributing significantly to the Ganga-Brahmaputra system.
- **World's Deepest Gorge:** Cuts through the [Yarlung Tsangpo Grand Canyon](#), one of the deepest gorges globally.
- **Highly Braided Channel:** In Assam, heavy sediment and reduced slope create multiple shifting river channels.
- **Delta Formation:** Merges with the Ganga to form the world's largest delta in Bangladesh.

## NGT APPROVE GREAT NICOBAR PROJECT



- The Great Nicobar Project is a **mega infrastructure development plan** valued at around ₹80,000 crore.
- It aims to transform Great Nicobar Island into a **strategic maritime and economic hub** in the [Indian Ocean Region](#) (IOR).

#### Location:

- Situated in **Great Nicobar Island**, the southernmost island of the Andaman and Nicobar archipelago.
- Lies close to the [Malacca Strait](#), one of the world's busiest shipping lanes.
- Spread over **166 sq km**, with diversion of around **130 sq km of forest land**.

#### Aim of the Project:

- The primary objective is to transform the island into a major economic and strategic hub. It aims to leverage the island's location near the Malacca Strait to boost India's maritime presence and [regional security](#).

#### Key Features:

- **International Transshipment Port:** A massive port facility designed to handle global shipping traffic.
- **Integrated Township:** Development of a modern urban center to support the project's activities.
- **Dual-Use Civil-Military Airport:** A strategic airfield intended for both commercial and defense purposes.
- **Power Infrastructure:** A 450-MVA hybrid gas and solar power plant to provide reliable energy.
- **Land Diversion:** The plan entails diverting 130 sq km of forest land and felling nearly one million trees.

## RAVI RIVER



- The Ravi River is one of the five rivers of the Indus river system, which give Punjab its name ("Land of Five Rivers").
- It is an eastern river under the [Indus Waters Treaty](#), allocated exclusively to India.

#### Origin:

- Originates near the [Rohtang Pass](#) in Himachal Pradesh.
- Rises at an elevation of around **4,400 metres above sea level**.
- Initially flows as two streams — **Budhil and Tantgari** — which later merge.

#### Course of the River:

- **Countries:** Flows through **India and Pakistan**.
- **Within India:**
  - Himachal Pradesh (mainly Chamba district).
  - Punjab (Gurdaspur, Amritsar districts).
  - Forms part of the **India–Pakistan international boundary** before entering Pakistan.
- **In Pakistan:**
  - Flows past **Lahore**.
  - Eventually joins the [Chenab River](#).

#### Tributaries of Ravi River:

- **Right Bank Tributaries:** Siul River, Baira River, Budhil River, and Tant Gari
- **Left Bank Tributaries:** Ujh River and Chirchind Nala.
- **Other Tributaries / Streams:** Bhadal River, Sewa River, Bein and Basantar.

**Ravi is a Tributary of:** The Ravi River ultimately joins the **Chenab River**, which is a major western tributary of the Indus River.

#### Key Features of the Ravi River:

- The river flows through both India and Pakistan, with nearly half its course lying within Indian territory.
- **Catchment area in India:** A significant drainage basin in Himachal Pradesh and Punjab supports irrigation and hydropower projects.
- Flow is sustained by [Himalayan snowmelt](#) in summer and intensified by heavy monsoon rains from June to September.
- Intense rainfall often leads to high discharge levels, causing seasonal flooding in downstream areas.
- **Controlled by major dams** like Chamera I, II, III and Ranjit Sagar (Thein) Dam – These projects regulate water flow for hydropower generation, irrigation, and flood management.
- **Shahpur Kandi Dam:** A multipurpose project on the Ravi River near the Punjab–J&K border, aimed at utilising surplus eastern river [waters for irrigation](#) and reducing flow into Pakistan.